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CHAPTER

Microbiology

The word microbe (microorganism) is used to describe an organism that is so small that, normally, it cannot be seen without the aid of a microscope. Viruses, Bacteria, Archaea, fungi, and protista are all included in this category.

Prokaryotes : Phylogenetic overview

Microbes are found in all three major kingdoms of life: the Bacteria, the Archaea and the Eukarya. The Bacteria and Archaea are prokaryotes, while all other microbes are **eukaryotes.**

Woese *et al.* noted that bacteria are distant from plants and animals and, by contrast, plants and animals are not so far from each other. Therefore, they established a new superior concept of domains over the kingdom, and proposed three domains, Bacteria, Archaea and Eukarya in 1991

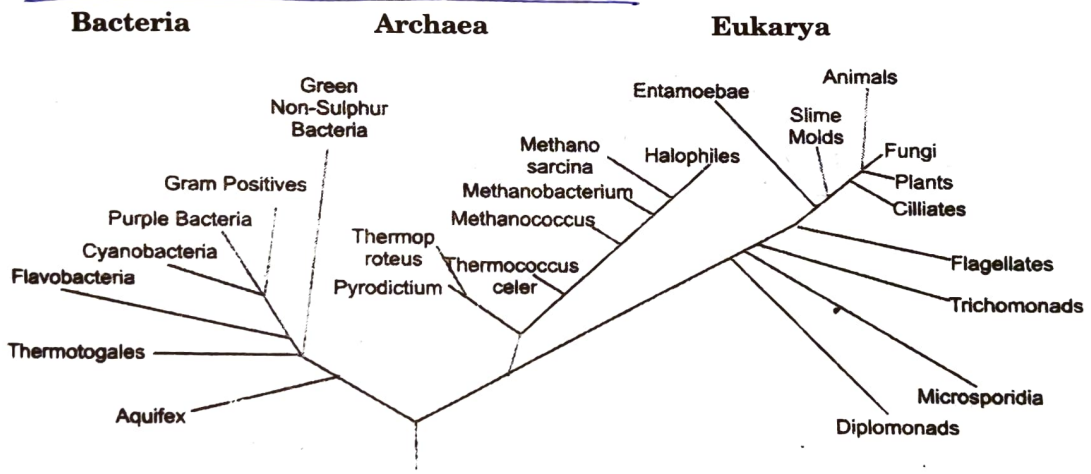


Fig : Universal phylogenetic tree derived from comparative sequencing of 16s or 18s ribosomal RNA

The domains **Archaea** and **Eukarya** are distinctly related to each other. Eukarya includes the organisms which possess glycerol fatty acyldiester as membrane lipids and eukaryotic rRNA. The domain Bacteria consists of such member's which have membrane lipids as diacyl glycerol diesters and eubacterial rRNA. The member of domain Archea consists of isoprenoid glycerol diester (or diglycerol tetraether) lipids in their membrane and archebacterial rRNA.

In a modern sense, bacteria, cyanobacteria, actinomycetes, etc. are distributed in the domain bacteria; methanogens extremely thermophilic organisms, extremely halophilic organisms, etc. are in the domain Archaea; and molds, yeasts, fungi, animals, plants etc. are in the domain Eukarya.

Structure of Bacterial Cell

Prokaryotic cells

Bacteria and Archaea are small, generally <1 mm–50 mm width diameter though cells up to 750 mm have been found (*Thiomargarita namibiensis*). They are single-celled microorganisms that belong to a group called prokaryotes so classified because their DNA is not enclosed within a nuclear membrane (Gr. pro before; *karyote*, nucleus). Their internal cells structure is simple (Fig.) with most of their cellular complexity associated with the cell surface structures. Prokaryotes lack mitochondria and chloroplasts, the organelles associated with energy production in eukaryote cells; and internal membrane structures associated with protein synthesis and processing,

such as the endoplasmic reticulum and Golgi apparatus. However internal membranes are found in some groups of prokaryotes associated with processes such as photosynthesis or methane oxidation. In spite of this simplicity in structure there is a large amount of variation in the appearance of the cells when observed under the microscope. Two shapes of cell predominate, the coccus (spherical) and the bacillus, or rod, but a wide range of other shapes (morphologies) also can be found as shown in *Table*

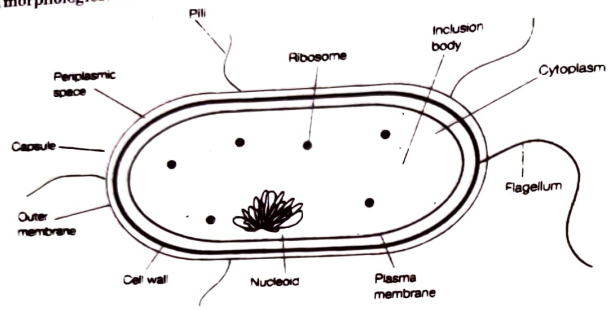


Fig. Diagram of a prokaryote cell (Gram-negative).

Table: Shapes of prokaryotes and characteristic associations of cells seen using light microscopy

Shape	Organization of cells	Examples
Cocci - spherical cells	In chains	*Streptococcus*
	In grape-like clusters	*Streptococcus*
	In pairs (diplococci)	*Neisseria*
	In cubes of cells (packets)	*Sarcina*
Rods/bacilli - rods may be very short or long	Singly	*Pseudomonas*
	In filaments of attached cells	*Bacillus sp.*
Tightly coiled flexible cells-(spirochaetes)		*Treponema*
Comma shaped - called a vibrio		*Vibrio*
Spirillum - long, curved rigid rods		*Rhodospirillum*
Appendaged - cells have stalk or tube-like extensions		*Rhodomicrobium*
Pleomorphic - bacteria vary in shape		*Corynebacteria*
Branching mycelium-like filaments		*Streptomyces*

Prokaryotes are ubiquitous. They are a highly successful and diverse group of organisms that can obtain energy and carbon from a wide range of sources and therefore can colonize every niche on our planet from deep ocean trenches to volcanic craters. In the 1970s, using DNA sequencing information, it was found that the group we know as the bacteria could be split into two, the Bacteria (Gr. *bakterion*, a small rod) and the Archaea (Gr. *Archaios*, ancient) and it appears that these two groups evolved away from each other very early in the history of living things. Members of the Bacteria include some of the more familiar bacteria such as *Escherichia coli* and *Staphylococcus aureus*, the prokaryotes that are best studied and understood. The Archaea are a very diverse group of organisms, which differ from the Bacteria in a number of features having, in particular, very different cell walls and membranes. This group includes archaea that are capable of existing in extreme environments, such as hot springs (examples: *sulfolobus* and *Pyrococcus*) and high salinity (*Halobacterium*), and the methanogens such as *Methobacterium*, which produce methane as a result of metabolism.

Table. Some differences between Bacteria and Archaea

Feature	Bacteria	Archaea
Cell wall	Muramic acid present	Muramic acid absent
Lipids	Ester-linked	Ether-linked
Methanogenesis	Absent	Possible
RNA polymerases	One	Several
Initiator tRNA	fMet	Met
Ribosomes	Sensitive to streptomycin and chloramphenicol	Resistant to streptomycin and chloramphenicol
	Resistant to diphtheria toxin	Sensitive to diphtheria toxin

(*i*) The cytoplasm and ribosomes

The cytoplasm of prokaryotes is aqueous, containing a cocktail of molecules, ribonucleic acid (RNA) and proteins necessary for the cell functions. The main structures found in the cytoplasm, common to all prokaryotes, are the **ribosomes.** The ribosomes consist of a small and a large subunit, made up of a complex of proteins and RNA, and are the sites of protein synthesis in the cell. The ribosomes in prokaryotic cells, although similar in shape and function to those of eukaryotic cells, are different in the nature of the proteins and RNAs that make up their structure. This has proved to be very useful to the human population as antibiotics that act by inhibiting protein synthesis in prokaryotes are not effective against eukaryotic protein synthesis, thus allowing selective toxicity. Archaeal ribosomes are the same size as those of the Bacteria but, in some features, they are similar to eukaryotic ribosomes in that they are resistant to the antibiotics streptomycin and chloram- phenicol, and sensitive to the action of diphtheria toxin.

(*ii*) Prokaryote DNA

Prokaryote DNA is located within the cytoplasm. It often consists of single chromosome which varies in size between different species of bacteria (the *E. coli* chromosome is 4.6×10^6 base pairs long.) The DNA is circular, tightly supercoiled and associated with proteins. Although the chromosome is not contained within a nuclear membrane, it is often seen as a discrete area within the cell in electron micrographs which may be referred to as the **nucleoid**. Many bacteria also contain small molecules of extra- chromosomal. DNA called **plasmids**.

(*iii*) Other internal prokaryote features

Some prokaryotes contain structures associated with specialist functions. Granular structures, called *inclusion bodies*, can often be seen under the light microscope. These granules are normally used for storage and may be either membrane bound, such as poly-β-hydroxybutyrate (PHB) granules, or found scattered in the cytoplasm, an example being polyphosphate granules (also called **metachromatic granules** as they change the color of basic dyes). **Lipid droplets** can also be seen in some prokaryotes. An interesting inclusion body, the gas **vacuole**, is found in *Cyanobacteria* and other photosynthetic bacteria that live in an aqueous environment. This protein-surrounded vacuole provides buoyancy, allowing the prokaryote to float near the surface of the water.

Although bacteria lack membrane bound organelles, invaginations of the plasma membrane called mesosomes are often seen in electron micrographs.

(*iv*) Endospores

A number of Bacteria, mainly studied in the *Bacillus* and *Clostridium* genera, produce a special reproductive structure called the **endospore**. This can be seen in the light microscope using either specialist spore stains such as malachite green or by phase-contrast microscopy. The spore has a number of layers surrounding the genetic material making it incredibly resistant to all kinds of environmental stress such as heat, UV irradiation, chemical disinfectants and drying. As a number of important pathogenic bacteria are spore producers, sterilization measures must therefore be designed to remove these hardy structures some of which can withstand boiling for several hours.

(v) Bacterial cell wall and surface

The cytoplasm of all Bacteria is enclosed within a **plasma (cytoplasmic) membrane** external to which, in most cases of Bacteria, is a rigid **cell wall** made up of sugars and amino acids called **peptidoglycan**. The role of the cell wall is to protect the cell from lysis resulting from osmotic pressure and it also gives shape to the cell. Some Bacteria, the mycoplasma, do not have cell walls and therefore are unable to survive outside an animal host which provides it with the right osmotic environment. Gram-negative Bacteria have an additional outer membrane containing lipopolysaccharide (LPS). External to this may be other layers of polysaccharide or protein making up a capsule or slime layer. Layers external to the cell wall may be referred to as the cell **envelope**.

A number of features are found on the surface of the Bacteria which aid the survival of the cell in the environment. Thin hair-like proteinaceous appendages, several micrometers long, called **pili** (s. pilus) or **fimbriae** (s. fimbria) are often seen projecting from the surface of the cell in electron micrographs. These structures are normally associated with the adhesion of Bacteria to surfaces although there is one particular type of pilus, the sex pilus, which is involved in the transfer of genetic material from one bacterium to another. Motile prokaryotes have one or more **flagella** (s. flagellum) which are slender, rigid structures up to 20 m long. Rotation of the flagellum in relation to the cell allows the movement of the bacterium. Normally, prokaryotes move in response to external stimuli; an example of this is **chemotaxis** when the microbe move towards nutrients or away from harmful substances as a result of chemical gradients.

Bacterial Cell Envelope and Cell wall Synthesis

The Gram stain

The Bacteria are frequently divided into two groups on the basis of their reaction to a stain devised by Christian Gram in 1884. The differential reaction to the staining procedure is because of the structure of the cell envelope in these two groups of bacteria. **Gram -positive** Bacteria have a single membrane called the **cytoplasmic** (or **plasma**) membrane, surrounded by a thick layer of **peptido-glycan** (20-80 nm). The **Gram-negative** Bacteria have only a thin layer of peptidoglycan (1-3 nm) but on the outside of this there is a further **outer membrane** which acts as an additional barrier.

The procedure for the Gram stain is as follows. Fixed cells are stained with a dark stain such as crystal violet, followed by iodine which complexes with the stain in the cell wall of the bacteria. Alcohol is added, which washes the dark stain of crystal-violet-iodine complex out of the thin-walled cells but not from those that have thick cell walls. Finally, a paler stain such as carbol fuschin, called a counter-stain, is added which stains the decolorized cells pink. This light stain is not seen on the cells that retained the first darker stain. The cells that retain the crystal-violet-iodine complex (with thick cell walls) are called Gram-positive and appear dark purple under light microscopy. The ones that lose the stain (with thin cell walls and an outer membrane) are called Gram-negative and stain pink or pale purple.

Lipid bilayer

The plasma membrane is a semi-permeable lipid bilayer which acts as a barrier between the cytoplasm and the surrounding environment. The membrane is made of phospholipids (*Fig.*) such as phosphatidyl choline, which consists of a polar headgroup attached via glycerol to two long chains of non-polar fatty acids. Such molecules which have both polar and non-polar groups are called **amphipathic**. The polar headgroups are **hydrophilic** (Gr. *hydro,* water; *philic,* loving) and the fatty acids are **hydrophobic** (Gr. *phobis,* hating). This means that in an aqueous environment phospholipid molecules orientate themselves to ensure that the polar headgroups are associated with the water molecules and the hydrophobic chains are tucked away in the interior of the bilayer away from the water. Hence, a lipid bilayer can be formed, called a membrane (*Fig.*) about 5-10 nm thick. However, recently it has been shown that sterol-like molecules called **hopanoids** are found in Bacterial membranes and methanotrophs contain a large number of sterols in their internal membrane systems.

The plasma (cytoplasmic) membrane

Embedded in the lipid bilayer are proteins of various functions (*Fig.*) including transport proteins, proteins involved in energy metabolism and receptor proteins that can detect and respond to chemical stimuli. **Integral** proteins are those that are fully associated with the membrane and may penetrate all the way through. These proteins therefore contain hydrophobic amino acids in the regions which are buried in the lipid.

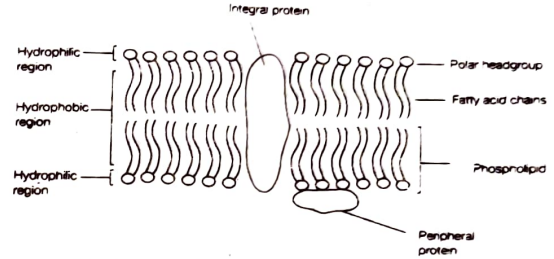


Fig. Diagram of the structure of the plasma membrane

Peripheral proteins are ones that are only loosely associated by charge interactions with the positively charged polar headgroups of the phospholipids or on integral proteins and can be removed from purified membranes by washing with salt solutions. The lipids and proteins are fully mobile and move in relation to each other. This widely accepted model of membrane structure is called the fluid mosaic model.

Transport across membranes

Three different types of energy-driven transport systems have been found in bacteria.

- Proton-motive force driven transport which involves a single membrane spanning protein. Energy is provided by the proton gradient across the cytoplasmic membrane. An example is the lactose permease protein in *E. coli.*
- ABC (ATP-binding cassette) transport systems which consist of three components.

  An example of this type of mechanism is the dipeptide transport system in *E. coli.*
- Group translocation is unique to bacteria and involves the chemical modification of the substrate during transport. The most studied system is the glucose phosphotransferase system in *E. coli* in which glucose is phosphorylated as it moves across the membrane. Group translocation of a number of sugars have been found.

The bacterial cell wall

Outside the plasma membrane is the **peptidoglycan** cell wall which is made of sugars and amino acids. This is sometimes called the murein layer. The role of the cell wall is to provide rigidity and strength, preventing the cell from osmotic lysis when placed in dilute environments.

It consists of long polymers of two sugar derivitives NAG and NAM with side chains of four alternating D- and L-amino acids attached to the NAM. Rigidity is achieved by crosslinks between the amino acid chains; normally from the third amino acid in one chain to the fourth amino acid in another chain. The nature of the amino acid side chains, and the links that join them, vary between bacterial species; however, the third amino acid is always a diamino acid (i.e. it has two amino groups, so it can form an extra peptide bond) and the fourth amino acid is normally D-alanine (D-Ala). In *E. coli* and most other Gram-negative Bacteria there is a direct link between the diamino acid, meso-diaminopimelic acid and D-Ala. In Gram-positive Bacteria the linkage often contains a bridge of amino acids: for example, in *Staphylococcus aureus,* there is a bridge of five glycines between D-Ala and L-lysine. Multiple crosslinks within and between chains makes peptidoglycan a very strong and rigid structure.

There are several unique features of peptidoglycan including:

- The presence of NAM; this sugar is not found in eukaryotic cells;
- The presence of D-amino acids; L-amino acids are normally found in proteins.

These features make the peptidoglycan a target for antimicrobial agents that destroy prokaryotic cells specifically, but do not harm eukaryotic cells; an example of this is the antibiotic **penicillin. Lysozyme,** a natural antibacterial agent found in tears and natural secretions, breaks down the β(1→4) linkage between NAM and NAG. Removal of the cell wall under conditions where the osmolarity of the medium is the same as the inside of the cell (isotonic solution) results in the formation of round **protoplasts** (Gram-positives) or **spheroplasts** (Gram-negatives) which survive as long as the isotonicity is maintained. These structures lyse if placed in a more dilute medium, illustrating the importance of peptidoglycan to the cells' survival.

Gram-positive cell walls also contain large amounts of another polymer, called **teichoic acid**, made up of glycerol or ribitol joined by phosphate groups D-Ala, glucose or sugars may be attached to the glycerol or ribitol and the polymers are attached either directly to the NAM in the peptidoglycan or to lipids in the membrane (in this case they are called **lipoteichoic** acids). The function of these molecules is unclear but they may have a role in maintaining the structure of the cell wall and in the control of **autolysis**.

Cell walls of a number of genera including *Mycobacterium, Corynebacterium* and *Nocardia* contain waxy esters of mycolic acids, which are complex fatty acids.

The periplasmic space

The outer membrane of Gram-negative Bacteria acts as an additional barrier protecting the peptidoglycan from toxic compounds such as lysozyme which act on the cell wall. It creates an aqueous space between the two membranes called the **periplasmic space** which is thought to have a gel-like structure with a loose network of peptidoglycan running through it.

The outer membrane

The outer membrane of Gram-negative Bacteria is made up of phospholipids but it also contains some unique features.

- Pores formed by proteins called **porins** such as OmpF and OmpC allow the passive diffusion of small molecules into the periplasmic space.
- An abundant small lipoprotein called Braun's lipoprotein that is covalently bound to the peptidoglycan and is embedded in the outer membrane by its hydrophobic lipid, therefore holding the peptidoglycan and outer membrane close together.
- LPS molecules are found in the outer membrane projecting into the surrounding medium.

Most importantly, the lipid A portion of the LPS molecule is frequently highly toxic to mammals. Called endotoxin, its presence in the blood stream, even at very low concentrations, leads to toxic shock and death. This is true of pathogenic as well as non-pathogenic Bacteria.

Extracellular layers

Different words are used to describe the layers of material often seen on the surface of cells. A well organized dense structure is called a **capsule** whereas if the material is diffuse and easily lost it is called **a slime layer. Glycocalyx is** often used to describe both capsules and slime layers that are made up of polysaccharides; however; some bacteria have proteinaceous capsules. These additional surfaces layers have a number of functions:

- they act as permeability barriers to the cell surface;
- they protect the Bacteria from phagocytosis;
- they protect the cell from desiccation;
- they aid in the attachment (adhesion) of prokaryotes to surfaces.

Synthesis of peptidoglycan

The stages in the synthesis of peptidoglycan are as follows.

1. UDP-*N*-acetylmuramic acid (NAM) is synthesized in the cytoplasm.
2. L-Ala is isomerized to D-Ala.
3. Two molecules of D-Ala are joined to form a dipeptide.
4. Amino acids are added sequentially to the UDP-NAM ending with the addition of D-Ala-D-Ala to form a UDP-NAM-pentapeptide (Fig.).
5. The NAM-pentapeptide is transferred to the bactoprenol-phosphate carrier located in the cytoplasmic membrane.
6. *N*-acetylglucosamine (NAG) is transferred from UDP-NAG to the NAM-pentapeptide to give the final peptidoglycan repeating subunit. Other peptide side chains may also be added at this point such as the pentaglycine interbridge found in *Staphylococcus aureus* peptidoglycan.
7. The repeating subunit is translocated to the external surface of the membrane.
8. The NAG-NAM-pentapeptide is transferred from bactoprenol onto the growing end of peptidoglycan chain by **transglycosylation** and the bactoprenol carrier is recycled back to the inside of the membrane. This stage is sensitive to the antibiotic vancomycin.
9. Finally, cross-links are formed between the peptide side chains by **transpeptidation.** These can be either directly, as in *Escherichia coii* where diaminopimelic acid (DAP) is linked to D-Ala on another side chain with the loss of the terminal D-Ala or, via a pentaglycine bridge as occurs in *Staphylococcus aureus.* It is this stage that is sensitive to the β-lactam antibiotics.

Insertion of peptidoglycan into existing cell walls

In order to protect the cell from lysis by turgor pressure within the cell, cross linked peptidoglycan must be continuous across the cell surface. Gaps in the peptidoglycan therefore need to be kept to the minimum and carefully. controlled by the cell. A carefully controlled process allows the creation of breaks in the existing peptidoglycan by a group of enzymes called **autolysins.** These can either break the polysaccharide chains or the peptide cross-links and are controlled by **autolysin inhibitors.** New peptidoglycan is then added onto the existing peptidoglycan. In some Bacteria, such as the Gram-positive cocci, only a few sites for the insertion of new material exist in the cell surface and the principal site is where the cell will eventually divide (the area of septum formation). In many rod-shaped Bacteria, both Gram-positive and -negative, although there is major insertion of peptidoglycan in the area of septum formation, there are also multiple sites around the cell.

Bacterial Movement and Chemotaxis

Motility

The bacterial flagellum is a long, thin (14 nm) structure that is free at one end and attached to the cell at the other end. The position and number of fiagella are often used as characteristics for classification. **Polar flagella** are positioned as one or both ends of the cell. Where flagella are found at various sites around the cell these are termed **peritrichous**, and where several are located at one end of the cell they are termed **lophotrichous.**

Flagella structure

Flagella are composed of subunits of a protein called **flagellin.** The portion of the flagellum embedded in the membrane is surrounded by two pairs or rings called the basal body. In Gram-negative Bacteria, an outer pair of rings is associated with the LPS and peptidoglycan layers of the cell wall, and an inner pair of rings is located within or just above the plasma membrane. In Gram-positive Bacteria, which lack the outer LPS layer, only inner pair of rings is present.

Flagellar movement

Flagella are rigid structures which do not flex but rotate. The rotary motion of the flagellum is driven by the basal body, which acts like a motor. The rings within the structure are thought to act together in generating rotational movement and the driving force for this comes from PMF. Dissipation of the proton gradient releases energy which causes rotation of the flagellum. The direction of flagellar rotation determines the type of movement.

Chemotaxis

Chemotaxis is the movement of an organism towards or away from a chemical. **Positive chemotaxis** is movement towards a chemical (attractant, i.e. a substrate). **Negative chemotaxis** is movement away from a chemical (repellent). Chemotaxis is thus a response to chemicals in the environment and requires that the cell has some form of sensory system. Prokaryote movement can be divided into runs where the organism moves in one direction (caused by counterclockwise flagellar rotation) and 'twiddles' where the organism randomly tumbles (caused by clockwise rotation of the flagella). Where there is no gradient of attractant or repellent the organism moves in a random way with a large number of twiddles; however, in the presence of a gradient, the runs become longer and twiddles less frequent as the organism experience; higher concentrations of attractant or repellent.

Methyl-accepting chemotaxis proteins (MCPs, also called **transducers**) are involved in translating the chemotactic signals from chemoreceptors to the flagellar motor.

In *E. coli* four types of MCP proteins have been identified - MCPs I, II, III and IV. Each MCP responds to different attractants and repellents. MCPs are so named because they become methylated or demethylated during chemotactic events. MCPs are transmembrane proteins that interact with attractants or repellents either directly or by way of chemoreceptors. A maximum of four methyl groups can be added to each MCP.

Bacterial Nutrition and Growth

Bacterial growth is measured by an increase in cell number; whereas growth of multicellular organisms is measured by an increase in their size/mass. All bacteria require a source of carbon to synthesize components of cytoplasm as well as a source of energy and electron in order to grow.

Nutritional aspects of bacterial growth

In addition to the need for carbon, hydrogen and oxygen, all organisms require sources of energy and electrons for growth to take place. On the basis of sources of carbon, energy, and hydrogen/electrons bacteria can be categorized into following types.

Based on carbon sources

1. Autotrophs → CO_2 sole or principal biosynthetic carbon source
2. Heterotrophs → Reduced, preformed, organic molecules from other organisms

Based on energy sources

1. Phototrophs → Light as their energy source
2. Chemotrophs → Oxidation of organic or inorganic compounds

Based on hydrogen /electron sources

1. Lithotrophs → Reduced inorganic molecules act as electron source
2. Organotrophs → Organic molecules act as an electron source

Major nutritional types of bacteria

Based on the primary source of carbon, energy and electrons, bacteria are placed in four major nutritional classes.

Photolithotrophic autotrophy (Photolithoautotrophy)

1. Source of Carbon → Carbon dioxide
2. Source of Energy → Light energy
3. Source of Electrons → Inorganic hydrogen/electron donor
4. Representative bacteria → Purple and green sulfur bacteria, Blue-green bacteria

Photoorganotrophic heterotrophy (Photoorganoheterotrophy)

1. Source of Carbon → Organic carbon source (CO_2 may also be used)
2. Source of Energy → Light energy
3. Source of Electrons → Organic hydrogen/electron donor
4. Representative bacteria → Purple nonsulfur bacteria, Green nonsulfur bacteria

Chemolithotrophic autotrophy (Chemolithoautotrophy)

1. Source of Carbon → Carbon dioxide
2. Source of Energy → Chemical energy source (inorganic)
3. Source of Electrons → Inorganic hydrogen/electron donor
4. Representative bacteria → Sulfur-oxidizing bacteria, Hydrogen bacteria, Nitrifying bacteria

Inorganic electron-donor	Groups	Representatives species
$NH_3 \xrightarrow{O_2} NO_2^-$	Ammonia oxidisers	*Nitrosomonas europaea*
$NO_2^- \xrightarrow{O_2} NO_3^-$	Nitrite oxidisers	*Nitrobacter winogradskyi*
$S^0, S_2O_3^{2-} \xrightarrow{O_2} SO_4^{2-}$	Sulphur oxidisers	*Thiobaillus thiooxidans*
$Fe^{2+} \xrightarrow{O_2} Fe^{3+}$	Iron bacteria	*Thiobacillus ferrooxidans*
$H_2 \xrightarrow{O_2} H_2O$	H_2 producers	*Aicaligenes eutrophus*
$CO \xrightarrow{O_2} CO_2$	Carboxidobacteria	*Pseudomanas carboxidovorans*

Chemoorganotrophic heterotrophy (Chemoorganoheterotrophy)

1. Source of Carbon → Organic carbon source
2. Source of Energy → Chemical energy source (organic)
3. Source of Electrons → Organic hydrogen/electron donor
4. Representative bacteria → Most nonphotosynthetic bacteria

Bacterial Growth

Unicellular microbial growth is autocatalytic such that the rate of growth is proportional to the cell concentration already present. During exponential growth the cell dry weight (or number) increases as follows

$$X_0 \rightarrow 2X_0 \rightarrow 4X_0 \rightarrow 8X_0 \ldots\ldots\ldots\ldots nX_0$$

where X is the cell dry weight ($g.l^{-1}$) and the interval is constant and termed the doubling time (t_d).

Consequently the number of doublings, n, after time t, is given by

$$n = \frac{t}{t_d}$$

It follows that the cell density (X_t) after time, t, will be related to the density orignally present (X_0) according to

$$X_t = X_0\, 2^n = X_0 2^{t/td}$$

Rearranging and taking logarithms yields

$$\ln (X_t/X_0) = \text{lag } 2^{(t/td)}$$

$$= \frac{t}{t_d} \log 2$$

Thus,

$$\frac{\ln X_t - \ln X_0}{t} = \frac{\ln 2}{t_d}$$

$$= \frac{0.693}{t_d}$$

This is more commonly expressed in terms of

$$2.303 \frac{(\log X_t - \log X_0)}{t} = \frac{0.693}{t_d}$$

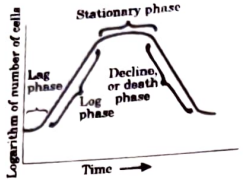


Fig. A standard bacterial growth curve

Synchronous Growth

Cells in the same stage of their growth cycle, i.e., growing synchronously. The synchrony generally lasts only a few generations, since even the daughters of a single cell soon get out of phase with one another. A population can be synchronized by manipulating the physical environment or the chemical composition of the medium. For example, the cells may be inoculated into a medium at a suboptimal temperature; if they are kept in this condition for some time; they will metabolize slowly but will not divide. When the temperature is subsequently raised, the cells will undergo a synchronized division. The most common method of synchronization takes advantage of the fact that the smallest cells in a log-phase culture are those which have just divided. When these cells are separated out by filtration or by differential centrifugation, they are reasonably well synchronized with each other.

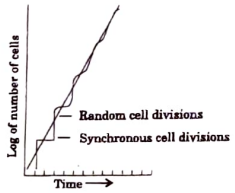


Fig. Synchronous growth of bacteria. The step like growth pattern indicates that all the cells of the population divide at about the same time.

Factors Affecting Bacterial Growth

Physical Factors

(i) pH

The acidity or alkalinity of a medium is expressed in terms of pH. It was invented by the Danish chemist Soren Sorenson to describe the limits of growth of microorganisms in various media.

According to their tolerance for acidity or alkalinity, bacteria are classified as :

- Acidophiles
- Neutrophiles
- Alkaliphiles

Acidophiles or acid-loving organisms, grow best at a pH of 0.1 to 5.4. Lactobacillus, which produces lactic acid, is an acidophile, but it tolerates only mild acidity, neutrophiles exist from pH 5.4 to 8.5. Most of the bacteria that cause disease in humans are neutrophiles. Alkaliphiles or alkali-loving (base-loving) organisms, exist from pH 7.0 to 11.5. Vibrio *cholera*, the causative agent of the disease Asiatic cholera, grows best at a pH of about 9. Al-caligenes faecalis, which sometimes infects humans already weakened by another disease, can create and tolerate alkaline conditions of pH 9 or higher. The soil bacterium Agrobacterium grows in alkaline soil of pH 12.

(ii) Temperature

According to their growth temperature range, bacteria can be classified as :

- Psychrophiles,
- Mesophiles
- Thermophiles

Psychrophiles or cold loving organisms, grow best at temperatures of 15° to 20°C, although some live quite well at 0°C. They can be further divided into **obligate psychrophiles,** such as Bacillus globisporus, which cannot grow above 20°C, and **facultative psychrophiles,** such as Xanthomonas pharmicola, which grows best below 20°C but also can grow above that temperature. Psychrophiles live mostly in cold water and soil.

Mesophiles, which include most bacteria, grow best at temperatures between 25° and 40°C.

Thermophiles, heat-loving organisms, grow best at temperatures from 50° to 60°C. Many are found in compost heaps, and a few tolerate temperatures as high as 110°C in boiling hot springs. They can be further classified as **obligate thermophiles,** which can grow only at temperatures above 37°C, or **facultative thermophiles,** which can grow both above and below 37°C. Bacillus stearothermophilus, which usually is considered an obligate thermophile, grows at its maximum rate at 65° to 75°C.

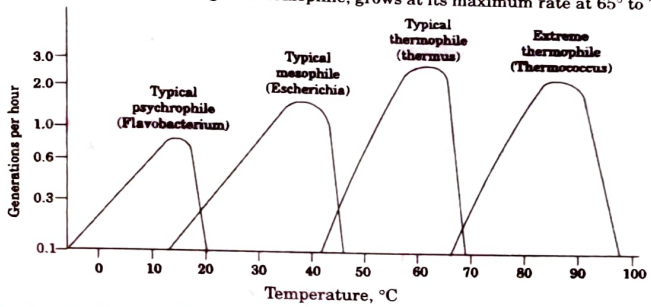


Fig. Growth rates of psychrophilic, mesophilic and thermophilic bacteria. Notice the overlap of temperature ranges at which these organisms can survive. Growth rates are much lower at the extreme ends of the ranges.

(iii) Oxygen

Obligate means that the organism must have the specified environmental condition. **Facultative** means that the organism is able to adjust to and tolerate the environmental conditions, but it can also live in other conditions.

Obligate aerobes, such as pseudomonas, which is a common cause of hospital-acquired infections, must have free oxygen for aerobic respiration, whereas **obligate anaerobes,** such as Bacteroides are killed by free oxygen.

Between the extremes of obligate aerobes and obligate anaerobes are the microaerophiles, the facultative anaerobes, and the aerotolerant anaerobes. **Microaerophiles** appear to grow best in the presence of a small amount of free oxygen. Microaerophiles such as Campylobacter, which can cause intestinal disorders, also are **capnophiles,** or carbon dioxide-loving organisms. They thrive under condition of low oxygen and high carbon dioxide concentration. **Facultative anaerobes** ordinarily carry on aerobic metabolism when oxygen is present, but they shift to anaerobic metabolism when oxygen is absent. Staphylococcus and Escherichia coli are facultative anaerobes; they often are found in the intestinal and urinary tracts, where only a small amount of oxygen is available.

Aerotolerant anaerobes can survive in the presence of oxygen but do not use it in their metabolism. Lacto bacillus, for example, always captures energy by fermentation, regardless of whether the environment contain oxygen.

(iv) Hydrostatic Pressure

Water in oceans and lakes exerts **hydrostatic pressure,** pressure exerted by standing water, in proportion to its depth. Such pressure doubles with every 10 m increase in depth. Bacteria that live at high pressures, but die if left in the laboratory for only a few hours at standard atmospheric pressure, are called **barophiles.** It appears that their membranes and enzymes do not simply tolerate pressure but require pressure to function properly. The high pressure is necessary to keep their enzyme molecules in **the proper three-dimensional configuration.**

(v) Osmotic Pressure

Cells in hyperosmotic environments lose water and undergo **plasmolysis,** or shrinking of the cell. In microorganisms with a cell wall, the cell or plasma membrane separates from the cell wall.

Nutritional Factors

(i) Carbon Sources

Photoautotrophic organisms reduce carbon dioxide to glucose and other organic molecules. Both autotrophic and heterotrophic organisms can obtain energy from glucose by glycolysis, fermentation, and the Krebs cycle.

(ii) Nitrogen Sources

Many microorganisms reduce nitrate ions (NO_3^-) to amino groups (NH_2) and use the amino groups to make amino acids. Some can synthesize all 20 amino acids found in proteins.

(iii) Sulfur and Phosphorus

Some organisms can synthesize sulfur-containing amino acids from inorganic sulfur and other amino acids Microorganisms obtain phosphorus mainly from inorganic phosphate ions (PO_4^{3-}). They use phosphorus (as phosphate) to synthesize ATP, phospholipids, and nucleic acids.

(iv) Trace Elements

Potassium, zinc, magnesium, and manganese are used to activate certain enzymes. Cobalt is required by organisms that can synthesize vitamin B_{12}. Iron is required for the synthesis of heme-containing compounds and for certain enzymes. Although little iron is required, a shortage severely retards growth. Calcium is required by Gram-positive bacteria for synthesis of cell walls and by spore-forming organisms for synthesis of spores.

(v) Vitamins

A vitamin is an organic substance that an organism requires in small amounts and that is typically used as a coenzyme. Microbes living in the human intestine manufacture vitamin K, which is necessary for blood clotting, and some of the B vitamins, thus benefiting their host.

Bacterial Genome : Bacterial Chromosome and Plasmid

Nucleoid

The nucleoid is the region in a prokaryotic cell that contains the genome. The bacterial nucleoid, which contains the dsDNA fibrils, lacks a limiting membrane. The bacterial chromosome is a single, continuous circular DNA molecule. Bacterial chromosome does not contain basic histone proteins, but low-molecular weight polyamines and magnesium ions may perform a function similar to that of eukaryotic histones. The bacterial chromosome is organized into independently supercoiled loops called *domains*. The domain organization enables the chromosomal DNA to undergo structural changes during different cellular processes (replication, transcription, and segregation) that take place simultaneously in a bacterial cell. Small nucleoid associated proteins (HU, H-NS, IHF, Fis, Lrp, and Dps) and the high-molecular-weight protein SMC (structural maintenance of chromosomes) facilitate compaction of chromosomal DNA by bending, bridging, and wrapping. In addition, SMC protein is involved in chromosome segregation.

Variations in prokaryotic genome structure

Most prokaryotes, like *E. coli,* contain a single circular DNA molecule that makes up their entire genome, recent studies have indicated that some prokaryotes contain as many as four linear or circular chromosomes. For example, *Vibrio cholerae,* the bacteria that causes cholera, contains two circular chromosomes. One of these chromosomes contains the genes involved in metabolism and virulence, while the other contains the remaining essential genes. An even more extreme example is provided by *Borrelia burgdorferi,* the bacterium that causes Lyme disease. It contains up to 11 copies of a single linear chromosome. Unlike *E. coli, Borrelia* cannot supercoil its linear chromosomes into a tight ball within the nucleoid; rather, these strands are diffused throughout the cell. The DNA molecules of Archaea share many similarities with eukaryotes. Archaeans are the only group of prokaryotes that use eukaryote-like histones to condense their DNA molecules. DNA molecules of Archaea can also be negatively supercoiled, positively supercoiled, or not supercoiled at all.

Prokaryotic versus Eukaryotic genome

Most prokaryotes reproduce asexually and are haploid, meaning that only a single copy of each gene is present. By contrast, eukaryotes that reproduce sexually generally contain multiple chromosomes and are said to be diploid, because two copies of each gene exist— with one copy coming from each of an organism's parents. Noncoding sequences account for an average of 12% of the prokaryotic genome, as opposed to upwards of 98% of the genetic material in eukaryotes. Another difference between prokaryotes and eukaryotes is that prokaryotic cells often contain one or more plasmids (i.e., extrachromosomal DNA molecules that are either linear or circular). Unlike eukaryotic chromosomes, most prokaryotic genomes are organized into polycistronic operons separated by only a few base pairs. The proteins encoded by each operon often collaborate on a single task, such as the metabolism of a sugar into by-products that can be used for energy. The organization of prokaryotic DNA thus differs from that of eukaryotes in several important ways. The most notable difference is the condensation process that prokaryotic DNA molecules undergo in order to fit inside relatively small cells.

Plasmids

Plasmids (term coined by J.Lederberg) are replicons that are maintained as discrete, extrachromosomal genetic elements in bacteria (also present in some yeast and other fungi). They are usually much smaller than the bacterial chromosome, varying from less than 5 kbp to more than several hundred kbp, though plasmids as large as 2 Mbp also occur in some bacteria. Plasmids usually encode traits that are not essential for bacteria viability, and replicate independently of the chromosome. Most plasmids are supercoiled, circular, double-stranded DNA molecules, but linear plasmids have also been demonstrated in genera *Borrelia* and *Streptomyces.*

Plasmids can be classified into *conjugative* and *non-conjugative* type. Conjugative plasmids code for functions that promote transfer of the plasmid from the donor bacterium to other recipient bacteria, but non-conjugative plasmids do not. Conjugative plasmids have genes for the synthesis of pili that help in the process of conjugation. The best studied conjugative plasmid is the **F-factor** (or F-plasmid) of *E.coli.* The bacterial cell contain F-factor are described as F^+. F^+ cell acts as donor, where as F^- cell (a cell lacking an F-plasmid) acts as recipient.

The average number of molecules of a given plasmid per bacterial chromosome is called its *copy number.* Some plasmids are present in the cell in only 1-3 copies, whereas others may be present in over 100 copies. Large plasmids (>40 kilobase pairs) are often *conjugative,* have small copy numbers, code for all functions required for their replication, and partition themselves among daughter cells during cell division in a manner similar to the bacterial chromosome. Plasmids smaller than 7.5 kilobase pairs usually are *non-conjugative,* have high copy numbers, rely on their bacterial host to provide some functions required for replication, and are distributed randomly between daughter cells at division.

Many plasmids control medically important properties of pathogenic bacteria, including resistance to one or several antibiotics or production of toxins. Many plasmids confer antibiotic resistance on the cells that contain them and are described as **R-plasmid** (resistance plasmid). Some R-plasmids have only a single resistance gene, whereas others have many. Many R-plasmids are conjugative plasmids.

Col plasmids are able to produce *bacteriocin* such as colicin, proteins that prevent growth of susceptible bacterial strains that do not contain a Col plasmid. Bacteriocins are proteins that can interact with sensitive bacteria and inhibit one or more essential processes, such as DNA replication, transcription, translation or energy metabolism. Bacteriocins produced by *E.coli* are termed *colicins.* There are many types of colicins, each designated by a letter (for example, colicin B) and each having a particular mode of inhibition of sensitive cells.

Type of plasmid	Gene functions
Resistance	Antibiotic resistance
Fertility	Conjugation and DNA transfer between bacteria
Killer	Synthesis of toxins that kill other bacteria
Degradative	Enzymes for metabolism of unusual molecules
Virulence	Pathogenicity

Host range

Most plasmids can exist in only a limited number of closely related bacteria; these are called *narrow host range* plasmids. Some plasmid like IncP of *E.coli* can be transferred to and maintained in bacteria of a large number of species. These are called *broad host range* plasmids.

Functions encoded by plasmids

Depending on their size, plasmids can encode a few or hundreds of different proteins. However, plasmids rarely encode gene products that are essential for growth, such as RNA polymerase, ribosomal subunits, or enzymes of the tricarboxylic acid cycle. Instead, plasmid genes usually give bacteria a selective advantage under only some conditions. Gene products encoded by plasmids include enzymes for the utilization of unusual carbon sources such as toluene, resistance to substances such as heavy metals and antibiotics, synthesis of antibiotics, and synthesis of toxins and proteins that allow the successful infection of higher organisms. Table given below lists a few traits that are encoded by naturally occurring plasmids.

Table : Some naturally occurring plasmids and the traits they carry

Plasmid	Trait
ColE1	Bacteriocin which kills *E.coli*
Tol	Degradation of toluene and benzole acid
Ti	Tumor initiation in plants
pJP4	2,4-D degradation
pSym	Nodulation on roots of legume plants
RK2	Resistance to ampicillin, tetracycline, and kanamycin

Table : List of some plasmid-coded traits

Trait	Organisms in which trait is found
Antibiotic resistance	*E.coli, Salmonella sp., Staphylococcus sp.*
Pilus synthesis	*E.coli, Pseudomonas sp.*
Tumor formation in plants	*Agrobacterium tumefaciens*
Nitrogen fixation (in plants)	*Rhizobium sp.*
Oil degradation	*Pseudomonas sp.*
Gas vacuole production	*Halobacterium sp.*
Insect toxin synthesis	*Bacillus thuringiensis*
Plant hormone synthesis	*Pseudomonas sp.*
Antibiotic synthesis	*Streptomyces sp.*
Increased virulence	*Yersinia enterocolitica*

Hfr formation

The F factor is an example of episome. An episome is a plasmid that can exist autonomously in a cell or can integrate into the bacterial chromosome. A bacterial cell that contains an integrated F factor is referred to as an **Hfr cell**. The frequency of insertion occurs at about 10^{-5}–10^{-7} per generation. Integration involves homologous recombination between two covalently closed circular DNA molecules forming one circular molecule containing both of the original DNA structures. It is thought that the insertion sequences (IS sequence) present in the F genome and those in the host chromosome serve as regions of homology for the insertional event.

Plasmids in eukaryotic microbes

Plasmids are not limited to bacteria. For example, plasmids also found in eukaryotic microbes like yeast. One yeast plasmid is called the 2 circle. The 2 circle is a 6.3 kb circular, extrachromosomal element found in the nucleus of most *Saccharomyces cerevisiae* strains. It is stably maintained at about 50 to 100 copies per haploid genome of the yeast cell. Like the host chromosomes, the 2 circle is coated with nucleosomes and replication is initiated by host replication enzymes once per cell cycle.

Recombination and Gene Transfer

Recombination is the process in which a new genotype is formed when the genetic materials of two different organisms or parents are combined. Changes that occur in DNA due to recombination results into redistribution of the information. It is a way of generating diversity within the gene pool of a population. Recombination process is of two types - general recombination (homologous) and site specific (non homologous) recombination.

General or Homologous Recombination

In prokaryotes, a gene called RecA protein is essential for homologous recombination pathway. The homologous recombination process begins with nick in one of the DNA molecule (Fig). The nicking takes place by the action of *enzyme* called endonucleases, present in the cell. The nicked strand is displaced by the action of special protein called helicases to form single strand. This single strand DNA pairs with the corresponding region of the homologous chromosomes, forming a heteroduplex. The enzyme RecA protein that results in the formation of the bridge between the two homologous DNA catalyzes this step. The chromosome then, rotates so that the two strands no longer cross each other and held together by covalent bonds. The endonucleases cleave the DNA strands and establish independent chromosome by breaking the heteroduplex strand.

This recombinant process is a natural mechanism that occurs inside the cell. Whether it leads to recombination or not depends on the variability of genes outside the recombination region.

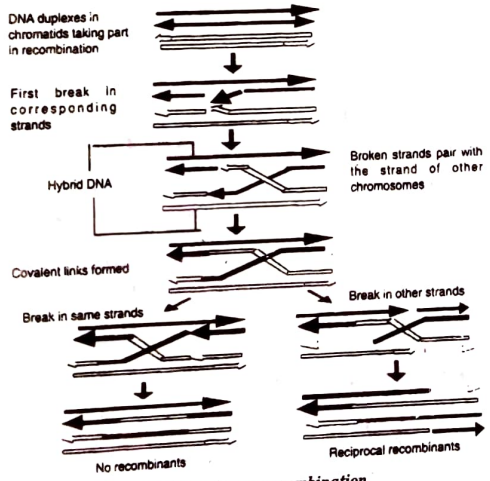


Fig. Homologous recombination

Site-specific and Non-Homologous Recombination

Site-specific recombination is independent of homologous recombination. In this process, a small segment of DNA strand gets inserted at a specific site on a large double stranded DNA, for example insertion of bacteriophage into the chromosome of *E. coli*. The process by which a gene moves from one place to other in the genome is called transposition.

Transposable Genetic Elements

Transposable genetic elements are segments of DNA that have the capacity to move from one location to another (i.e. jumping genes).

Properties of Transposable Genetic Elements

1. *Random movement* – Transposable genetic elements can move from any DNA molecule to any DNA other molecule or even to another location on the same molecule. The movement is not totally random; there are preferred sites in a DNA molecule at which the transposable genetic element will insert.
2. *Not capable of self replication* – The transposable genetic elements do not exist autonomously (exception – some transposable phages) and thus, to be replicated they must be a part of some other replicon.
3. *Transposition mediated by site-specific recombination* – Transposition requires little or no homology between the current location and the new site. The transposition event is mediated by a transposase coded for by the transposable genetic element. Recombination that does not require homology between the recombination molecules is called site-specific or nonhomologous recombination.
4. *Transposition can be accompanied by duplication* – In many instances transposition of the transposable genetic element results in removal of the element from the original site and insertion at a new site. However, in some cases the transposition event is accompanied by the duplication of the transposable genetic element. One copy remains at the original site and the other is transposed to the new site.

Types of Transposable Genetic Elements

1. Insertion sequences (IS) – Insertion sequences are transposable genetic elements that carry no known genes except those that are required for transposition.

   Insertion sequences are given the designation IS followed by a number, e.g. IS1. Insertion sequences are small stretches of DNA that have at their ends repeated sequences, which are involved in transposition. In between the terminal repeated sequences there are genes involved in transposition and sequences that can control the expression of the genes but no other nonessential genes are present. The importance of insertion sequences is in -

   Mutation: The introduction of an insertion sequences into a bacterial gene will result in the inactivation of the gene.

   Plasmid insertion into chromosomes: The sites at which plasmids insert into the bacterial chromosome are at or near insertion sequence in the chromosome.

   Phase Variation: The flagellar antigens are one of the main antigens to which the immune response is directed in our attempt to fight off a bacterial infection. In *salmonella* there are two genes which code for two antigenically different flagellar antigens. The expression of these genes is regulated by an insertion sequences. In one orientation one of the genes is active while in the other orientation the other flagellar gene is active. Thus, *salmonella* can change their flagella in response to the immune systems' attack. Phase variation is not unique to *salmonella* flagellar antigens. It is also seen with other bacterial surface antigens. Also the mechanism of phase variation may differ in different species of bacteria (e.g. *Neisseria*, transformation).
2. Transposons (Tn) – Transposons are transposable genetic elements that carry one or more other genes in addition to those which are essential for transposition.

   Transposons are given the designation Tn followed by a number. The structure of a transposon is similar to that of an insertion sequence. The extra genes are located between the terminal repeated sequences. In some instances (composite transposons) the terminal repeated sequences are actually insertion sequences. The importance of transposons is that many antibiotic resistance genes are located on transposons. Since transposons can jump from one DNA molecule to another, these antibiotic resistance transposons are a major factor in the development of plasmids. These multiple drug resistance plasmids have become a major medical problem because the indiscriminate of antibiotics have provided a selective advantage for bacteria harboring these plasmids.

Exchange of Genetic Information

In bacteria recombination occurs as one-way gene transfer from donor to recipient. This occurs by three mechanisms – (*i*) transformation, (*ii*)transduction, and (*iii*) conjugation.

Transformation

Transformation is gene transfer (naked DNA) for donor cell to uptake by a recipient cell. Certain bacteria (e.g. *Bacillus, Haemophilus, Neisseria, Pneumococcus*) can take up DNA from the environment and this DNA is incorporated into the recipient's chromosome.

1. *Factors Affecting Transformation*

(*a*) *DNA size:* Double stranded DNA of at least 5×10^5 daltonsis required. Thus tranformation is sensitive to presence of nucleases in the environment.

(*b*) *Competence of the recipient:* Some bacteria are able to take up DNA naturally. However, these bacteria can take up DNA a particular time in their growth cycle when they produce a specific protein called a competence factor. At this stage the bacteria are said to be competent. Other bacteria are not able to take up DNA naturally. However in these bacteria competence can be induced *in vitro* by treatment with chemicals like $CaCl_2$.

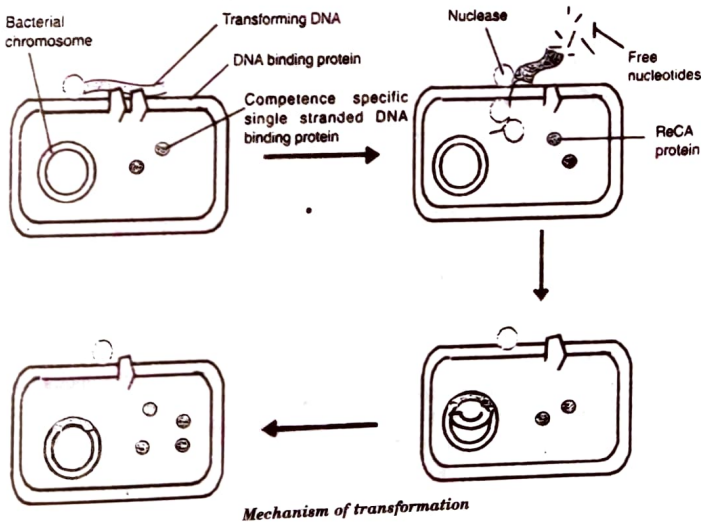
2. *Steps in Transformation*

(*a*) *Uptake of DNA:* uptake of DNA by gram +ve and Gram –ve bacteria differs. In Gram +ve bacteria the DNA is taken up as a single stranded molecule and the complementary strand is made in the recipient. In contrast, Gram- bacteria take up double stranded DNA.

(*b*) *General Recombination:* After the donor DNA is taken up, a reciprocal recombination event occurs between the chromosome and the donor DNA. This recombination requires homology between the donor DNA and the chromosome and results in the substitution of DNA between the recipient and the donor. Because of the requirement for homology between the donor and host DNA, only DNA from closely related bacteria are expected to successfully transform.

3. *Significance*

Transformation occurs in nature and it can lead to increased virulence. In addition transformation is widely used in recombinant DNA technology.



Mechanism of transformation

Transduction

Transduction is the transfer of genetic information from a donor to a recipient by way of a bacteriophage. The phage coat protects the DNA in the environment so that transduction, unlike transformation, is not affected by nucleases in the environment. Not all phages can mediate transduction. In most cases gene transfer is between members of the same bacterial species. However, if a particular phage has a wide host range than transfer between species can occur. The ability of a phage to mediated transduction is related to the life cycle of the phage.

1. Types of Transduction

(a) *Generalized Transduction:* Generalized transduction is transduction in which potentially any bacterial gene from the donor can be transferred to the recipient. The mechanism of generalized transduction is illustrated in. Phages that mediate generalized transduction breakdown host DNA into smaller pieces and package their DNA into the phage particle by a "head-pull" mechanism. Occasionally one of the pieces of host DNA is randomly packaged into a phage coat. If a recipient cell is infected by a phage that contains donor DNA, donor DNA enters the recipient. In the recipient a generalized recombination event can occur which substitutes the donor DNA and recipient DNA.

(b) *Specialized Transduction:* Specialized transduction is transduction in which only certain donor genes can be transferred to the recipient. Specialized transduction is mediated by lysogenic or temperate phage and the genes that get transferred will depend on where the prophage has inserted in the chromosome. During excision of the prophage, occasionally an error occurs where some of the host DNA is excised with the phage DNA. Only host DNA on either side of where the prophage has inserted can be transferred (i.e. specialized transduction). After replication and release of phage and infection of a recipient, lysogenization of recipient can occur resulting in the stable transfer of donor genes.

Conjugation

1. *Mating Types in Bacteria*

Conjugation is mating type of recombination process in bacteria.

(a) *Donor:* The ability of a bacterium to be a donor is a consequence of the presence in the cell of an extra piece of DNA called the F factor or fertility factor or sex factor. The F factor is a circular piece of DNA that can replicate autonomously in the cell, it is an independent replicon. Extrachromosomal pieces of DNA that can replicate autonomously are called plasmids. The F factor has genes on it that are needed for its replication and for its ability to transfer DNA to a recipient. F factor has genes that produce a sex pilus (F pilus) on the surface of the bacterium. This pilus is important in the conjugation process.

(b) *Recipient:* The ability to act as a recipient is a consequence of the lack of the factor.

2. Physiological States of the F Factor

(a) *Autonomous (F^+):* In this state the F factor carries only those genes necessary for its replication and for DNA transfer. There are no chromosomal genes associated with the F factor in F^+ strains. In crosses of the type $F^+ \times F^-$ the F^- becomes F^+ while F^+ remains F^+. Thus, the F factor is infectious. In addition, there is only low level transfer of chromosomal genes.

(b) *Integrated (Hfr):* In this state the F factor has integrated into the bacterial chromosome via a recombination event. In crosses of the type Hfr × F^- the F^- rarely becomes Hfr and Hfr remains Hfr. In addition, there is a high frequency of transfer of donor chromosomal genes.

(c) *Autonomous with chromosomal genes (F'):* In this state the F factor is autonomous but carries some chromosomal genes. F' factors are produced by excision of the F factor from an Hfr. Occasionally, when the F factor is excising from the Hfr chromosome, donor genes on either side of the F factor can be excised with the F factor generating an F'. F' factors are named depending on the chromosomal genes that they carry. In crosses of the type $F^+ \times F'$ the F^- become F' while F' remains F'. In addition there is high frequency of transfer of those chromosomal genes on the F' and low frequency transfer of other donor chromosomal genes.

3. Mechanism of Conjugation

(a) $F^+ \times F^-$

(i) *Pair formation:* The tip of the sex pilus comes in contact with the recipient and a conjugation bridge is formed between the two cells. It is through this bridge the DNA will pass from the donor to the recipient. Thus, the DNA is protected from environmental nucleases.

(ii) *DNA transfer:* The plasmid DNA is nicked at a specific site called the origin of transfer and is replicated by a rolling circle mechanism. A single strand of DNA passes through the conjugation bridge and enters the recipient where the second strand is replicated.

(iii) This process explains the characteristics of $F^+ \times F^-$ crosses. The recipient becomes F^+, the donor remains F^+ and the there is low frequency of transfer of donor chromosomal genes.

Hfr × F^- Crosses

(i) Pair Formation

(ii) *DNA transfer:* The DNA is nicked at the origin of transfer and is replicated by a rolling circle mechanism. But the DNA that is transferred first is the chromosome. Depending upon where in the chromosome the F factor has integrated and in what orientation, different chromosomal genes will be transferred at different times. Only when the entire chromosome is transferred will the F factor will be transferred. Since shearing forces separate the mating pairs it is rare that the entire chromosome will be transferred.

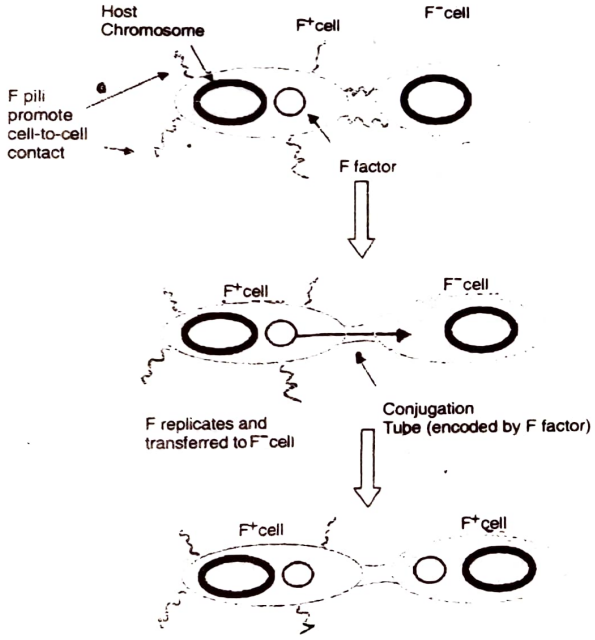


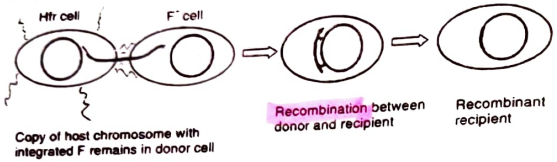
Fig. $F^- \times F^+$ cross

(c) F' X F' Crosses

(*i*) Pair formation.

(*ii*) DNA transfer – This process is similar to F⁺ X F⁻ crosses. However, since the F⁺ has some chromosomal genes on it these will also be transferred.

(*iii*) This mechanism explains the characteristics of F' X F⁻ crosses. The F⁻ becomes F', the F' remains F⁺ and there is high frequency transfer of donor genes on the F' but low frequency transfer of other donor chromosomal genes.



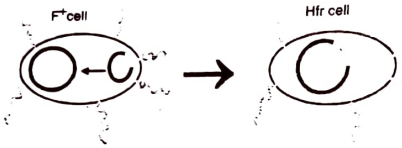


Fig. Hfr × F' cross

Bacterial Taxonomy

Classification of prokaryotes : The main aim of taxonomy is to set up a **classification:** an ordering of prokaryotes into groups based on common properties. This classification can then be used, to identify individual bacterial species or strains **(identification)** and name them **(nomenclature)**. Classification is also used to study the relationships of different groups of microorganisms based on shared phenotypic properties and probably a common evolutionary history. A wide range of morphological, biochemical and molecular characters are measured and the information derived from this is used to create a hierarchical classification system, into which individual strains of microorganisms can be placed.

The most important rank in the taxonomic hierarchy is called the **species,** which from the time of Linnaeus (1707-1778), has been named in a similar way. Related species are grouped into a **genus** (p1. genera), which is usually based around some distinct phenotypic character, and the generic name is used as part of a binomial species name. So, for example, *Escherichia coli* is in the genus *Escherichia* named after Escherich, who first isolated the bacteria in 1885. The specific epithet *'coli'* is in this case an indication as to where the organism was isolated (from i.e. the colon). However, note that the binomial name is a label not a description; it does not have to describe the species in any way. Note too that the genus name has a capital initial letter but the specific epithet (which cannot be used alone) does not, and that the whole binomial name is Latinized and therefore should be italicized or underlined. Normally, once the name of the organism has been written in full, the genus name can be reduced to a letter, or letters, as long as there can be no confusion with other genera. This *Escherichia coli* will subsequently be written as *E. coli.*

The higher taxonomic categories into which the organism can be ranked are, sequentially, family, order, class, division and kingdom (domain). Similarly, species may be divided into subspecies and a culture of a particular organism is usually referred to as a **strain**. Strain is also used to describe a sub-group of a species that shares some feature such as one type of O-antigen or mutation. There are also informal names, not italicized, which are sometimes useful for describing groups of organisms such as the green-sulfur bacteria or the spirochetes.

The authority on prokaryotic classification is *Bergey's Manual of Systematic Bacteriology,* which contains as near as possible complete listings of all prokaryotic species and their distinguishing characteristics.

General Methods of Classifying Bacteria

Three methods are used for arranging bacteria into taxa:

(*i*) **Intuitive Method.**

The trouble with this method is that the characteristics of an organism that seem important to one person may not be so important to another and different taxonomists may arrive at very different groupings. However some classification schemes based on the intuitive method have proved to be quite useful

(*ii*) **Numerical Taxonomy.**

Scientist determine many characteristics (usually 100 to 200) for each strain studied giving each characteristic equal weight. Then using a computer he or she calculates the % similarity (%S) of each strain to every other strain.

For any two strains, $\%S = \frac{NS}{NS + ND}$

where NS = number of characteristics that are the same (positive or negative) for the two strains and

ND = number of characteristics that are different.

Those groups having a high %S to each other are placed into groups; those groups having a high %S to each other are in turn placed into larger groups.

(*iii*) **Genetic Relatedness.**

The third and most reliable method of classification is based on the degree of genetic relatedness between organisms. This method is the most objective of all and is based on the most fundamental aspect of organisms, their hereditary material (DNA). In the 1960s the development of that branch of science known as molecular biology provided techniques by which the DNA of one organism could be compared with that of other organisms. At first only crude comparisons could be made. Based on mol% G + C values. It is true that two organisms of the same or similar species that are very closely related will have very similar mol% G + C values and it is also true that two organisms having quite different mol% G + C values are not very closely related. *However, it is important to realize that organisms that are completely unrelated may have similar mol% G + C values.* Therefore much more precise methods of comparison were needed — namely methods by which the DNA molecules from various organisms could be compared with respect to the sequence of that component nucleotides. This sequence is the most fundamental characteristics of an organism. Modern techniques have now mode it possible to make such a comparison.

General feature of important bacterial group

1. Rickettsias

Most live in an obligate intracellular association parasitic or mutualistic, with eukaryotic hosts (vertebrates or arthopods); a few can be grown on moderately complex bacteriological media containing blood.

There cell wall contain muramic acid. While glutamate is oxidised with ATP generations. Mainly there are rod-shaped, coccoid, and often pleomorphic that stain gram-negative and lack flagella. The parasitic species are associated with the reticuloendothelial and vascular endothelial cells or erythrocytes of vertebrates and often with various organs of arthropods, which may acts as vectors or primary hosts. The mutualistic species are found in insects. Important Genera are: *Bartronella, Grahamella* and *Rochalimaea*.

It is named after Howard T. Ricketts who identified this microorganisms as the causal organism of typhus and Rocky Mountain spotted fever. They are small, Gram-negative coccobacilli which are obligate intracellular parasites, grow in the cytoplasm of infected cells of mammals and arthropods. Rickettias can be cultured only in embryonated eggs and grow at 35°C.

2. Chlamydias

These are nonmotile, obligate parasite, coccoid bacteria multiply with in membrane bounded vacuoles in the cytoplasm of the cells of humans, other mammals, and birds (Arthropods do not act as host). The multiplication occurs by means of a unique development cycle, which divide by fission. They are pathogenic. The specialised techniques are involved for their cultivation.

These are small (0.2–1.0 m) rod or cocoid, Gram-negative having both DNA and RNA. They are non-motile, obligate parasite, cocoid bacteria live inside the host cells. They are pathogenic. The cell walls are devoid of muramic acid and glucose is not oxidized with ATP generation, hence parasite depends upon the host ATP cells. The specialized techniques are involved for their cultivation. They are present in two different forms during their life cycle. A large, metabolically active reticulate body which is non-infectious with flexible cell wall and a smaller form are present which is called elementary body. This body causes infection and can attack other cells of the same host or transmitted to new hosts. They multiply by a process of attachment of elementary body to the host cell wall followed by phagocytosis, conversion, reproduction, condensation and release of the elementary and reticulate bodies by description of the wall material. The reticulate body divides by binary fission.

The Mycoplasmas (or Mollicutes) : The Cell Wall-Less Bacteria

This group consists of very small prokaryotes totally devoid of cell walls, consist of only a plasma membrane and incapable of synthesis of peptidoglycan and its precursors. These are resistant to penicillins and sensitive to lysis by osmotic shock, detergents and alcohols. These are pleomorphic, varying in shape from spherical or pear shaped structures (0.3-0.8 m in dia) to branched or helical filaments. Usually nonmotile, but some show gliding motility in liquid surfaces. They are Gram-negative. Most species require sterols and fatty acids in their growth media for growth, mostly these are facultatively anaerobic, but few are obligate anaerobes, and under favourable conditions, most species form colonies that have a characteristic of "fried egg" appearence. All are the parasites, commensals or saprophytes, but several are pathogenic to human, animals, plants and insects. The mol% G+C of the DNA ranges from 23-40.

Archaeobacteria

General Characteristics

1. Greek, *archaios,* ancient and *bakterion* a small rod.
2. They usually prefer extreme aquatic and terrestrial habitats. They are often present in anaerobic, hypersaline, or high temperature environments.
3. They can stain either gram positive or gram negative.
4. Morphologically, they may be spherical rod shaped, spiral lobed, plate shaped irregularly shaped, or pleomorphic. They range in diameter from 0.1 to over 15 m in length.
5. They multiply either by binary fission, budding, fragmentation, or other mechanisms.
6. Physiologically they are either aerobic, facultatively anaerobic, or strictly anaerobic. Nutritionally they range from chemolithoautotrophs to organotrophs. Some are mesphiles; others are hyperthermophiles that can grow above 100°C.

Archaeobacterial Cell Walls

Archaeobacterial *do not have the muramic acid and D-amino acids which is the characteristic of eubacterial peptidoglycon.* All archaeobacteria resist attack by lysozyme and β-lactum antibiotics such as penicillin. Gram positive archaeobacteria can have a variety of complex polymers in their walls. *Methanobacterium* and some other methanogens have walls containing pseudomurein a peptidoglycan like polymer that has L-amino acids in its cross links, N - acetyltaloszminuronic acid (instead of N-acetylmuramic acid, and) β (1 → 3) glycosidic bonds instead of β(1 → 4) glycosidic bonds. *Methanosarcina* and *Halococcus* contain complex polysaccharides similar to the chondroitin sulfate of animal connective tissue. Other heteropolysaccharides are also found in gram positive walls.

Some methanogens *(methanolobus), Halobacterium,* and several extreme thermophiles *(Sulfolobus, thermoproteus,* and *Pyrodictium)* have glycoproteins in their walls. In contrast, other methanogens *(Methanococcus, Methanomicrobium* and *Methanogenium)* and the extreme thermophie *Desulfurococcus* have protein walls.

Archaeobacterial Lipids and Membranes

Polar lipids are present in archaeobacterial membranes : phospholipids, sulfolipids and glycolipids. From 7 to 30 % of the membrane lipids are nonpolar lipids which usually are derivatives of squalene. These lipids can be combined in various ways to yield membranes of different rigidity and thickness. For example, the C_{20} diethers can be used to make a regular bilayer membrane. A much more rigid monolayer membrane may be constructed of C_{40} tetraether.

Archaeobacterial Genetics and Molecular Biology

Chromosome of archaeobacteria is a single closed DNA circle, however their genomes are significantly smaller than the normal eubacterium E. coli DNA has a size of about 2.5×10^9 daltons whereas *Thermoplasma acidophilum.* DNA is about 2.5×10^9 daltons and Methanobacterium thermoautotrophicum DNA is 1.1×10^9 daltons. The variation in G + C content is great from about 21 to 68 mol% and is another sign of archaeobacterial diversity. Archaeobacteria have few plasmids.

Bacterial Toxin

A toxin is a specific substance, often a metabolic product of the organism that damages the host. Toxins can even induce diseasc in the absence of the organism that produced them. Toxin produced by organisms can be divided into two main categories : Exotoxins and endotoxins.

Endotoxins Traditionally, bacterial toxins have been split into two groups on the basis of whether they are associated with the bacterial cell, **endotoxins**, or released into the medium by growing bacteria, **exotoxins.** Endotoxin activity is due to the lipid A portion of LPS, which is an integral part of the outer membrane of Gram-negative bacteria. The endotoxin is normally released from Gram-negative cells in the bloodstream after lysis as a result of the action of the host immune system or by the action of some antibiotics. Once released, the LPS activates the complement cascade by the alternative pathway and induces the release of cytokines and other defense mediators from macrophages and other cells. These bioactive compounds are a normal part of the host defense mechanisms and help to rid the body of infection, but when induced by LPS, the amounts of these are much higher than normal and cause toxicity to the host itself. This is called **septic** shock which has the symptoms of fever, circulatory collapse and organ failure, and eventually death. Components of Gram-positive cells, including lipoteichoic acids and peptidoglycan, can also induce septic shock.

Exotoxins Traditionally, the word exotoxin has been used to describe bacterial proteins produced by both Gram-negative and Gram-positive bacteria that are toxic to host cells. Generally, the word exotoxin is shortened to toxin, when describing bacterial proteins, as not all exotoxins are necessarily secreted into the surrounding medium but may be held on the surface of the cell or in the periplasmic space. The exotoxins are generally heat labile; there are a number of heat-stable toxins such as the *E. coli* ST enterotoxin. They are water soluble, so can act at a site distant from the original site of infection. In some cases, such as botulism, there is no need for colonization of the host by the bacterium at all, the toxin alone is enough to cause the disease (toxemia). The nomenclature of toxins is varied :

- Some toxins, especially those that are the clear cause of disease symptoms, are named after the bacteria that produces them such as cholera toxin (produced by *Vibrio cholerae*) or diphtheria toxin (*Corynebacterium diphtheriae*).
- Others are named after their sites of action such as neurotoxins (acting on nerve cells), enterotoxins (acting on the gut) or leukotoxins (active against white blood cells). Toxins that are active against a number of cell types may just be described as cytotoxins.
- Some toxins are just given a letter or number such as *Pseudomonas aeruginosa* toxin A or *Staphylococcus aureus* α-toxin.
- Toxins may be named on the basis of their activities such as phospholipase C (produced by *Listeria monocytogenes)* and adenylate cyclase produced by *Bordatella pertussis.*

Some bacteria, especially the Gram-positives, may produce a number of different toxins, others may only produce one. Toxin production is generally very carefully controlled at the level of transcription in bacteria and, in some cases, only a subset of bacteria within a strain will actually secrete a toxin. Most toxins are either membrane-damaging toxins or toxins that act inside the cell. A small third group, called **superantigens**, interact with T-helper cells, causing over-induction of the immune system and symptoms similar to shock.

The role of toxins in the disease process is a matter of some debate. However, as discussed in they may play a number of roles including:

- the release of nutrients from cells;
- destruction of cells associated with the immune defense;
- escape from the phagolysosome by intracellular bacteria.

Membrane damaging toxins

These cytolytic toxins cause lysis by destroying the integrity of the plasma membranes of the host cell. Usually they are active against a wide range of cell types and can be divided into two groups on the basis of their mechanisms of action: enzymic action and pore forming. A large number of different Gram-positive bacteria produce enzymes such as phospholipases and sphingomyelinases, which break down the lipid components on the membrane. This destabilizes the membrane, eventually leading to cell lysis. The second group of toxins includes the thiol-activated cytolysins, streptolysin O (produced by *Streptococcus pyogenes*) and listeriolysin O (produced by *L. monocytogenes*) and *Staph. aureus* α-toxin. These toxins form pores in the host cell membranes which results in the leakage of nutrients and essential ions from the cell and eventually cell lysis.

Intracellularly acting toxins

The most potent group of toxins known to man are the bacterial exotoxins that: act on intracellular targets. These include proteins such as botulinum toxin which inhibits the release of neurotransmitters in the peripheral nervous system cholera toxin that acts in the gut to induce watery diarrhea and diphtheria toxin which inhibits protein synthesis *(Table)*. Although these toxins cause a very disparate set of symptoms at different sites in the body they do tend to have a common **bipartite** structure. One part of the molecule is responsible for binding to specific receptors in the host cell membrane, called the B-portion while a separate part of the molecule, the A-portion carries the toxic enzymic activity These toxins are, therefore, frequently called A–B type toxins. Diphtheria toxin is one of the simplest of the A–B toxins consisting of just one A and one B portion. These are synthesised together as a single molecule, then cleaved to give two peptides joined by a disulfide bridge. In contrast, cholera toxin consists of five B subunits surrounding an active A_1–A_2 molecule, which is synthesized separately.

The B portion is responsible for the cell specificity of most of these toxins. It will bind only to cells that have the correct receptor. Hence, in the case of botulinum toxin, the receptor glycoprotein or glycolipid is only found in the peripheral neurones whereas diphtheria toxin binds to growth factor precursor found on many cell types. After binding to a receptor, the toxin-receptor complex is internalized in the cell either by endocytosis, followed by translocation of the A portion from the endocytic vacuole into the cytoplasm, or by translocation of the A portion across the host plasma membrane, directly into the cytoplasm.

In many cases the enzymic activity of the A-B-type toxins is ADP-ribosylation of a target site in the cell. The A-portion catalyzes the removal of the ADP-ribosyl group from NAD and the attachment of that group to a cellular protein. The target proteins and the consequences of the ADP-ribosylation are various, as shown in *Table*, and the varied nature of these explains why toxins cause a wide range of different symptoms. Diphtheria toxin, for example, inhibits protein synthesis, with one molecule of the toxin enough to kill the cell.

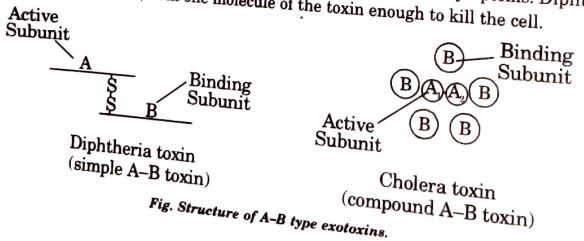


Fig. Structure of A–B type exotoxins.

Table : Some examples of bacterial exotoxins that act inside the cell

Bacteria	Toxin	Site of action	Mode of action	Symptom/role in disease
Vibrio cholerae *E coli*	Cholera toxin LT toxin	Intestine	ADP-ribosylation of Gs (stimulatory) protein which regulates adenylate cyclase	Water secretion into the intestine resulting in watery diarrhea
Corynebacterium Diphtheriae	Diphtheria toxin	Many cell types	ADP-ribosylation of EF-2 leads to inhibition of protein synthesis	Cell death, general organ damage
Pseudomonas aeruginosa	Exotoxin A			
Bordatella pertussis	Pertussis toxin	Many cell types	ADP-ribosylation of Gi (inhibitory) protein which regulates adenylate cyclase	Adherence to cells, cell damage, fluid secretion
Shigella dysenteriae *E coli*	Shiga toxin Vero (Shiga-like) toxin	Many cell types	RNA glycosidase enzyme modifies 28S rRNA in 60S ribosome subunit. Inhibits protein synthesis	Cell death
Costridum tetani	Tetanus toxin	Neurons in central nervous system	Metallopeptidase inhibits release of neurotransmitters	Spastic paralysis
Cootulinum	Botulinal toxin	Peripheral neurons	Metallopeptidase inhibits release of neurotransmitters	Flaccid paralysis

In contrast, cholera toxin modifies Gs protein, a regulatory protein of adenylate cyclase, causing it to be permanently switched to the production of cAMP. In the gastrointestinal tract, where the toxin acts, the most significant effect of the rise in cAMP is the production of an ion imbalance, leading to massive water loss from the cell This is seen as copious watery diarrhea, causing dehydration and death, if not treated.

Other enzymic activities are also associated with the A portion of A-B type toxins. Shiga toxin, produced by *Shigella dysenteriae,* is a ribonuclease which inactivates the 60S subunit of the ribosome. The neurotoxins, botulinum toxin, which causes flaccid paralysis, and tetanus toxin, which induces spastic paralysis, are both zinc-requiring endopeptidases. The different symptoms caused by each toxin are probably as a result of the different cell specificities of the toxins. Botulinum toxin acts on the peripheral nervous system whereas tetanus toxin interferes primarily with the central nervous system.

Control of Micro-organisms

Chemical Agents :

The potency or effectiveness, of a chemical antimicrobial agent is affected by time, temperature, pH, and concentration. The death rate of organisms is affected by the length of time the organisms are exposed to the antimicrobial agent. A pH that increases the degree of ionization of a chemical agent often increases its ability to penetrate a cell. Such a pH also can alter the contents of the cell itself. Increasing concentration may increase the effects of most antimicrobial chemical agents. High concentrations may be **bactericidal** (killing), whereas lower **concentrations** may be **bacteriostatic** (growth-inhibiting).

Both ethyl and isopropyl alcohol are exceptions to the rule regarding increasing concentrations.

Mechanisms of action of Chemical Agents

Reactions That Affect Proteins

In denaturation, hydrogen and disulfide bonds are disrupted, and the functional shape of the protein molecule is destroyed. When treated with mild heat or with some dilute acids, alkalis, or other agents for a short time, proteins are temporarily denatured. Denaturation is bactericidal if it permanently alters the protein such that the protein's normal state cannot be restored. Denaturation is bacteriostatic if it temporarily alters the protein.

Reactions That Affect Other Cell Components

Other cell components affected by chemical agents include nucleic acids and energyproducing systems. Alkylating agents can replace hydrogen on amino or alcohol groups in nucleic acids. Certain dyes, such as crystal violet, interfere with cell wall formation. Some substances, such as lactic acid and propionic acid end products of fermentation, inhibit fermentation and thus prevent energy production in certain bacteria, molds, and some other organisms.

Reactions That Affect Viruses

Inactivation can be effected by destroying either the viruses nucleic acid or their proteins. Alkylating agents, such as ethylene oxide, nitrous acid, and hydroxylamine act as chemical mutagens - they alter DNA or RNA.

Viruses sometimes remain infective even after their proteins are denatured.

Specific Chemical Antimicrobial Agents

Soaps and Detergents

Soaps contain alkali and sodium and will kill many species *of Streptococcus, Micrococcus,* and *Neisseria* and will destroy influenza viruses. Many pathogens that survive washing with soap can be killed by a disinfectant applied after washing.

Detergents are said to be *cationic* if they are positively charged and *anionic* if they are negatively charged. Cationic detergents are used to sanitize food utensils. Although not effective in killing endospores, they do inactivate some viruses. Anionic detergents are used for laundering clothes and as household cleaning agents. They are less effective sanitizing agents than cationic detergents, probably because the negative charges on bacterial cell walls repel them.

Acids and Alkalis

Soap is a mild alkali, and its alkaline properties help destroy microbes. A number of organic acids lower the pH of materials sufficiently to inhibit fermentation. Lactic and propionic acids retard mold growth in breads and other products. Benzoic acid and several of its derivatives are used to prevent fungal growth in soft drinks, catsup, and margarine. Sorbic acid and sorbates are used to prevent fungal growth in cheeses and a variety of other foods. Boric acid, formerly used as an eyewash, is no longer recommended because of its toxicity.

Heavy Metals

Heavy metals used in chemical agents include selenium, mercury, copper, and silver. Even tiny quantities of such metals can be very effective in inhibiting bacterial growth. Silver nitrate was once widely used to prevent gonococcal infection in newborn infants. Hospitals replaced silver nitrate with antibiotics such as erythromycin.

Organic mercury compounds, such as merthiolate and mercurochrome, are used to disinfect surface skin wounds. Such agents kill most bacteria in the vegetative state but do not kill spores. They are not effective against *Mycobacterium*. Merthiolate is generally prepared as a **tincture** (tingk'chur), that is, dissolved in alcohol. The alcohol in a tincture may have a greater germicidal action than the heavy metal compound. Thimerosal, another organic mercury compound, can be used to disinfect skin and instruments and as a preservative for vaccines. Phenylmercuric nitrate and mercuric naphthenate inhibit both bacteria and fungi and are used as laboratory disinfectants.

Selenium sulfide kills fungi, including spores. Preparations containing selenium are commonly used to treat fungal skin infections. Shampoos that contain selenium are effective in controlling dandruff.

Copper sulfate is used to control algal growth.

Halogens

Hypochlorous acid, formed by the addition of chlorine to water, effectively controls micro-organisms in drinking water and swimming pools. It is the active ingredient in household bleach and is used to disinfect food utensils and dairy equipment. It is effective in killing bacteria and inactivating many viruses.

Iodine also is an effective antimicrobial agent. Tincture of iodine was one of the first skin antiseptics to come into use. *Iodophors,* slow-release compounds in which the iodine is combined with organic molecules. Betadine and Isodine are used for surgical scrubs and on skin where an incision will be made.

Bromine is sometimes used in the form of gaseous methyl bromide to fumigate soil that will be used in the propagation of bedding plants.

Alcohols

When mixed with water, alcohols denature protein. They are also lipid solvents and dissolve membranes. Ethyl and isopropyl alcohols can be used as skin antiseptics. Alcohol disinfects but does not sterilize skin because it evaporates quickly and stays in contact with microbes for only a few seconds. It kills vegetative micro-organisms on the skin surface but does not kill endospores, resistant cells, or cells deep in skin pores.

Phenols

Phenol and phenol derivatives called *phenolics* disrupt cell membranes, denature proteins, and inactivate enzymes. Amphyl, which contains amylphenol, destroys vegetative forms of bacteria and fungi and inactivates viruses. It can be used on skin, instruments, dishes, and furniture. The orthophenylphenol in Lysoi gives it similar properties. *Cresols* is found in creosote, a substance used to prevent the rotting of wooden posts, fences, rail-road ties, and such.

Oxidizing Agents

Oxidizing agents disrupt disulfide bonds in proteins and thus disrupt the structure of membranes and proteins. Hydrogen peroxide (H_2O_2), which forms highly reactive superoxide (O^{-2}), is used to clean puncture wounds. When hydrogen peroxide breaks down into oxygen and water, the oxygen kills obligate anaerobes present in the wounds. Hydrogen peroxide is quickly inactivated by enzymes from injured tissues. A recently developed method of sterilization that uses vaporized hydrogen peroxide can now be used for small rooms or areas, such as glove boxes and transfer hoods. Another oxidizing agent, potassium permanganate, is used to disinfect instruments and, in low concentrations, to clean skin.

Alkylating Agents

Alkylating agents disrupt the structure of both proteins and nucleic acids. Because they can disrupt nucleic acids, these agents may cause cancer. Formaldehyde, glutaraldehyde, and β-propiolacetone are used in aqueous solutions. Ethylene oxide is used in gaseous form.

Formaldehyde inactivates viruses and toxins without destroying their antigenic properties. Glutaraldehyde kills all kinds of micro-organisms, including spores, and sterilizes equipment exposed to it for 10 hours. Beta-propiolacetone destroys hepatitis viruses, as well as most other microbes, but penetrates materials poorly. It is, however, used to inactivate viruses in vaccines.

Gaseous ethylene oxide has extraordinary penetrating power. Used at a concentration of 500 mg/1 at 50°C for 4 hours, it sterilizes rubber goods, mattresses, plastics, and other materials destroyed by higher temperatures.

Dyes

The dye acridine, which interferes with cell replication by causing mutations in DNA, can be used to clean wounds. Methylene blue inhibits growth of some bacteria in cultures. Crystal violet (gentian violet) blocks cell wall synthesis, possibly by the same reaction that causes this dye to bind to cell wall material in Gram staining. It effectively inhibits growth of Gram-positive bacteria in cultures and in skin infections. It can be used to treat protozoan (*Trichomonas*) and yeast (*Candida albicans*) infections.

Other Agents

Certain plant oils have special antimicrobial uses. Thymol, derived from the herb thyme is used as a preservative, and eugenol, derived from oil of cloves, is used in dentistry to disinfect cavities. Other agents are used as food preservatives. They include sulfites and sulfur dioxide, used to preserve dried fruits and molasses; sodium diacetate, used to retard mold in bread; and sodium nitrite, used to preserve cured meats and some cold cuts. Foods containing nitrites should be eaten in moderation because the nitrites are converted during digestion to substances that may cause cancer.

Physical Antimicrobial Agent

Principles and Appications of Heat Killing

Heat is preferred agent of sterilization for all materials not damaged by it. It rapidly penetrates thick materials not easily penetrated by chemical agents. The **thermal death point** is the temperature that kills all the bacteria in 24-hour -old broth culture at neutral pH in 10 minutes. The **thermal death time** is the time required to kill all the bacteria in a particular culture at a specified temperature. The decimal reduction time, also known as the **DRT** or **D value,** is the length of time needed to kill 90 percent of the organisms in a given population at a specified temperature. The temperature is indicated by a subscript: $D_{80°C}$.

Dry Heat

Dry (oven) heat penetrates substances more slowly than moist (steam) heat. It is usually used to sterilize metal objects and glassware and is the only suitable means of sterilizing oils and powders. Objects are sterilized by dry heat when subjected to 171°C for 1 hour 160°C for 2 hours or longer, of 121°C for 16 hours or longer, depending on the volume.

An open flame is a form of dry heat used to sterilize inoculating loops and the mouths of culture tubes by incineration and to dry the inside of pipettes.

Moist Heat

Moist heat, because of its penetrating properties, is a widely used physical agent. Boiling water destroys vegetative cells of most bacteria and fungi and inactivates some viruses, but it is not effective in killing all kinds of spores. The effectiveness of boiling can be increased by adding 2 percent sodium bicarbonate to the water. However, if water is heated under pressure, its boiling point is elevated, so temperatures above 100°C can be reached. This is normally accomplished by using an autoclave (aw' to-klav), in which a pressure of 15 lb/in^2. above atmospheric pressure is maintained for 15 to 20 minutes, depending on the volume of the load. At this pressure, the temperature reaches 121°C, which is high enough to kill spores as well as vegetative organisms and to disrupt the structure of nucleic acids in viruses. In this procedure it is the increased temperature, and not the increased pressure, that kills micro-organisms.

Pasteurization

Pasteurization, a process invented by pasteur to destroy organisms that caused wine to sour, does not achieve sterility. It does kill pathogens, especially *Salmonella* and *Mycobacterium,* that might be present in milk, other dairy products, and beer. *Mycobacterium* used to cause many cases of tuberculosis among children who drank raw milk. Milk is pasteurized by heating it to 71.6°C for at least 15 seconds in the flash method or by heating it to 62.9°C for 30 minutes in the holding method. Certain strains of bacteria of the genus Listeria were found in pasteurized milk and cheeses. This pathogen causes diarrhea and encephalitis and can lead to death in pregnant women.

Refrigeration

Many fresh foods can be prevented from spoiling by keeping them at 5°C (ordinary refrigerator temperature). However, storage should be limited to a few days because some bacteria and molds continue to grow at this temperature, strains of *Clostridium botulinum* have been found growing and producing lethal toxins in a refrigerator when the organisms were deep within a container of food, where anaerobic conditions exist.

Freezing

Freezing at –20°C is used to preserve foods in homes and in the food industry. Although freezing does not sterilize foods, it does significantly slow the rate of chemical-reactions so that microorganisms do not cause food to spoil.

Drying

Drying can be used to preserve foods because the absence of water inhibits the action of enzymes. Many foods, including peas, beans, raisins, and other fruits, are often preserved by drying. Yeast used in baking also can be preserved by drying.

Freeze- Drying

Freeze-drying, or lyophilization (li-of"i-li-za'shun), is the drying of a material from the frozen state. This process is used in the manufacture of some brands of instant coffee; freeze-dried instant coffee has a more natural flavor than other kinds.

Organisms are rapidly frozen in alcohol and dry ice or in liquid nitrogen and are then subjected to a high vacuum to remove all the water while in the frozen state. Rapid freezing allows only very tiny ice crystals to form in cells, so the organisms survive this process.

Ultra-violet Light

Ultra-violet (UV) light consists of light of wavelengths between 40 and 390nm, but wavelengths in the 200-nm range are most effective in killing microorganisms.

Ionizing Radiation

X-rays, which have wavelengths of 0.1 to 40nm, and gamma rays, which have even shorter wavelengths, are forms *of ionizing radiation,* so named because it can dislodge electrons from atoms, creating ions. (Longer wavelengths comprise *nonionizing radiation.*) These forms of radiation also kill microorganisms and viruses. Many bacteria are killed by absorbing 0.3 to 0.4 millirads or radiation; polioviruses are inactivated by absorbing 3.8 millirads. A rad is a unit of radiation energy absorbed per gram of tissue; a millirad is one-thousandth of a rad. Humans usually do not become ill from radiation unless they are subjected to doses greater than 50 rad.

Microwave Radiation

Microwave radiation has wavelength's of approximately 1 mm to 1 m. Microwave oven frequencies are tuned to match energy levels in water molecules. In the liquid state, water molecules quickly absorb the microwave energy and then release it to surrounding materials as heat. Thus, materials that do not contain water, such as plates made of paper, china, or plastic, remain cool while the moist food on them becomes heated.

Strong Visible Light

Strong visible light, which contains light of wavelengths from 400 to 700 nm (violet to red light), can have direct bactericidal effects by oxidizing light-sensitive molecules such as riboflavin and porphyrins (components of oxidative enzymes) in bacteria.

Sonic and Ultrasonic Waves

Ultrasonic waves, or waves with frequencies above 15,000 cycles per second, can cause bacteria to cavitate. **Cavitation** (kav"i-ta'shun) is the formation of a partial vacuum in a liquid-in this case, the fluid cytoplasm in the bacterial cell. Bacteria so treated disintegrate, and their proteins are denatured. Enzymes used in detergents are obtained by cavitating the bacterium *Bacillus subtills*. The disruption of cells by sound wave is called sonication (son"i-ka'shun).

Filtration

Filtration is the passage of a material through a filter, or straining device. *Membrane filters,* thin disks with pores that prevent the passage of anything larger than the pore size, are widely used today. They are usually made of nitrocellulose and have the great advantage that they can be manufactured with specific pore sizes from 25 m to less than 0.025 m.

High-efficiency particulate air (HEPA) filters are used in the ventilation systems of areas where microbial control is especially important, such as in operating rooms, burn units, and laminar Flow transfer hoods in laboratories. HEPA filters also capture organisms released in rooms occupied by petients with tuberculosis or in laboratories where especially dangerous microbes are studied. These filters remove almost all organisms larger than 0.3 m in diameter. Used filters are soaked in formalin before they are disposed of.

Osmotic Pressure

High concentrations of salt, sugar, or other substances create a hyperosmotic medium, which draws water from microorganisms by osmosis. **Plasmolysis** (plaz-mol'i-sis), or loss of water, severely interferes with cell function and eventually leads to cell death. The use of sugar in jellies, jams, and syrups or salt solutions in curing meat and making pickles plasmolyzes most organisms present and prevents growth of new organisms. A few halophilic organisms, however, thrive in these conditions and cause spoilage, especially of pickles and some fungi can live on the surface of jams.

Applications of Microorganisms

Antibiotics

Antibiotics (anti = against, bios = life) are chemical substances secreted by some micro-organisms which inhibit the growth and development of other microbes.

Broad Spectrum Antibiotics :

They can control the growth of several (wide range, i.e., gram + ve & gram-ve) organisms e.g. tetracyclines, chloramphanicol.

Narrow Spectrum Antibiotics :

They are effective against selected species of bacteria, (gram + ve or gram-ve) e.g., penicillin, streptomycin.

Antibiotic	Antibacterial	Producing micro-organism
Penicillin-G		*Penicillin sp.*
Cephalosporin		*Acremonium sp.*
Streptomycin		*Streptomyces sp.*
Tetracycline		*Streptomyces sp.*
Chloramphenisol		*Cephalosporium sp.*
Bacitracin		*Bacillus sp.*
	Antitumor	
Actinomycin-D		*Streptomyces sp.*
Mitomycin-C		*Streptomyces sp.*
Bleomycin		*Streptomyces sp.*
Adriamycin		*Streptomyces sp.*
Daunomycin		*Streptomyces sp.*
	Antifungal	
Griseofulvin		*Penicillin sp.*
	Food Preservative	
Natamycin		*Streptomyces sp.*
Nisin		*Streptomyces sp.*
	Antiprotozoal	
Daunorubicin		*Streptomyces sp.*
	Antituberculosis	
Refamycin		*Nocardia sp.*
	Antiamoebic	
Tetracyclin		*Streptomyces sp.*
Fumagillin		*Aspergillus sp.*

1. **Penicillins**

   Penicillins are a group of β-lactam containing bacterial antibiotics. Natural Penicillin (G and V) are effective against several gram negative bacteria. They inhibit the bacterial cell wall (i.e. peptoglycan) synthesis and cause cell death. Earlier *Penicillium notatum* was used for large scale production of penicillin but now *Penicillium chrysogenum* and its improved mutant strains are used.

2. **Cephalosporins**

   Cephalosporins are broad spectrum antibiotics with low toxicity. Cephalosporin-C was developed by fungus, *Cephalosporium*. Mutants of *C. acremonium* have been developed for improved production.

3. **Aminoglycosides**

   Aminoglycosides are oligosaccharide antibiotics. Aminoglycosides are very potent antibiotics and act against Gram-positive and Gram-negative bacteria, besides mycobacteria. Aminoglycosides are produced by, *Actinomyces sp.* e.g. streptomycin by *S.griseus*

4. **Tetracyclins**

   Tetracyclins are broad spectrum antibiotics which are effective against gram-positive and gram-negative bacteria besides mycoplasm chlomydias, rickettsias. Tetracyclines are used to combat stomach ulcers against *Helicobacter pylori*. The first tetracyclin antibiotic was produced from *Streptomyces aureofaciens*. Genetically engineered strains of *S. rimosus* are developed for increased synthesis of oxytetracyclins.

5. **Macrolides**

   Macrolides are a group of antibiotics with large lactone rings or macrocyclic lactone ring. Macrolides are produced by Actinomycetes e.g. *Erythromycin* by *S. erythreus*

6. **Aromatic Antibiotics**

   Chloramphenicol

   Chloramphenicol are broad spectrum antibiotics that can act against Gram-positive and Gram-negative bacteria, besides rickettsias, actinomycetes and chlamydias. Chloramphenical is produced by *Streptomyces venezuelae* and *S. omiyanesis*.

Viruses

Historical Review

In 1676, Dutch naturalist. **Antony van Leeuwenhoek** discovered an important secret of nature—**the existence** of minute, invisible organisms (*micro-organisms* or *microbes*) which can be observed, **not by naked eyes, but** only under the microscopes. He had observed certain micro-organisms (bacteria, protozoans, yeast, **etc.**) in pond water and other liquids under his microscopes. He called these organisms **"the little animalcules".** This discovery earned the title of *"Father of Microbiology"* for **Leeuwenhoek,** although the foundation of *"Microbiology"* as a separate branch of biological studies was laid long afterwards in middle of 19th century by French scientist. **Louis Pasteur.**

Only a few years later in 1892, Russian biologist, **D.J. Ivanovsky,** similarly suspected that the mosaic disease of tobacco leaves is caused by an invisible ultramicrospic pathogen much smaller than bacteria. To confirm his suspicion, he extracted the juice of some diseased leaves, filtered the juice through fine porcelain bacterial filters (such filters were made in Pasteur's laboratory) to filter out all bacteria from the juice and, then, rubbed the filtrate upon healthy leaves; the leaves became infected and developed the disease. Existence of pathogens smaller than bacteria was thus established. Later, these pathogens were named *"viruses"*, meaning poison, and the credit of their discovery went to **Ivanovsky.**

Investigating the tobacco mosaic disease, another Dutch scientist, **Beijerinck** (1898) confirmed Ivanovsky's observations To the bacteria-free filtrate that contained the tobacco mosaic virus (TMV), **Beijerinck** gave the term of *"contagium vivum fluidum"*, meaning *"living contagious fluid"*.

Nature and Distribution of Viruses

In 1935, **W.M. Stanley** achieved success in isolating and crystallizing tobacco mosaic virus (TMV) and shared the 1946 Nobel Prize for this work with **Northrop** and **Sumner** who had earlier crystallized an enzyme for the first time. The crystals prepared by him were found to be viable infectants, i.e., these were capable of infecting tobacco leaves as usual. Hence, it was established that viruses are not organisms, but only nonliving (=inanimate or inert) ultramicroscopic particles capable of infecting live cells and causing serious infectious diseases. These particles are now called **"virions".**

Invention *of electron microscopy by* **Knoll** and **Ruska** (1934), just a year before Stanley's work, trigerred a fast growth in our knowledge about viruses durring the last 5-6 decades under a separate branch of study, called **"Virology".**

Structural Features of Viruses

Size of Viruses

Virus particles (=virions) vary widely in size. The smallest viruses (e.g. virus of foot and mouth disease), measuring about 20 millimicrons (20 nm or m = 0.02 micrometres) are about the size of largest protein molecules (haemycanin). The **largest viruses** (e.g. poxviruses), measuring about 350 nm (=0.35 micrometres) are as large as the smallest bacteria (*Mycoplasma*).

Shape and Symmetry of Viruses

Most **plant viruses** are rod-shaped or filamentous (e.g. TMV) while the **animal viruses** may be filamentous (e.g. influenza virus) or roughly spherical (e.g. polio virus). Filamentous virions are cylindrical structures with a helical symmetry. In most animal viruses, the helix is highly coiled into a subspherical structure. The spherical virions are polyhedral structures like diamonds, with a typical icosahedral symmetry, each having 20 equilateral triangular sides or facets and 12 vertices or corners.

Poxviruses and rabies virus are unique in shape and symmetry. **Poxviruses** are brick-shaped and **rabies virus** is bullet-shaped. These viruses are complex in structure and symmetry.

Viruses of bacteria (**bacteriophages**) are also unique in shape and structure. Each virion is a tedpole-shaped structure having two main parts—a polyhedral **head** and a springlike cylindrical or conical **tail** with a thin collar in between. The free, flattened end of the tail bears a large endplate or baseplate from which six slender hooklike legs, called tail fibres, and six small pegs or spikes project in different directions. Certain bacteriophages lack the endplate, tail fibres and pegs.

Composition and Structure of Viruses

Nucleic acids and proteins—After crystallization of TMV by **Stanley,** it became easy to find out the chemical composition of viruses. **Bawden** and **Pirie** (1937) also isolated TMV in crystalline form and demonstrated that this virus contained RNA and proteins. Similarly, **Northrop** (1938) isolated a bacteriophage and demonstrated that it contained a nucleic acid and proteins. Thus, it was firmly established that viruses are **nucleoprotein particles.** In spite of being composed of the two most vital biopolymers, viruses are nonliving objects, because these lack a cellular organization and protoplasm. These particles were, then, named **"virions".**

It was soon discovered that a single long and coiled, filamentous molecule of a nucleic acid (either DNA or RNA) forms about 6% central, **core part** of a virion, while the 94% remaining part is a thick, impregnable, protective sheath, formed of protein molecules, around the nucleic acid filament. The sheath was named **capsid.** Hence, virions were also called **"nucleocapsids".**

Viral genome— It may be a single or double-stranded, and linear or circular, molecule of either DNA or RNA. Possesion of only one type of nucleic acid (DNA or RNA), functioning of RNA as "hereditary blueprint", single-stranded DNA molecules and double-stranded RNA molecules are characteristics never found in "living systems" and, hence, unique to the viruses.

Viral capsid— The capsid of a virion is formed of identical repeated subunits, called **structural units** in helical and **capsomeres** in icosahedral capsids. A structural unit is a single polypeptide chain, whereas a capsomere is comprised of multiple polypeptide chains of similar or different kinds.

Viral envelope—In many viruses, there is a thin envelope around the capsid. The envelope structurally resembles typical cell membranes having a phospholipid bilayer with embedded proteins and surface projections or spikes of gylcoproteins, called **peplomers.** Spikes may be of one or two types in shape and composition and may have **agglutinins** (which agglutinate RBCs) or **neuraminidase enzyme,** or both.

Viral enzymes—Besides the enzymes of viral envelope, certain **auxillary enzymes** are found associated with the genome in several types of viruses. These enzymes catalyse replication and transcription of viral genome after a virus has entered its host cell.

Some important enzymes of this category are listed in the following chart—

Enzyme	Viruses in which found	Function of enzyme
1. DNA-dependent DNA polymerase	Poxvirus, Hepatitis B virus	Replication of double-stranded DNA (dsDNA) genome
2. DNA-dependent RNA polymerase	Poxvirus and certain bacteriophages	Transcription of mRNA from dsRNA
3. dsRNA transcriptase	All dsRNA viruses (Reoviruses and Rotoviruses)	Transcription of mRNA from dsDNA
4. Single-stranded RNA (ssRNA) transcriptase	All viruses with ssRNA genome of reverse polarity to mRNA (+ RNA), *i.e.*, genome of —RNA : Viruses of rabies, mumps, measles, influenza, etc.	Transcription of + RNA from–RNA and replication of–RNA
5. Reverse transcriptase (RNA-dependent DNA polymerase)	RNA oncoviruses (cancer-causing viruses), AIDS virus	Transcription of DNA from RNA genom

Classification of Viruses

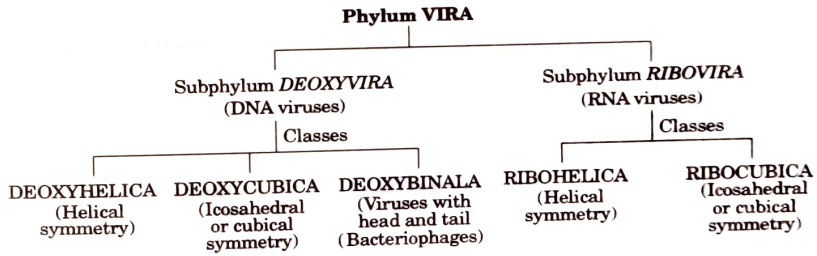
No evolutionary or phylogenetic relationships exist between viruses. A natural system of classification cannot therefore, be devised for viruses.

Holmes (1948) proposed **Order Virales** for the viruses, and divided it into three groups on basis of the category of hosts—

1. **Zoophaginae**—Animal viruses
2. **Phytophaginae**—Plant viruses and
3. **Phaginae**—Viruses of bacteria (*Bacteriophages*), blue-green algae (*Cyano-phages*), and fungi (*Mycophages*).

In 1962, **Lwoff, Home and Tournier** devised a system (now called LHT system) of classifying viruses mainly on bases of nature of genome, capsid symmetry, number of structural units/capsomeres in the capsid, size of virions, etc. This system was largely adopted by the **International Committee on Taxonomy of Viruses (ICTV).**

Under the LHT system, viruses are placed under a separate phylum, **"Vira"**. Vira is divided into two subphyla and each subphylum is further divided into classes as follows—



The various classes have been further divided into orders, suborders and families under the LHT system.

Plant Viruses

Virologists have visualized more than 180 types of plant viruses. Like the TMV, majority of plant viruses possess a helical symmetry and a single-stranded RNA (ssRNA) genome. A few plant viruses possess a double-stranded **RNA** (dsRNA) or double-stranded DNA (dsDNA) genome.

Phage Viruses

These include bacteriophages, cyanophages and mycophages. These are shaped like tadpoles or vertebrate sperms (spermatozooids) with a **head** and a **tail**. The genome, contained in the head, is usually double-stranded DNA (dsDNA). A few spherical or filamentous phages, possessing ssDNA, or dsRNA, or ssRNA have also been discovered. Tailed phages use their tails as "hypodermic syringes" for injecting their genome into their host cells through the rigid host cell wall.

In respect of host cell-phage relationship, there are two types of bacterio-phages—virulent or lytic and temperate or lysogenic. **Lytic bacteriophages** multiply in host bacterium which, then, undergoes complete lysis (degeneration) to release the resulting daughter phages. In case of **lysogenic bacteriophages,** the viral genome gets integrated into bacterial genome and is, now, called as *"provirus"* or *"prophage"*. This phenomenon is called **lysogeny**, and the host bacterium is called **lysogenic**. The prophage codes for new proteins which impart some new properties to the bacterium. This is called **lysogenic** or **phage conversion**. At times of binary fission, or any other stress condition, the prophage may seprate from bacterial chromosome and become virulent. Lysogenic bacteriophages may, therefore, act as carriers of genes from one bacterium to others. This is called **transduction.**

Animal Viruses

About 600 or so known animal viruses have been segregated into about one and a half dozen families, of these, the following families include the viruses of important human diseases:

Chart of Important Families of Viruses of Human Diseases

	Families	Capsid sym metry/ virion diameter (nm)	Naked or Enveloped	Nature of genome	Important examples	Diseases caused
VIRUSES WITH DNA GENOME (DEOXYRIBOVIRUSES)						
1.	Herpesviridae	icosahedral; 100-200	Enveloped	Double-stranded DNA (dsDNA)	Herpes simplex Varicella- Zoster (VZ) virus	Fever blisters chickenpox
2.	Poxviridae	Complex; 250-300	Enveloped	dsDNA	Variola. Vaccinia	Smallpox, Cowpox
VIRUSES WITH RNA GENOME (RIBO VIRUSES)						
3.	Picornaviridae	Icosahedral; 20-30	Naked	Single-strandedRNA (ssRNA)	Poliovirus Rhinovirus Coxsackie Hepatitis virus	Poliomyelitis common cold Myocarditis Hepatitis
4.	Reoviridae	Icosahedral; 70-80	Naked	Double-stranded(RNA) (dsRNA)	Reovirus Rotovirus	Respiratory tract infection Gasteroenteritis
5.	Togaviridae	Icosahedral; 40-60	Enveloped	ssRNA	Alphavirus Flavivirus Pestivirus	Encephalitis Yellow fever Hog cholera
6.	Orthomyxoviridae	Helical (coiled, spherical) 80-120	Enveloped	ssRNA with RNA transcriptase enzyme	Influenza A, B, C viruses	Influenza
7.	Paramyxoviridae	Helical (coiled, spherical) 150-300	Enveloped	ssRNA with RNA transcriptase enzyme	Pneumovirus Paramyxovirus Morbilivirus	Pneumonia Mumps, Parainfluenza Measles
8.	Rhabdoviridae	Helical (coiled) 60×180	Enveloped	ssRNA with RNA transcriptase enzyme	Lyssavirus Vesiculovirus	Rabies Vesicular stomatitis
9.	Retroviridae	Helical (coiled) 100	Enveloped	ssRNA with reverse transcriptase enzyme	Oncoviruses HIV	Cancer AIDS

Viroids and Prions

Viroids

In 1971, **T.O. Diener** reported that the *"spinde tuber disease"* of potato, causing gnarling and cracking of the tuber, is caused by multiplication of a pathogen much smaller than viruses and present in tuber cells themselves. This pathogen was found to be a free (naked), short and circular RNA strand of about 300 nucleotides. **Diener** named it **"viroid".** It was found that viroids are infectious because, after multiplying in a plant cell, these disperse in the tissue and invade new cells. About a dozen plant diseases have since been shown to be caused by viroids. It is also possible that the causative agents of some more plant and animal diseases turn out to be viroides.

Prions

While searching for the cause of a mental disease, "kuru" (= *laughing death*), widespread in a tribe of New Guinea, **D.C. Gajdusek** (1968) discovered that the disease is caused by a *"Slow virus injection"*. He shared the 1976 Nobel Prize for this work. Later, he suspected that other neurological disorders, like *"mad cow disease"* in cows and human *dementia* and *Parkinson's disease* might also be of this category. **Prusiner** (1985) speculated that *Creulzfeldt-Jacob disease* (a type of insanity in young people) and other diseases of this category are due to certain extremely minute (4-6 nm in diameter) protein particles which are coded by the genome of certain brain cells, and act as infectious pathogens. These particles have been named "prions".

Virus Multiplication

Obligatory parasitism—Outside cells viruses are nonliving, inactive particles but, after entering into live cells, these multiply fast by replication, like organisms. Thus, these represent obligatory parasitism : these can be defined as **"inanimate obligatory parasites".** Obviously, these have their own *"hereditary blueprint (=genome)"* for replication, but no metabolic machinery to use the genome. After entering into host cells, therefore, they inactivate the cellular genome and employ the cellular metabolic machinery to obey their genome for their own replication. Thus, their genome is the basic infectious material.

Host-cell Cycle of Viruses

The different stages from the contact of a virus with its host cell to the release of its copies (progeny viruses) from the host cell constitute the **host-cell cycle** of a virus. It may also be called **replicative** or **parasitic cycle** of the virus, but not its life cycle, because growth never occurs in a virus; simply the components of its several copies are synthesized and assembled in the host cell.

Early studies on viral replication began with bacteriophages (Fig). The complete host-cell cycle comprises following six phases —

1. **Adsorption (attachment of virion with host cell)**—As a virion comes in contact with a suitable (=susceptible) host cell, it becomes attached upon host cell surface due to interaction between its attachment proteins and specific receptor proteins (glycoproteins or lipoproteins) of host cell membrane.

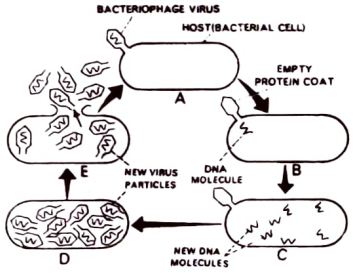


Fig. Host-cell cycle of bacteriophage

2. **Penetration**—As mentioned before, a bacteriophage virion leaves its capsid outside and injects only its nucleic acid filament (dsDNA) into the host bacterium by using its tail as a *"hypodermic syringe"* : the tail fibers bend, and the springlike tail compresses to inject the viral genome through a puncture in the rigid wall of the bacterium.

Most **plant viruses** enter whole into the host cells at points of injuries upon leaf surfaces, or these are inoculated into plant cells by arthropod vectors.

Animal viruses also enter whole into their host cells. Three types of penetration mechanisms have been described in their case —

(*i*) **Direct passage**—Nonenveloped virions *(e.g.,* poliovirus) reach into host cell cytoplasm by simply pushing through the host cell membrane.

(*ii*) **Fusion**—The envelope of some enveloped viruses *(e.g.* viruses of mumps, pneumonia, chickenpox, etc.) fuses with host cell membrane, becoming continuous with it and, thus, releasing the nucleocapsid into host cell cytoplasm.

(*iii*) **Endocytosis (=viropexis)**—The virion, in this case, is actively engulfed by the host cell by a process resembling phagocytosis (= cell-eating), so that it is enclosed in a vacuole or vesicle when it reaches into the cytoplasm of host cell.

3. **Uncoating**—Within the host cell, all of a virion, except its genome and enzymes associated with the genome, is digested by lysosomal enzymes of the host cell.
4. **Biosynthesis (=Eclipse phase)**—This phase includes replication of viral genome and synthesis of viral proteins, as well as, the enzymatic proteins required for inactivation of host cell genome, and for initiation, regulation and control of viral synthesis, assembly and release. Replication of viral genome of most DNA viruses occurs in host cell nucleus. Contrarily, the genome of most RNA viruses is replicated in host cell cytoplasm. Viral proteins are always synthesized in host cell cytoplasm.
5. **Maturation**—This comprises assembly of viral components into progeny virions. It occurs in host cell nucleus *(e.g.* herpesvirus) or cytoplasm *(e.g.* poxviruses). In case of enveloped viruses, the envelope is respectively derived from nuclear and cell membranes.
6. **Release**—Usually, quite a large number of progeny viruses are formed in the host cell. In case of bacteriophages, progeny viruses are released by lysis of host bacterium. In case of animal viruses, progeny viruses are generally released by budding from host cell surface.

The host-cell cycle is completed in about 15 to 30 minutes in case of bactriophages, but in 15 to 30 hours in case of animal viruses. The progeny viruses, released from host cells, attack fresh host cells in the infected tissues.

Natural Defenses Against Viral Infections

1. **Interferons**—In 1957, **Isaacs** and **Lindenmann** discovered that vertebrate cells, infected with viruses, produce a diffusible antiviral protein which they named **interferon.** Later, interferons were found to be a class of glycoproteins. Interferon diffuses out from an infected cell, enters into similar cells in the neighbourhood and imparts a defense mechanism to these cells against viral infections.
2. **Body immunity**—Viruses are effectively poisonous, *i.e.,* these are good antigens to induce active production of specific antibodies (by lymphocytes) and immune reactions in the body. Thus, viral infections establish prolonged and effective immunity in the body against reinfection by the respective viruses.

Immunoprophylaxis—Vaccines

Vaccines have been extensively used to build a strong and prolonged antiviral immunity in the body against possible viral infections. The vaccines contain inactivated (=attenuated) virions themselves, which cannot cause the concerned diseases, but still act as antigens to induce antibody production by the immune system of body.

Chemotherapy—Antiviral drugs

Since viruses are strictly intracellular parasites, chemotherapeutic agents and antibiotics which help in the treatment and control of bacterial diseases are ineffective for viral diseases. Certain antiviral drugs have, however, been prepared since 1960. Some important antiviral drugs are **Thiosemicarbazone** for treatment and prevention of smallpox, **Ribavirin (=Virazole)** for influenza. **Deoxyuridines** for herpes, **Foscarnet** for hepatitis B, etc.

AIDS and HIV

Body immunity

Every one of us is surrounded by a large variety of innumerable invisible disease-producing microbes **(=pathogens)** and poisonous chemicals **(=antigens).** Different pathogens and antigens keep on entering into our bodies through various means (inhaled air, ingested food and drinks, cuts and wounds, bites of insects and other animals, etc.), threatening our existence. However, nature has gifted us with a highly efficient, built in defence system to effectively resist the attack of these tiny foes and poisons. This resistance of our body to pathogens and antigens is called **body immunity.**

Body immunity is of two main types—innate and acquired. **Innate immunity** is genetic and, hence, hereditary. **Acquired immunity** is acquired by us during our lifetime against particular pathogens and antigens that enter into our bodies. A class of proteins, called **antibodies,** are formed under acquired immunity system, to inactivate and destroy these pathogens and antigens. The sentinels of this defense system are a type of white blood cells (WBCs), called **lymphocytes.** These cells are of different types to play different roles in acquired immunity. One type of lymphocytes, called $\mathbf{T_H}$ ($=T_4$) **lymphocytes,** play the crucial role of arousing other components of this immunity system to activity and of inducing the production of antibodies. **These T_4 lymphocytes are the target cells of AIDS virus.**

Thus, AIDS is a viral disease in which our acquired immunity defense system is gradually shattered and the patients, therefore, die of multiple infectious diseases. That is why, it has been named as **"Acquired Immunodeficiency Disease".** It is a deadly disease indeed, one that permits all other infectious diseases to fearlessly destroy our bodies.

Brief history of AIDS Virus

First indication of AIDS disease came to light in U.S.A. in 1981 amongst a group of young homosexuals who were addicts of heroin and other narcotics, and had lost immunity competence to pneumonia and skin cancer. **Luc Montagnier** (1983) of Pasteur Institute in Paris, and **Robert Gallo** (1984) of N.I.H., U.S.A. discovered that a new type of *retrovirus,* yet unknown to scientists, is the causative pathogen of AIDS. In 1986, the International Committee on Virus Nomenclature decided to name the AIDS virus as *Human Immunodeficiency Virus* (HIV).

Structure of HIV

HIV is a spherical, enveloped virus of about 90-120 nm diameter (Fig.). Its nucleocapsid is icosahedral. Its genome consists of a single-stranded RNA filament segmented into two indentical filaments and associated with a **"reverse transcriptase enzyme"**. The envelope is comprised of a lipid bilayer derived from host cell membrane, and projecting, knoblike glycoprotein spikes with pedicles formed of virus-coded glycoproteins.

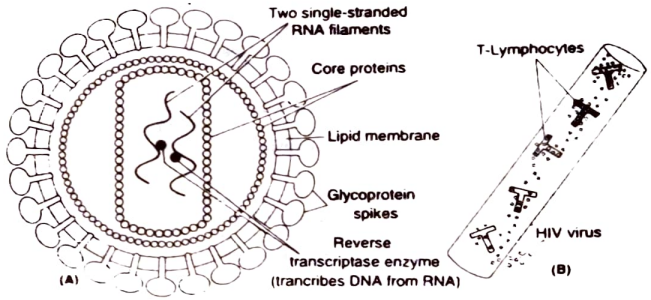


Fig. Virus of AIDS (HIV); B. Attack of HIV upon T_4 lymphocytes

Pathogenesis

T_4 ($=T_H$) lymphocytes are the principal target cells of **HIV.** After uncoating in host cell, the reverse transcriptase enzyme of the virus first catalyzes transcription of viral RNA into a single-stranded DNA and, then catalyzes doubling of ssDNA into a double-stranded DNA (dsDNA) filament. The dsDNA becomes integrated with the host cell DNA as a *"provirus",* causing a latent infection. From time to time, the provirus becomes active and codes for production of progeny virions which are released from the host cell into blood plasma by budding and infect new lymphocytes. Eventually, the T_4 lymphocytes die and disintegrate. Thus, the T_4 lymphocyte count in blood gradually diminishes. From the blood plasma, the virus also permeates into other body fluids like semen, vaginal fluids, saliva, tears, urine, sweat, breast milk, etc.

Clinical symptoms

Generally, a person infected with **HIV** does not show any apparent symptoms for a number (3 to 12) of years. Ten to fifteen percent infected people may, however, suffer from mild fever, headache and uneasiness for sometime after a few weeks of infection. Soon, these symptoms disappear, and infected persons continue transmitting the virus to other persons.

Gradual destruction of more and more T_4 lymphocytes through several years ultimately leads to a **full blown AIDS disease** during which a patient suffers from persistent fever and diarrhoea, loss of appetite and weight, weakness and fatigue, body pain, dry cough, white lesions in mouth and alimentary canal, and warts upon anus and genitals. Eventually, the patient loses immunity and dies of **multiple infectious diseases.**

Transmission and Epidemiology

HIV is present in blood and other body fluids of infected persons. Hence, it is transmitted from one to another person through activities involving exchange or transfer of body fluids. Thus, **HIV** is primarily transmitted through sexual activities. That is why, it is predominant amongst male homosexuals. Virus-laden **T_4 lymphocytes** are transmitted with semen through mucosal tears during anal sex. Unprotected vaginal or oral sex, sharing of syringes and needles by intravenous drug addicts, use of unsterilized syringes by doctors and health workers, etc., are also responsible for transmission of **HIV.**

Another potential mode of transmission of **HIV** is through transfusion of infected blood and blood products, and tissue and organ transplantations from infected to healthy persons. Also, infected mothers are most likely to transmit the virus to their children during pregnancy, or through breast milk after birth.

Normal family and social contacts, like hugging, dry kissing, hand shake, coughing and sneezing, sharing of food, toilet, bed, etc., and insect bites do not transmit **HIV** infection. Thus **AIDS** is infectious, but not a contagious disease like common cold, influenza, polio, measles etc.

Although **AIDS** is hardly a two decades old disease, it has spread virually all over the world, becoming **pandemic.** According to the estimates of **World Health Organization (WHO),** about 37 million people in the world harbour **HIV** and about 8500 new cases are detected every day. **HIV** cases in the world are, therefore, expected to be around 45 million by the turn of this century.

HIV infection in our country was first detected in 1986 in some female sex workers of Tamil Nadu (Madras). Since then, about 2.5 million cases have been detected, and infection is fast spreading.

Prophylaxis and Prevention

No vaccine to provide people a passive prophylactic immunity against **HIV** prior to infection has yet been developed, although scientists all over the world are actively engaged in an effort to find out a proper vaccine. The **prevention of AIDS.** therefore, presently rests upon general precautions, such as safer sex by changing sexual attitudes and extensive use of condoms, strict screening of blood and blood products used for transfusions and tissue and organ transplantations, use of perfectly sterilized and preferably disposable syringes and extensive health education to arouse general awareness amongst people about the high risk involved in **HIV infection.** To arouse such an awareness, **WHO** has announced **December, 1** to be observed as **World's AIDS Day.**

Diagnosis and Treatment

Certain genes of HIV genome code for synthesis of some antigens which are released into the blood by infected T_4 lymphocytes. The immune system of body produces antibodies against these antigens. Based on these antigens and antibodies, HIV infection is detected by **serological tests.** Scientists have developed kits to make the serological tests simple. **Elisa** (enzyme-linked immunosorbent assay) **kit** is, perhaps, least cumborsome and least expensive and, hence, most commonly used.

The most distressing fact about **AIDS** is that there is as yet no cure for it. There is no effective drug to save the immune system of AIDS patient from collapse. Hundreds of AIDS patients are, therefore, dying every day in the world.

About 30 drugs have been reported as potential drugs for treatment of AIDS, yet the disease is still incurable. **Zidovudine (Azidothymidine—AZT)** has been licensed, but has been found to simply delay the progress of AIDS, not cure it. A Chinese herbal medicine, named **XQ-9302, Amphotericine B** and its methyl ester, and administration of **AZT** with protease inhibitors, hold promise of AIDS cure, but only future will tell.

Usefulness of Viruses

Viruses are not without virtues. Their usefulness is as follows—

1. Many characteristics of viruses make these an ideal experimental material for scientific investigations.
2. Genomes of a variety of viruses (*e.g.* retroviruses, most bacteriophages and certain DNA viruses) become incorporated into host cell genome and may modify the host-cell in some respects. These viruses have been widely used as vectors for introducing foreign genes into bacteria and other cells for purposes of **biotechnology, genetic engineering, gene therapy,** and preparation of **antibiotic drugs** for prevention and treatment of bacterial diseases like diarrhoea, colitis, dysentery, cholera, plague, enteritis, etc.
3. Insect-killing viruses are likely to play a significant role in our war against insect pests in near future.

EXERCISE – I

MCQ TYPE QUESTIONS

1. Which of the following fixes CO_2 in carbohydrates
   (*a*) *Bacillus* (*b*) *Rhizobium*
   (*c*) *Nitrobacter* (*d*) *Rhodospirillum*
2. Which of the following compounds are decomposed during putrefaction
   (*a*) Proteins (*b*) Fats
   (*c*) Carbohydrates (*d*) None
3. Mesosome in a bacterial cell is
   (*a*) Plasmid
   (*b*) Connection between two cells
   (*c*) Plasma membrane infolded for respiration
   (*d*) None of these
4. Bacterial flagella do not show ATPase activity and 9+2 organisation ,these are chemically
   (*a*) Flagellin (*b*) Pilin
   (*c*) Tubulin (*d*) Bacterin
5. The nitrifying bacteria are
   (*a*) Autotrophic (*b*) Saprophtic
   (*c*) Parasitic (*d*) Chemosynthetic
6. Branched chain lipids occur in the cell membranes of
   (*a*) *Archaebacteria* (*b*) *Mycoplasma*
   (*c*) *Actinomycetes* (*d*) *Streptomyces*
7. A bacterium is capable of withstanding extreme heat, dryness and toxic chemicals. This indicates that it is probably able to form
   (*a*) A thick peptidoglycan wall
   (*b*) Endospores
   (*c*) Endotoxins
   (*d*) Endogenous buds
8. Infoldings of the plasma membrane of gram positive bacteria gives rise to
   (*a*) Clathrin (*b*) Chondritin
   (*c*) Chondrioides (*d*) Chromatin
9. The process in which viruses are involved in sexual reproduction of bacteria is called
   (*a*) Transduction (*b*) Transcription
   (*c*) Transformation (*d*) Translation
10. Identify the correct pair of events when temperate phages infect bacteria.
    I. No prophages are formed.
    II. Bacterial cell undergoes many divisions
    III. Bacterial cell undergoes immediate lysis
    IV. Prophages are formed
    The correct pair is
    (*a*) I, II (*b*) II, III
    (*c*) III, IV (*d*) II, IV
11. The bacteria grown in the medium containing S^{35} as one source of sulphur show its incorporation into?
    (*a*) DNA (*b*) Protein
    (*c*) RNA (*d*) None of these
12. Sex factor in bacteria is?
    (*a*) F-replicon
    (*b*) Chromosomal replicon
    (*c*) RNA
    (*d*) Sex pilis
13. A free living anaerobic bacterium capable of N2 fixation in soil is?
    (*a*) *Rhizobium* (*b*) *Azotobacter*
    (*c*) *Streptococcus* (*d*) *Clostridium*
14. Which of the following groups of plants are highly useful in increasing soil fertility?
    (*a*) Red algae (*b*) Fungi
    (*c*) Bacteria (*d*) Bryophytes
15. Curing of tea leaves is brought about by the activity of?
    (*a*) Viruses (*b*) Fungi
    (*c*) Bacteria (*d*) Mycorrhiza
16. Retting of fibers is caused by?
    (*a*) *Bacillus* (*b*) *Clostridium*
    (*c*) *Nitrobacter* (*d*) *Rhizobium*
17. Which of the following is a free living nitrogen fixing bacteria present in the soil?
    (*a*) *Pseudomonas* (*b*) *Rhizobium*
    (*c*) *Azotobacter* (*d*) *Nitrosomonas*
18. For retting of jute the fermenting microbe used is?
    (*a*) *Methophilic bacteria* (*b*) Butyric acid
    (*c*) *Helicobacter pylori* (*d*) *Streptococcus lactin*
19. Which bacteria is responsible for the reduction of nitrates into nitrogen, (denitrifying bacteria) in soil?
    (*a*) *Nitosomonas* (*b*) *Pseudomonas*
    (*c*) *Rhizobium* (*d*) *Clostridium*
20. Which of the following is symbiotic nitrogen fixes?
    (*a*) *Streptomyces* (*b*) *Anabena*
    (*c*) *Frankia* (*d*) *Rhizobium*
21. Milk is changed into curd by?
    (*a*) *Acetobacter aceti*
    (*b*) *Bacillus megatherium*
    (*c*) *Xanthomonas citri*
    (*d*) None of these

22. Which does not help in N_2 fixation?
(a) Anabaena (b) Nostoc
(c) Oscillatoria (d) Rhizobium

23. Which of the following is a non symbiotic fertilizer?
(a) VAM (b) Azotobacter
(c) Anabaena (d) Rhizobium

24. The main role of bacteria in carbon cycle involves-
(a) Photosynthesis
(b) Assimilation of nitrogen compounds
(c) Chemosynthesis
(d) Digestion or breakdown of organic compounds

25. Which of the following is a flowering plant with nodules containing filamentous nitrogen fixing bacteria
(a) *Casuarina equisetifolia*
(b) *Crotalaria juncea*
(c) *Cycas revolute*
(d) *Cicer arietinum*

26. The function of leghaemoglobin in the root nodules of legumes
(a) Expression of nif gene
(b) Inhibition of nitrogenise activity
(c) Oxygen removal
(d) Nodule differentiation

27. Organisms called methanogens are most abundant in
(a) Hot spring (b) Sulphur rock
(c) Cattle yard (d) Polluted stream

28. Mycolic acid is present in the cell wall of bacteria causing
(a) Tetanus (b) Cholera
(c) diphtheria (d) tuberculosis

29. Tetanolysin is produced by
(a) *Mycobacterium leprae*
(b) *Clostridium botulinum*
(c) *Clostridium tetani*
(d) *none of the above*

30. Which is the cause of Anthrax disease
(a) Virus (b) Bacteria
(c) Mycobacteria (d) Algae

31. The disease pneumonia is caused by
(a) Virus (b) Cyanobacteria
(c) Bacteria (d) Cold

32. The gram negative bacteria detect and responsed to chemicals in their surroundings by:
(a) Lipopolysaccharide (b) Muramic acid
(c) Porins (d) Volutin granules

33. Tobacco mosaic virus(TMV) genes are:
(a) Single stranded RNA
(b) Double stranded DNA
(c) Proteinaceous
(d) Double stranded DNA

34. The tailed bacteriophages are:
(a) Motile on surface of bacteria
(b) Non motile
(c) Motile on surface of plant leaves
(d) Actively motile in water

35. Which of the following pair of diseases is caused by virus?
(a) Rabies, mumps
(b) cholera, tuberculosis
(c) Cholera, tuberculosis
(d) AIDS, syphilis

36. Human immunodeficiency virus (HIV) has a protein coat and a genetic material which is
(a) Double stranded RNA
(b) Double stranded DNA
(c) Single stranded DNA
(d) Single stranded RNA

37. Cauliflower mosaic virus contains
(a) ssRNA (b) dsRNA
(c) dsDNA (d) ssDNA

38. Which of the following statements is not true for retrovirus?
(a) DNA is not present at any stage in the life cycle of retroviruses
(b) Retroviruses carry gene for RNA-dependent DNA polymerase
(c) The genetic material in mature retroviruses is RNA
(d) Retroviruses are causative agents for certain kinds of cancer in man

39. The protein coat of virus is called
(a) Capsomeres (b) Prions
(b) Viriod (d) Capsid

40. When a bacteriophage, is in lytic phase, carry some of the bacterium partially digested chromosome with it to another host cell, the process is called
(a) Generalised transduction
(b) Conjugation
(c) Transformation
(d) Specialised transduction

41. Viral diseases have no cure because
(a) Viruses have no cell wall
(b) Viruses can multiply repeatedly within the host cell
(c) Presence of capsid
(d) Viruses possesses no cytoplasm

42. Viroids have
(a) Single stranded RNA not enclosed by protein coat
(b) Single stranded DNA not enclosed by protein coat
(c) Double stranded DNA enclosed by protein coat
(d) Double stranded RNA enclosed by protein coat

43. Small pox is caused by a kind of
(a) Bacteria (b) Virus
(c) Protozoa (d) Helminthes

44. Virus parasite in bacteria are
(a) Bacteriophage (b) Viroid
(c) Cyanophage (d) Mycoplasma

45. Which disease is caused by virus?
(a) Polio (b) Leprosy
(c) TB (d) Malaria

46. Enveloped virus infects/enters the host cell through
(a) Direct penetration
(b) Injecting nucleic acid
(c) Endocytosis
(d) Endocytosis through chemical stimulators

47. Which of the following is not true regarding viruses?
(a) They can be cultured in artificial media lacking living cells.
(b) They do not have protein synthesizing machinery of their own.
(c) They may contain RNA as the genetic material.
(d) They can be crystallized.

48. Coliphage T_2 has
(a) ds RNA (b) ss RNA
(c) ds DNA (d) ss DNA

49. Which one is absent in viruses?
(a) Replication (b) Protein synthesis
(c) Energy liberation (d) Mutation

50. Which of the following statements is true about viruses?
(a) All viruses known to man are obligate parasites.
(b) Some viruses have cellular structure and are saprophytes.
(c) Viruses are filterable facultative parasites.
(d) Viruses are capable of performing metabolic activities on their own.

51. A bacteriophage with radioactive DNA and protein when infects a bacterium, the radioactivity inside the bacterium will be located
(a) In DNA
(b) In protein
(c) Both in DNA and protein
(d) In all parts of the bacterial cell

52. Which has double stranded RNA?
(a) Cauliflower Mosaic Virus
(b) TMV
(c) α X 174
(d) Reovirus

53. A virus containing ssRNA which acts as a template for DNA synthesis is known as
(a) Polio virus (b) Retro virus
(c) Pox virus (d) Adeno virus

54. During lytic life cycle of a virulent DNA phage, which of the following does not occur?
(a) Host cell produce large number of virions
(b) The host cell lyse
(c) New phages are released
(d) The phage DNA gets integrated into the host chromosome

55. Which of the following plant virus has DNA in it?
(a) Tobacco mosaic virus
(b) Potato mosaic virus
(c) Tomato mosaic virus
(d) Cauliflower mosaic virus

56. Which of the following bacteria requires Na+ for growth ?
(a) Photosynthetic bacteria, cyanobacteria
(b) Certain marine bacteria, photosynthetic bacteria
(c) Cyanobacteria, certain marine bacteria
(d) All of these

57. Which is the odd option for Rhodospirillum rubrum
(a) Phototrophic (b) Organotrophic
(c) Heterotrophic (d) Chemotropic

58. Minimal nutritional requirements for Lactobacillus is
(a) Inorganic salts, organic carbon, inorganic nitrogen
(b) Atmospheric N2, 2 or more amino-acids, 2 or more vitamins
(c) Inorganic N2, 2 or more amino-acids, 2 or more vitamins
(d) Both (a) and (c)

59. In turbidostat, if the cell density becomes too high, the dilution rate will be
(a) Increased (b) Constant
(c) Decreased (d) All of these

60. For about quantitative measurement of bacterial growth, all are right except
(a) Cell count is directly determined by using an electronic particle counter
(b) Cell mass is directly measured by weighing or by a measurement of cell nitrogen
(c) Cell mass is measured indirectly by turbidity
(d) Cell activity is measured or determined indirectly by relating the degree of biochemical activity to the size of the population

61. The properties which made γ and X-rays as ionizing radiations is
(a) Having energies of more than 10eV
(b) Having energies of more than 5eV
(c) Having energies below 1eV
(d) All of these

62. Sterilizing agents include
(a) Dry heat (b) Ethylene oxide
(c) Alcohol (d) Both (a) & (b)

63. Which of the following can be reliably used for hand washing ?
(a) Isopropyl alcohol (b) Cresol
(c) Both (a) & (b) (d) Lysol

64. Sporicidal disinfectant is following except
(a) Glutaraldehyde
(b) Formaldehyde
(c) Ethylene oxide
(d) Benzalkonium chloride

65. Sterilization of culture media containing serum is by
(a) Autoclaving (b) Micropore filter
(c) Gamma radiation (d) Centrifugation

66. Odd option for sporicidal
(a) Lysol (b) glutaraldehyde
(c) Both (a) & (b) (d) formaldehyde

67. Composition of nutrient broth
(a) Beef extract – 3g, peptone – 15g, H_2O – 1000 ml
(b) Beef extract – 3g, peptone – 7g, H_2O – 1000 ml
(c) Beef extract – 5g, peptone – 1g, H_2O – 1000ml
(d) Beef extract – 3g, peptone – 3g, agar – 3.5 g, H_2O – 1000 ml

68. Match the following

1. Rhodospirillum rubrum (anaerobic)	A. Organotrophic
2. E.Coli	B. Phototrophic
3. Desulfovibrio desulfuricans	C. Chemotropic, Lithotrophic
4. Pseudomonas pseudoflava (H_2 supplied)	D. Lithotrophic

(a) 1-A; 2-C; 3-D; 4-B (b) 1-B; 2-A; 3-C; 4-D
(c) 1-D; 2-B; 3-A; 4-C (d) 1-D; 2-C; 3-B; 4-A

69. Match the following Vitamin with requirements for some bacteria

1. Riboflavin	A. Leuconostoc dextranicum
2. Niacin	B. Bacteroides melaninogenicus
3. Pyridoxine	C. Clostridium tetani
4. Vit. K	D. Lactobacillus spp
5. Folic acid	E. Brucella abortus

(a) 1-C; 2-B; 3-E; 4-D; 5-A
(b) 1-C; 2-E; 3-D; 4-B; 5-A
(c) 1-D; 2-E; 3-C; 4-B; 5-A
(d) 1-A; 2-E; 3-D; 4-B; 5-C

70. Match the following Bacteria with optimum temperature for growth

1. Thermus aquaticus	A. 37°C
2. Streptococcus thermophilus	B. 70-72°C
3. Corynebacterium diphtheria	C. 40-45°C
4. Vibrio marinus strain MP-1	D. 15°C

(a) 1-D; 2-A; 3-B; 4-C (b) 1-B; 2-C; 3-A; 4-D
(c) 1-B; 2-A; 3-C; 4-D (d) 1-D; 2-C; 3-A; 4-B

71. Match the following method with growth expressed

1. Microscopic count	A. Same as for nitrogen determination
2. Plate count	B. Colony – forming units/ml
3. Turbidimetric measurement	C. Number of cells/ml
4. Dry weight determination	D. OD

(a) 1-C; 2-B; 3-D; 4-A (b) 1-C; 2-D; 3-A; 4-B
(c) 1-C; 2-A; 3-B; 4-D (d) 1-B; 2-C; 3-D; 4-A

72. Match the following medium components with their composition, for cultivation of E.Coli

1. $NH_4H_2PO_4$	A. 5g
2. Glucose	B. 1000 ml
3. H_2O	C. 0.2g
4. $MgSO_4, 7H_2O$	D. 1g

(a) 1-D; 2-C; 3-B; 4-A (b) 1-B; 2-C; 3-A; 4-D
(c) 1-B; 2-A; 3-D; 4-C (d) 1-D; 2-C; 3-A; 4-B

73. Match the following bacterial spores with their destruction time by dry heat

Organism	Destruction time (at 140°C)/min
1. Clostridium botulism	A. 5-15
2. Bacillus anthracis	B. Upto 180
3. Cl. welchii	C. 15-60
4. Cl. tetani	D. 5

(a) 1-C; 2-D; 3-B; 4-A (b) 1-B; 2-C; 3-D; 4-A
(c) 1-C; 2-B; 3-D; 4-A (d) 1-B; 2-A; 3-C; 4-D

NUMERICAL TYPE QUESTIONS

1. Bacterial cell divides in every minute it takes one hour to fill up a cup. How much time will be taken to fill half the cup

2. How many bacteria are produced in four hours if a bacterium divides once in half an hour?

3. A bacterium divides every 35 minutes. If a culture containing 10^5 cells/ml is grown for 175 minutes, what will be the cell concentration/ml after 175 minutes?

EXERCISE – II

(QUESTIONS FROM PREVIOUS GATE EXAMS)

MCQ TYPE QUESTIONS

2016

1. Bacteria with two or more flagella at one or both ends are called
(a) amphitrichous (b) peritrichous
(c) lophotrichous (d) atrichous

2. Which family of viruses has single stranded DNA?
(a) Herpesviridae (b) Poxviridae
(c) Retroviridae (d) Parvoviridae

3. Which of the following are TRUE for *Treponema pallidum*?
P. It is the causative agent of syphilis
Q. It is a spirochete
R. It is a non-motile bacterium
S. It is generally susceptible to penicillin
Choose the correct combination.
(a) P, Q and R only (b) P, Q and S only
(c) P, R and S only (d) Q, R and S only

4. Which combination of the following statements is **CORRECT** for cyanobacteria?
P. They can perform oxygenic photosynthesis
Q. Usually filamentous forms are involved in nitrogen fixation
R. Nitrogen fixation occurs in heterocysts
S. They cannot grow in a mineral medium exposed to light and air
(a) P, Q and R (b) P, S and R
(c) Q, R and S (d) P, Q and S

2015

5. Which one of the following organisms is used for the determination of phenol coefficient of a disinfectant?
(a) *Salmonella typhi* (b) *Escherichia coli*
(c) *Candida albicans* (d) *Bacillus psychrophilus*

6. Match the microorganisms in Group I with their fermentation products in Group II.

Group I	Group II
P. *Leuconostoc mesenteroids*	1. Cobalamin
Q. *Rhizopus oryzae*	2. Sorbose
R. *Gluconobacter suboxydans*	3. Dextran
S. *Streptomyces olivaceus*	4. Lactic acid
	5. Butanol

(a) P-5, Q-4, R-2, S-1 (b) P-5, Q-3, R-2, S-4
(c) P-3, Q-4, R-1, S-2 (d) P-3, Q-4, R-2, S-1

2014

7. Gram-positive bacteria are generally resistant to complement-mediated lysis because
(a) thick peptidoglycan layer prevents insertion of membrane attack complex into the inner membrane
(b) Gram-positive bacteria import the membrane attack complex and inactivate it
(c) membrane attack complex is degraded by the proteases produced by the Gram-positive bacteria
(d) Gram-positive bacteria cannot activate the complement pathway

8. Which of the following essential element(s) is/are required as major supplement to enhance the bioremediation of oil spills by the resident bacteria?
(a) Sulfur
(b) Nitrogen and phosphorus
(c) Iron
(d) Carbon

2013

9. Phylum proteobacteria is subdivided into α-, β-, δ-and ε-proteobacteria based on
(a) G + C content
(b) 23S rRNA sequences
(c) tRNA sequences
(d) 16S rRNA sequences

10. Match the commercial microbial sources in **Group I** with the products in **Group II.**

Group I
P. *Corynebacteriumlilium*
Q. *Klebsiellaoxytoca*
R. *Aspergillusniger*
S. *Alcaligeneseutrophus*

Group II
1. 2,3-Butane di-ol
2. Poly-β-hydroxybutyric acid
3. Glutamic acid
4. Citric acid

(a) P-3,Q-1,R-2,S-4
(b) P-3,Q-1,R-4,S-2
(c) P-1,Q-3,R-2,S-4
(d) P-1,Q-3,R-4,S-2

11. Match the cell structures in **Group I** with the organismsin **Group II.**

Group I	Group II
P. Endospores	1. *Methanobacterium*
Q. Bipolar flagella	2. *Treponema*
R. Pseudomurine in cell wall	3. *Spirillum*
S. Periplasmic flagella	4. *Clostridium*

(a) P-4, Q-3, R-1, S-2 (b) P-4, Q-3, R-2, S-1
(c) P-3, Q-4, R-1, S-2 (d) P-4, Q-1, R-3, S-2

2011

12. The lipopolysaccharides present in bacterial cell wall has lipid A which is connected to
(a) O-polysaccharide
(b) core polysaccharide
(c) both with O-polysaccharide and core polysaccharide
(d) rhamnose-mannose disaccharide

13. Gas vacuoles are present in
(a) *Anabaena flos-aquae*
(b) *Bacillus subtilis*
(c) *Acanthurus nigrofuscus*
(d) *Mycobacterium tuberculosis*

2010

14. Under stress conditions bacteria accumulate
(a) ppGpp (Guanosine tetraphosphate)
(b) pppGpp (Guanosine pentaphosphate)
(c) both ppGpp pppGpp
(d) either ppGpp or pppGpp

15. Which one of the following DOES NOT belong to the domain of Bacteria?
(a) Cyanobacteria
(b) Proteobacteria
(c) Bacteroids
(d) Methanobacterium

16. Match Group I with Group II

Group I	Group II
P. Staphylococcus aureus	1. Biofilms
Q. Candida albicans	2. Bacteriocins
R. Mycobacterium tuberculosis	3. Methicillin resistance
S. Lactobacillus lactis	4. Isoniazid

(a) P-1, Q-4, R-2, S-3 (b) P-2, Q-3, R-1, S-4
(c) P-3, Q-1, R-4, S-2 (d) P-1, Q-2, R-4, S-3

17. Match the products in Group I with the microbial cultures in Group II used for their industrial production

Group I	Group II
P. Gluconic acid	1. *Leuconostoc mesenteroids*
Q. L – Lysine	2. *Aspergillus niger*
R. Dextran	3. *Brevibacterium flavum*
S. Cellulase	4. *Trichoderma reesei*

(a) P-2, Q-1, R-3, S-4 (b) P-1, Q-3, R-4, S-2
(c) P-2, Q-3, R-1, S-4 (d) P-3, Q-2, R-4, S-1

2009

18. Match the products in Group 1 with their producer organisms in Group 2:

Group 1	Group 2
P. Ethanol from glucose	1. Aspergillus niger
Q. Probiotics	2. Leuconostoc mesenteroides
R. Citric acid	3. Saccharomyces cerevisiae
S. Sauerkraut	4. Bifidobacterium

(a) P-1, Q-3, R-2, S-4 (b) P-3, Q-4, R-1, S-2
(c) P-3, Q-4, R-2, S-1 (d) P-1, Q-4, R-3, S-2

2008

19. Match the products in group 1 with their producer organisms given in group 2

Group 1	Group 2
P. Ethanol	1. *Streptomyces orientalis*
Q. L-Lysine	2. *Saccharomyces cerevisiae*
R. Biopesticide	3. *Corynebacterium glutamicum*
S. Vancomycin	4. *Bacillus thuringiensis*

	P	Q	R	S
(a)	2	3	4	1
(b)	3	4	1	2
(c)	4	1	2	3
(d)	2	1	4	3

2007

20. A bioremedial solution to reduce oxides of nitrogen and carbon in flue gases is to integrate flue gas emission to
(a) micro-algal culture (b) fish culture
(c) mushroom culture (d) sericulture

NUMERICAL TYPE QUESTIONS

2014

1. A bacterium belonging to cocci group has a diameter of 2 m. The numerical value of the ratio of its surface area to volume (in m^{-1}) is _____

ANSWERS

EXERCISE – I

MCQ Type Questions

1. (d)	2. (a)	3. (c)	4. (a)	5. (d)	6. (a)	7. (b)	8. (c)	9. (a)	10. (d)
11. (b)	12. (a)	13. (d)	14. (c)	15. (c)	16. (b)	17. (c)	18. (b)	19. (b)	20. (c)
21. (d)	22. (c)	23. (b)	24. (d)	25. (a)	26. (c)	27. (c)	28. (d)	29. (c)	30. (b)
31. (c)	32. (c)	33. (a)	34. (a)	35. (a)	36. (d)	37. (c)	38. (a)	39. (d)	40. (a)
41. (b)	42. (a)	43. (b)	44. (a)	45. (a)	46. (b)	47. (a)	48. (c)	49. (c)	50. (a)
51. (a)	52. (d)	53. (b)	54. (d)	55. (d)	56. (d)	57. (d)	58. (d)	59. (a)	60. (d)
61. (a)	62. (d)	63. (c)	64. (d)	65. (a)	66. (a)	67. (a)	68. (b)	69. (b)	70. (b)
71. (a)	72. (c)	73. (c)							

Numerical Type Questions

1. 59 minutes 2. 256 3. 32×10^5 cells

EXERCISE – II

MCQ Type questions

1. (c)	2. (d)	3. (b)	4. (a)	5. (a)	6. (d)	7. (a)	8. (b)	9. (d)	10. (b)
11. (a)	12. (b)	13. (a)	14. (c)	15. (a)	16. (c)	17. (c)	18. (b)	19. (a)	20. (a)

Numerical Type Questions

1. (3 to 3)

EXPLANATIONS

EXERCISE – I

MCQ TYPE QUESTIONS

1. *Rhodospirillum* is photosynthetic bacteria.
2. Anaerobic breakdown of proteins is known as putrefaction.
3. In many bacteria (especially gram positive) the plasma membrane gives rise to infoldings called mesosome. They have respiratory enzymes like succinic dehydrogenase, cytochrome oxidase for respiration.
4. The bacterial flagellum is long, filamentous and protoplasmic appendage the arise in the cell envelope. In the bacterial flagella instead of 9+2 arrangement of tubulin there is simply a single filament of globular protein called flagellin.
5. Nitrifying bacteria such as *Nitrosomonas* and *Nitrobacter* manufacturing their organic food utilizing chemical energy in oxidation of some inorganic compound. Hence these bacterias are chemosynthetic.
6. Branched chain lipids of cell membranes of archaebacteria enable them to tolerate high temperature and high acidity.
7. During unfavourable condition highly resistant single spore is formed in the bacterial cell which is known as endospore.
8. Respiratory enzyme containing infolding of plasma membrane in bacteria called chondrioid.
9. In the process of transduction, virus is used as a carrier in the transfer of DNA from one bacterial cell to other.
10. Temperate phages are basically bacteriophages. It follows lysogenic pathway of development. The phage integrates its genetic material with the host cell. It may be done by physical incorporation of the phage genome into host genome, or the prophage maybe integrated as a stably maintained plasmid.
11. Sulphur is responsible for amino acids composition. The tertiary structure of proteins is stabilized by disulphide bridges.
12. The Fertility factor allows genes to be transferred from one bacterium carrying the factor to another bacterium lacking the factor by conjugation. The F factor is carried on the F episome, the first episome to be discovered. Unlike other plasmids, F factor is constitutive for transfer proteins due to the gene *traJ*. The F plasmid belongs to a class of conjugative plasmids that control sexual functions of bacteria with a fertility inhibition (Fin) system.
13. *Clostridium*
Clostridium is a saprophytic, anaerobic nitrogen fixing bacteria and has the ability to fix atmospheric nitrogen into ammonia.

14. Bacteria,

Bacteria cause nitrogen fixation.

15. Bacteria

Bacteria cause the curing of tea and coffee by which bitterness is removed and flavor is added.

16. *Clostridium* dissolves the peptic material and isolates the fibers.

17. *Azotobacter* (aerobic) is a free living, nitrogen fixing bacteria. This bacteria independently fix atmospheric nitrogen in their body and convert to organic nitrogen compounds. When they die, the organic nitrogen compounds of their body are made available to higher plants through the activities of other bacteria.

18. In this process, bacteria decompose the pectin of the plant to liberate fibers. The process of fiber preparation is known as retting. This process is carried out by *Clostridium butyrium*.

19. Some bacteria such as *Thiobacillus denitrificans, Pseudomonas aoruginosa* and *Micrococcus denitrificans* also occur in the soil which converts the nitrate and ammonia into atmospheric free elemental nitrogen.

20. Actinomycetes are filamentous bacteria, inhabitant of soil. The genus *Frankia* causes nitrogen fixing nodules to form in older tree roots (non legume plants).

21. *Lactobacillus casei* and *Streptococcus lactis* at 40 C.

22. Oscillatoria- They lack heterocyst, therefore they cannot take part in N2 fixation. *Oscillatoria* is a genus of filamentous cyanobacterium which is named for the oscillation in its movement.

23. Azotobacter- Non- symbiotic bacteria are also called free living nitrogen fixing bacteria i.e. Azotobacter. They are agriculturally important.

24. digestion or breakdown of organic compounds- the main role of bacteria is as a decomposer. The bacteria decompose complex organic molecule into simple ones which can be used in the carbon cycle.

25. *Casuarina equisetifolia*- Frankia is a filamentous nitrogen fixing bacteria which is symbioant to the root of Casuarina equisetifolia

26. Oxygen removal- LHB is Oxygen scavenger

Leghemoglobin (also leghaemoglobin or legoglobin) is a nitrogen or oxygen carrier, because naturally occurring oxygen and nitrogen interact similarly with this protein; and a hemoprotein found in the nitrogen-fixing root nodules of leguminous plants. It is produced by legumes in response to the roots being colonized by nitrogen-fixing bacteria, termed rhizobia, as part of the symbiotic interaction between plant and bacterium.

27. Cattle yard- Methanogens are archaebacteria abundant in cattle yard and paddy fields.

28. Tuberculosis- *Mycobacterium tuberculosis* releases mycolic acid. It is changed to Mycoides involved in causing the disease.

29. *Clostridium tetani*- tetanolysis is a neurotoxin *Clostridium tetani* is a rod-shaped, anaerobic bacterium of the genus species *Clostridium*. Like other *Clostridium* genus species, it is Gram-positive, and its appearance on a gram stain resembles tennis rackets or drumsticks. *C. tetani* is found as spores in soil or in the gastrointestinal tract of animals.

30. Anthrax is an acute disease caused by the bacterium *Bacillus anthracis*. Like many other members of the genus *Bacillus*, *Bacillus anthracis* can form dormant endospores (often referred to as "spores" for short, but not to be confused with fungal spores) that are able to survive in harsh conditions for decades or even centuries. Such spores can be found on all continents, even Antarctica. When spores are inhaled, ingested, or come into contact with a skin lesion on a host, they may become reactivated and multiply rapidly.

31. Pneumonia is primarily due to infections caused by bacteria and less commonly by fungi and parasites. Although there are more than 100 strains of infectious agents identified, only a few are responsible for the majority of the cases.

32. Porins- It is a protein found in the cell of gram negative bacteria which function as channels for the entry and exit of hydrophilic low molecular weight substances. Unlike other membrane transport proteins, porins are large enough to allow passive diffusion

33. Tobacco mosaic virus is a ribovirus and contains a single stranded RNA. Retroviruses have two copies of single stranded RNA. Double stranded RNA is found in Reovirus and tumour virus.

34. Bacteriophages are the viruses that infect bacteria. The base plate and the tail fibres of bacteriophage anchor the bacterial cell wall at a specific site. The tail fibres help viruses in coming in contact with specific sites on bacteria. It is called landing. Thus the bacteriophages are motile on the surface of bacteria.

35. Rabies (hydrophobia) is caused by a virus named *Rabies* virus. Mumps is an infectious viral disease causing fever, difficulty in opening mouth and painful swelling of the parotid glands which lie below the lobe of ear. It is caused by *paramyxovirus*, which comes out in tongue saliva of infected person. Cholera and tuberculosis and bacterial diseases caused by *vibrio cholera* and *mycobacterium tuberculosis* respectively. Typhoid and tetanus are bacterial diseases caused by *salmonella typhi* and *clostridium tetani* respectively. AIDS caused by HIV (human immunodefficiency virus).Syphilis is caused by *Treponema pallidum*.

36. HIV virus is a retrovirus, which contains single stranded RNA, surrounded by protein coat (core shell) as genetic material. It causes AIDS.

37. Cauliflower Mosaic Virus contains dsDNA.It is circular and shows semi-discontinuous type of replication.

38. Retrovirus is a group of RNA virus. This name comes from the fact that these viruses can convert their RNA back into a DNA copy using an RNA dependent DNA polymerase enzyme known as reverse transcriptase. Retrovirus can cause cancer to animals.

39. The mature virus particle(Singular unit virion), consist of an outer, inert, non-genetic protein coat or shell which surrounds and protects the genetic material or nucleic acid forming the core. The protein coat is called the capsid. The capsid is made up of numerous small subunits called capsomeres.

40. When certain pro viruses turn into virulent or lytic viruses thereby causing lysis of the host cell, they may carry with them segments of host's DNA. In such virus particles carrying a segment of DNA from one host affects another host which is different than the first one; the latter may inherit some of the properties of the former host due to the transfer of DNA segment through infecting phase. This phenomena of transfer of genetic material form one bacterial strain to another is called transduction. It is also known as generalised transduct.

41. Viruses have the strong reproductive capacity within the host boy, thus when viral disease occurs, several copies of that virus are produced in the body and cure by taking medicines is almost impossible. Only, body's own immunity can destroy them. Viruses cause a number of infectious human diseases like common cold, yellow fever, rabies, cancers, AIDS etc.

42. Viroids are a group of pathogens of some higher plants that are smaller or simpler than any of the known virus. They consist of single stranded linear or circular RNA molecules without a protein coat. Viroid does not produce a protein as it does not possess an initiation codon, viroid particle can multiply by both RNA dependent and DNA dependent replication.

43. Small pox is a communicable disease caused by Variola virus and clinically characterized by a sudden onset of fever, headache, backache, vomiting followed by appearance of rash on the third day of illness. Small pox is now an eradicated disease in India.

44. Bacteriophage are the parasites of bacteria. The base plate and the tail fibres of bacteriophage anchor the phages virion to host cell wall at a specific site where the tip of the phage tail happens to touch e.g., T2, T4 etc.

45. Polio is an acute inflammation of the spinal cord due to an enterovirus infection (25nm, small, RNA virus. As infantile paralysis of the limbs served by the damaged lower motor neurons may result, though there is no specific treatment, adequate protection can be provided with immunization. TB and leprosy are bacteria disease cause by Mycobacterium tuberculosis and Mycobacterium leprae respectively, whereas malaria is protozoan disease caused by Plasmodium species.

46. Viruses are acellular or non cellular particles the contains an inner core of genetic material DNA or RNA that is covered by non genetic protein "coat" called a capsid. The protein coat is made up of protein subunits called capsomeres. Together, the capsid and the nucleic acid core form the nucleocapsid. The nucleocapsid may be either enveloped or naked. Plant viruses are generally naked but anima viruses may or may not possess envelop. Plant viruses generally enter into the host through wounds as they cannot penetrate the cuticle of the host cell, so the entire virus comes in contract of host cell. The enveloped viruses inject their own nucleic acid inside the host cell.

47. Viruses are ultramicroscopic infectious particles which either contain DNA or RNA surrounded by a protein coat. They can grow, multiply and undergo mutations but only within living cells. Attempts to culture viruses outside the cells in different types of culture media have failed. The viruses are not affected by antibiotics.

48. Coliphage T_2 is T even phage which has hexagonal head, a rigid tail with contractile sheath and tail fibres with ds DNA.

49. Virus is completely inert outside its host cell. Inside its host cell, virus initiates the synthesis of viral proteins and undergoes replication. Energy liberation takes place within the mitochondria and within the cytoplasm. Virus lacks cell organelles, so, energy liberation cannot take place in virus

50. Viruses are sub microscopic organisms, generally less than 200 m, which are obligate parasites i.e., can live inside a living host only. They have either RNA or DNA and can pass through bacterial filters. They have characteristic mode of multiplication i.e. ,once a virus enters into the host cell, it takes control of whole biochemical machinery of the host cell and orders the metabolic machinery to synthesize their own (viral) components.

51. A bacteriophage with radioactive DNA and protein when infects the bacterium, the radioactivity inside the bacterium will be located in bacterial DNA rather than bacterial protein. This experiment was done by Hershey and Chase. It proves that DNA is the genetic material.

52. Reoviruses are a group of viruses, including some that affect the gastrointestinal system (such as rotavirus), and some that cause respiratory infections. Their genetic material is double-stranded RNA.

53. Retrovirus is a ssRNA containing virus that converts its RNA into DNA by means of the enzyme reverse transcriptase. This enables it to become integrated into its host DNA. Retroviruses can cause cancer to animals.

54. During lytic phase of viral reproduction, the viral genome, after entering the host cell, starts multiplication of its genome and synthesis of its proteins with the help of host cell machinery. The genome and proteins combine to form new viruses which are released by the cell rupturing. While in lysogenic phase, the viral genome gets integrated into the host genome.

55. Most of the animal viruses are DNA viruses with a few exceptions having RNA, example, rabies virus, polio virus etc. and most of the plant viruses are RNA viruses with a few having DNA. Cauliflower mosaic virus has double stranded DNA in it.

56. Most bacteria do not require Na^+ but certain marine bacteria, cyanobacteria and photosynthetic bacteria do require it. For those members of the archaeobacteria known as the 'red extreme halophiles', the requirement is astonishing, they can't grow with less than 12 to 15% NaCl.

57. All organisms require a source of energy. Some rely on chemical compounds for their energy and are designated as chemotrophs. Others can utilize radiant energy (light) and are called phototrophs. Both chemotrophs and phototrophs exist among bacteria.

E.g.	Energy
Rhodospirillum rubrum	phototrophic
Anaerobic conditions	chemotrophic
Aerobic conditions	

58. Minimal nutritional requirements of some heterotrophic bacteria

Salmonela typhi : Inorganic salts, organic carbon, inorganic N2, 1-amino-acid

Azospirillum : Inorganic salts, organic carbon, atmospheric N2, brasilense

E.Coli : Inorganic salts, organic carbon, inorganic N2

Lactobacillus : Inorganic salts, organic carbon, inorganic N2, 2 or more AA's and 2 or more vitamins.

59. A second type of culture apparatus is the 'turbidostat'. Here a photoelectric device continuously monitors the cell density within the culture vassel and controls the dilution rate to maintain the cell density at a constant value. If the density becomes too high, the dilution rate is increased. If the density becomes too low, the dilution rate is decreased.

60. Cell activity; indirectly by relating the degree of biochemical activity to the size of the population.

61. Gamma and X-rays which have energies of more than about 10eV are called ionizing radiations because they have enough energy to knock electrons away from molecules and ionize them; when such radiations pass through cells, they create free-hydrogen radicals, hydroxyl radicals and some peroxides, which in turn can cause different kinds of intracellular damage.

62. Sterilization agents are Heat, Ionization radiations, sterilant liquids, filtration, gases (ethylene oxide, formaldehyde gas).

63. Skin disinfectants are

Chlorhexidine

Alcohols – ethyl, isopropyl alcohols

Cresol, iodine, Chloroxylenol, Tincture iodine, Hexachlorophene

64. Sporicidal agents are; ethylene oxide, formaldehyde, glutaraldehyde, halogens Benzalkonium chloride is quaternary ammonium antiseptic which are not sporicidal.

Phenol and hexachlorophene are poor sporicidal.

65. Most of culture media are sterilized by Autoclaving at 121°C for 30 min at 15 lb pressure/inch^2

Steam under pressure (Autoclaving / steam sterillizer) used for sterilizing dressings, instruments, media etc.

66. Sporicidal agents are; ethylene oxide, formaldehyde, glutaraldehyde, halogens Benzalkonium chloride is quaternary ammonium antiseptic which are not sporicidal.

67. Composition of Nutrient broth :

Beef extract – 3g; peptone – 15g; water – 1000 ml

68. Nitrosomonas europaea, E.Coli, pseudomonas pseudoflava all are chemotrophs; but Chromantium okenii is a phototroph. Chromantium okenii uses H_2S as its electron donor, oxidizing it to elemental sulfur.

$H_2S \rightarrow S + 2e^+ + 2H^+$

All organisms require a source of electrons for their metabolism. Some organisms can use reduced inorganic compounds as electron donors and are termed lithotrophs (some may be chemolithotrophs others photolithotrophs). Other organisms use organic compounds as electron donors and are called organotrophs. E.g. E.Coli, pseudomonas pseudoflava.

All organisms require a source of energy. Some rely on chemical compounds for their energy and are designated as chemotrophs. Others can utilize radiant energy (light) and are called phototrophs. Both chemotrophs and phototrophs exist among bacteria.

E.g.	Energy
Rhodospirillum rubrum	phototrophic
Anaerobic conditions	chemotrophic
Aerobic conditions	

69. Riboflavin – Cl. Tetani
Niacin – Brucel'a abortus
Pyridoxine – lactobacillus spp
Vit. K – Bacteroides melaninogenicus
Folic acid – Leuconostoc dextranicum

70.

S. No.	Bacteria	Temperature		
		Minimum	Optimum	Maximum
1.	Thermus aquaticus	40	70-72	79
2.	Streptococcus thermophilus	20	40-45	50
3.	Corynebacterium diphtheria	15	37	40
4.	Vibrio marinus strain MP-1	–1	15	20

71. *Microscopic count*

Application – enumeration of bacteria in vaccines and cultures

Growth is expressed – No. of cells / ml

Plate count

Enumeration of bacteria in milk, H_2O, foods, soil, etc.

Colony forming units/ml

Turbidimetric measurement

Micrological assay, estimation of cell crop in broth

Optical density (OD) absorbance

Dry weight determination :

Measurement of cell crop from heavy culture suspensions to be used for research in metabolism – mg dry weight of cells / ml

72.

1. $NH_4H_2PO_4$	*b*. 1000 ml
2. Glucose	*a*. 5g
3. H_2O	*d*. 1g
4. $MgSO_4.7H_2O$	*c*. 0.2g

73.

Organism	Destruction time (at 140°C)/min
1. Clostridium botulism	*c*. 15-60
2. Bacillus anthracis	*b*. Upto 180
3. Cl. Welchii	*d*. 5
4. Cl. tetani	*a*. 5-15

NUMERICAL TYPE QUESTIONS

1. The number of bacteria becomes double after each division

2. As we know that bacterium divides after every half an hour, i.e., 30 minutes. Therefore, number of divisions in 4 hours are 240/30 = 8. Since, one bacterium on division produces 2 cells so cell concentration after 240 minutes will be (2)8 = 256.

3. 32×10^5 cells- As we know that bacterium divides after every 35 minutes through simple mitotic division therefore no of divisions are 175/35=5. Since 1 bacterium on division produces 2 cells. The cells conc. after 175 minutes will be $10^5 \times 2^5 = 32 \times 10^5$.

EXERCISE – II

MCQ TYPE QUESTIONS

1. There are 4 types of flagellar distribution on bacteria-

1. Monotrichous - Single polar flagellum- Example: *Vibrio cholerae*
2. Amphitrichous- Single flagellum on both sides- Example: *Alkaligens faecalis*
3. Lophotrichous- Tufts of flagella at one or both sides- Example: *Spirillum*
4. Peritrichous- Numerous falgella all over the bacterial body- Example: *Salmonella Typhi*

2. The virus family *Parvoviridae* contains 3 genera: the *parvoviruses* , the *dependoviruses*, and the *densoviruses* (which only infect insects). Parvoviridae is the only family of DNA viruses that infect humans that is single stranded. The DNA strand packaged in the virion can be either negative or positive sense, though in case of some strains, it is always negative.

3. Treponema pallidum is a spirochaete bacterium with subspecies that cause treponemal diseases such as syphilis, bejel, pinta, and yaws. It is a Gram-negative bacteria which is spiral in shape. It is generally susceptible to penicillin.

4. Cyanobacteria are photosynthetic, and so can manufacture their own food. Cyanobacteria get their name from the bluish pigment phycocyanin, which they use to capture light for photosynthesis. They also contain chlorophyll a, the same photosynthetic pigment that plants use. Nitrification cannot occur in the presence of oxygen, so nitrogen is fixed in specialized cells called heterocysts. These cells have an especially thickened wall that contains an anaerobic environment. e.g.,: Filamentous form of Nostoc.

5. Either *Salmonella typhi* or *Staphylococus aureus*.

6.

Group I	Group II
P. *Leuconostoc mesenteroids*	3. Dextran
Q. *Rhizopus oryzae*	4. Lactic acid
R. *Gluconobacter suboxydans*	2. Sorbose
S. *Streptomyces olivaceus*	1. Cobalamin

7. Gram positive bacteria have a thick layer of peptidoglycan as their cell wall which provides strength and support to the bacteria. Complement-mediated lysis and attack system is based on the formation of complement proteins which form membrane attack complex that insert into the plasma membrane of the cell and undergo cell lysis. But, due to the thick protection of gram positive bacteria by peptidoglycan on the outer cell membrane, complement proteins cannot enter the cell and cause lysis.

8. Oil spills in seas and oceans can be remedied by the use of certain special kinds of bacteria which degrade polyhydrocarbons. Such bacteria are utilized in the process of bioremediation to prevent serious impacts on marine environment caused by spillage. Such bacteria require nitrogen and phosphorus as major supplement for enhancing the rate of degradation of saturated and unsaturated hydrocarbons.

9. The phylum Proteobacteria is the largest and most diverse group of bacteria. They have many different shapes and ways of living. Some are free-living, some are parasitic, and others form mutualistic relationships with other organisms, such as those found in the rumen. Many Proteobacteria are responsible for diseases in animals and humans. The largest and most diverse Bacterial phylum is the Proteobacteria. All are Gram-negative and show relationship based on 16s RNA comparison.

10. Mostly, Glutamic acid is produced by the species of Corynebacterium particularly *Corynebacterium lilium*. *Klebsiella oxytoca* produces butanol and in this case 2, 3-butane-diol. Citric acid is a common product of fermentation of the fungi *Aspergillus niger*. *Alkaligenes eutrophus* commonly produces butyric acids along with other derivatives through its fermentation.

11. Clostridium is a genus of Gram-positive bacteria, belonging to the Firmicutes. They are obligate anaerobes capable of producing endospores. Spirillum is genus of spiral-shaped bacteria of the family Spirillaceae. Spirillum is microbiologically characterized as a gram-negative, motile helical cell with tufts of whiplike flagella at each end. Treponemes are helically coiled, corkscrew-shaped cells, 6 to 15 μm long and 0.1 to 0.2 μm wide. They have an outer membrane which surrounds the periplasmic flagella, a peptidoglycan-cytoplasmic membrane complex, and a protoplasmic cylinder. Multiplication is by binary transverse fission. Methanobacteria are methane producing microorganisms (methanogens) and have pseudomurine in their cell wall.

12. The Gram negative cell envelope contains an additional outer membrane composed by phospholipids and lipopolysaccharides which face the external environment. The highly charged nature of lipopolysaccharide confers an overall negative charge to the Gram negative cell wall. Lipopolysaccharide is comprised of a hydrophilic polysaccharide and a hydrophobic component known as lipid A which is responsible for the major bioactivity of endotoxin. Lipopolysaccharide can be recognized by immune cells as a pathogen-associated molecule through Toll-like receptor 4. Lipid A usually gets attached to the core polysaccharide of Gram negative cell wall.

13. Gas-vacuoles consisting of collections of gas-cylinders, and continuity of the plasma membrane with lamellae, are observed at all stages of growth of Anabaena flos-aquae. It is known to produce endotoxins, the toxic chemicals released when cells die. Once released (lysed), and ingested, these toxins can damage liver and nerve tissues in mammals.

14. Stress is a stringent response that occurs in bacteria due to amino acid starvation or fatty acid limitation. Other reasons involve heat shock or iron limitation. This response is signalled by p/ppGpp as an alarmone. It causes diversion of resources away from growth of cells towards the quenching of stress by promoting amino acid synthesis for enhancing survival. It continues until nutrients are restored. p/ppGpp causes inhibition of RNA synthesis in cases of AA shortage. It results in AA conservation by decreasing the translational machinery. Another mechanism of p/ppGpp action is the upregulation of genes for AA uptake and biosynthesis. It functions by influencing overall gene expression pattern for survival in stress conditions.

15. Besides cyanobacteria, all the others named were typically sharing the common feature belonging to bacteria domain – i.e, oxygen intolerance. However, cyanobacterium is a phylum which obtains energy through photosynthesis by performing oxygenic photosynthesis and reducing the atmosphere into an oxidising one. The ability to undergo oxygenic reactions distinctly distinguishes cyanobacteria from other bacterial domains. Thereby, cyanobacteria play a very important role in initiating biodiversity and are believed to evolve into plant chloroplast and eukaryotic algae by the process of endosymbiosis. Cyanobacterium due to its stand-out property of oxygenic photosynthesis in an oxygenic environment is thus distinct from other preferably an-oxygenic bacterial domains.

16. Methicillin-resistant Staphylococcus aureus (MRSA) are a type of staphylococcus or "staph" bacteria that are resistant to many antibiotics. Staph bacteria, like other kinds of bacteria, normally lives on skin and in nose, usually without causing problems. MRSA is different from other types of staph because it cannot be treated with certain antibiotics such as methicillin (as it is methicillin resistant).

    Biofilms formed by Candida albicans are resistant towards most of the available antifungal drugs. Therefore, infections associated with Candida biofilms are considered as a threat to immunocompromised patients. Also, the fungal pathogen Candida albicans is frequently associated with catheter-based infections because of its ability to form resilient biofilms.

    Isoniazid is an organic compound that is the first-line medication in prevention and treatment of tuberculosis. Isoniazid is a prodrug and must be activated by a bacterial catalase-peroxidase enzyme that in Mycobacterium tuberculosis is called KatG.

    Bacteriocins are the bioactive peptides produced by certain bacteria like lactobacillus lactis that inhibit the growth of other bacteria. It is a gram positive bacterium used in the production of milk and cheese.

17. Microbial species such as Aspergillus Niger have been utilized in many researches of gluconic acid fermentation.

    Strains of Brevibacterium flavum are used for the synthesis of L- lysine.

    Dextran is synthesized from sucrose by certain lactic-acid bacteria, the best-known being Leuconostoc mesenteroides and Streptococcus mutans.

    Trichoderma reesei is an industrially important cellulolytic filamentous fungus. T. reesei has the capacity to secrete large amounts of cellulolytic enzymes (cellulases and hemicellulases).

18. **P>3:** Saccharomyces cerevisiae is used extensively in batch fermentations to convert sugars to ethanol for the production of beverages and biofuels.

Q>4: Probiotics are live microorganisms thought to be beneficial to the host organism. Lactic acid bacteria (LAB) and bifidobacteria are the most common types of microbes used as probiotics.

R>1: In production technique of citric acid, cultures of A. niger are fed on a sucrose or glucose containing medium to produce citric acid.

S>2: Sauerkraut (sour cabbage) is finely shredded cabbage that has been fermented by various lactic acid bacteria, including Leuconostoc, Lactobacillus, Pediococcus etc.

19. **P>2:** Saccharomyces cerevisiae is used extensively in batch fermentations to convert sugars to ethanol for the production of beverages and biofuels.

Q>3: Industrially, L-lysine is usually manufactured by a fermentation process using Corynebacterium glutamicum

R>4: Bacillus thuringiensis (or Bt) is a Gram-positive, soil-dwelling bacterium, commonly used as a biological pesticide.

S>1: Streptomyces orientalis produces the antibiotic Vancomycin, which is used for the prevention and treatment of infections caused by Gram-positive bacteria, especially staphylococci.

20. The advantages of using a microalgal-based system in integration with flue gas emission are that:

- High purity CO_2 gas is not required for algal culture. Flue gas containing varying amounts of CO_2 can be fed directly to the microalgal culture. This will simplify CO_2 separation from flue gas significantly.
- Some combustion products such as NO_x or SO_x can be effectively used as nutrients for microalgae. This could simplify flue gas scrubbing for the combustion system.
- Microalgae culturing may yield high value commercial products. Sale of these high value products can offset the capital and the operation costs of the process.
- The envisioned process is a renewable cycle with minimal negative impacts on environment.

NUMERICAL TYPE QUESTIONS

1. Gram-positive anaerobic cocci (GPAC) are a heterogeneous group of organisms defined by their morphological appearance which are spherical-shaped bacterium.

Given, diameter of the cocci = 2μm

Therefore, radius = 1μm

Hence, surface area: volume of the cocci

$= 4\pi r^2/(4/3\pi r^3) = 3/r = 3\mu m$

■■

CHAPTER 6

Immunology

Immune System

The Immune system (I.S) is a remarkable adaptive defence system that has evolved in vertebrates to protect them from invaders pathogenic microgranisms and cancer. It is able to generate enormous variety of cells and molecules capable of specifically recognizing and eliminating an apparently limitless variety of foreign invaders. These cells and molecules act together in an exquisitely adaptable dynamic network whose complexity rivals that of nervous system.

In 1885, Pasteur administered the first vaccine to a human - the rabies vaccine. The experimental work of Emil Von Behring and Shibasaburo Kitasate in 1890 provided the first insight into the mechanism of immunity and earned noble prize in medicine in 1901. In 1930, a single substance called an antibody was shown. Immunity may be Active or Passive and Innate or Adaptive.

Active and Passive Immunity

Active immunity is the adaptive immunity that is induced by natural exposure to a pathogen (natural) or by vaccination (artificial). In this kind of immunity a person develops his own immune response to a microbe. Immunity resulting from transfer of antibodies or immune cells from an immune to a non-immune individual is known as passive immunity. It may be *naturally* (involves the transfer of antibodies from one host to another) or *artificially* (when antibodies or lymphocytes that have been produced outside the host are introduced into a host) acquired.

Active immunity	Passive immunity
Exposure to antigen	No exposure of antigen
Immunity achieved by injecting antigens Ag reactive T-cells	Immunity achieved by injecting Ab or
Activation of immune system	No immune system activation
Immune state develops over a period of weeks	Immunity develops immediately
Immunological memory develops	No immunological memory develops

Innate versus Adaptive systems

A number of cell types and defense molecules are usually present at the site of invasion or migrate to the site. These constitute the 'first line of defense'. This is called the '**innate immune system**' since it is present at birth and changes, little throughout the life of the individual. The cells and molecules of this innate system are mainly responsible for the first stages of expulsion of the microbe and may give rise to inflammation. Phagocytes are important cells in the innate immune system since they ingest and kill microbes.

The second line of defense is the '**adaptive immune system**' and this is brought into action while the innate immune system is dealing with the microbe and especially if it is unable to remove the invading microbe. The key difference betweent the two systems is that the adaptive system shows far more specificity and remembers that a particular microbe has previously invaded the body. This leads to a more rapid expulsion of the microbe on its second and third time of entry. The cells, molecules and characteristics of innate and adaptive immune systems are shown in *Table*.

Table : The innate and adaptive immune systems

Characteristics	Cells	Molecules
Natural immunity		
Responds rapidly	Phagocytes (PMNs and macrophages)	Cytokines
Has some specificity	Natural killer cells	Complement
No memory	Mast cells	Acute phase proteins
	Dendritic cells	
Adaptive immunity		
Slow to start	T and B cells	Antibodies
Highly specific		Cytokines
Memory		

Innate Immunity

The non-specific component of immunity, also known as innate immunity (Latin: *innasci*—to be born in) is not directed against any particular pathogen but is a general defence mechanism as the human body must constantly defend itself from microbial invasion. This defence mechanism is active right from the time a child is born (hence innate). The specificity of innate immunity is relatively low as it lacks the ability to distinguish one microbe from another.

The Components of Innate Immunity

The principal components of innate immunity are :

(*a*) mechanical and chemical barriers, (*b*) phagocytosis, (*c*) fever, (*d*) inflammation and

(*e*) acute-phase proteins.

(*a*) Mechanical Barriers

A mechanical or physical barrier refers to the various physical hindrances blocking the entry of the microbes into the host body. Mechanical barriers include the skin and the mucous membrane. These barriers act as the first line of defence against infection.

Skin

The outer layer of epidermis is coated with a tough protein called keratin that does not support viral replication or penetration by bacteria.

Mucous Membrane

(*a*) the lavaging action of physiological fluids, for example tears and saliva, that assist in flushing microbes from the body;

(*b*) the trapping action of mucous-coated hair in the anterior chambers of the nose;

(*c*) the expulsive effects of coughing and sneezing, which protect the respiratory and gastrointestinal tracts; and

(*d*) the cool temperature of the upper respiratory tract which inhibits replication by many viruses.

(*b*) Chemical Barriers

Acidic Gastric Secretions

The physiochemical environment in the stomach appears to be extremely inhospitable to invading pathogens. The secretion of hydrochloric acid by gastric parietal cells maintains the pH of the stomach at 2.0, which kills most microorganisms (except some resistant ones such as *Hepatitis A virus, Picornaviruses* and typhoid bacilli).

Low pH of the Skin

Sebum, secreted by the sebaceous glands present in the dermis, contains organic acids.These organic acids maintain the pH of the skin in the range of 3-5. This low pH inhibits or retards the growth of most microorganisms present on the surface of the skin.

Lysozyme

Lysozyme is a hydrolytic enzyme present in all mucous secretions, including tears, saliva and nasal secretion. It can lyse Gram-positive bacteria by clea ing the peptidoglycan layer found in the bacterial cell wall.

Gastric and Duodenal Enzymes

A large array of enzymes, including proteases and lipases, digests a variety of structural and metabolic chemical components of pathogens.

Antibodies and Inhibitors

The mucous secreted by gastric and intestinal cells usually contains IgA molecules as well as non-specific inhibitors of viral infections(for example, sialic acid found in mucous inhibits attachment of influenza virus particles to cells). Antibodies (IgA) are also found in tears and saliva. Secretory IgA protects the body surface against invading pathogenic microbes.

Interferons

The name interferons refers to a group of proteins produced by virus-infected cells that induce a generalized antiviral state in neighbouring un-infected cells. These proteins also augment innate immunity.

Complement Proteins

Complement proteins are a group of serum proteins that circulate in an inactive state in the plasma. A variety of specific and non-specific mechanisms activate these proteins. The activated forms of these proteins damage the invading pathogen.

Antimicrobial Peptides

All insects and mammals, including humans, secrete a number of antimicrobial peptides, such as defensins, for their protection. The human body is protected by 1 μm thick biofilm of defensins, that protects the external surface of the body from microbial assault.

(*c*) Phagocytosis

Phagocytosis of invading microorganisms is another important innate defence mechanism.

When bacteria or other invading parasites penetrate the skin or the mucous membrane, phagocytes, such as neutrophils, blood monocytes and tissue macro-phages, surge towards the site of infection. These phagocytes engulf the bacteria to form a large intracellular vesicle, called **phagosome**, containing the bacteria. Then, the involuntary guest trapped within the phagosome is destroyed by fusing it with "granules" (lysosomes) found in the cytoplasm of the phagocytes. These granules discharge their contents (enzymes and reactive oxygen species) inside the phagocytic vacuole, thereby degrading the bacteria.

(*d*) Fever

Fever is the condition of an abnormally high body temperature, accompanied by increased pulse rate and dry skin. It provides a non-specific defence against disease. Fever is a physiological response to infection.

Fever is beneficial to the host because it inhibits the growth of temperature-sensitive pathogens. Also, increased cell metabolism encourages rapid tissue repair and phagocytosis.

(*e*) Inflammation

Inflammation is the reaction of living tissue to either an injury or an infection. Inflammation is characterized by heat (*calor*), redness (*rubor*), swelling (*tumor*) and pain (*dolor*). It is a non-specific response of the body to injury.

The process of inflammation may be initiated by a variety of tissue products such as histamine, bradykinin, serotonin and prostaglandins released by a number of cells (such as mast cells and basophils, found in most tissues). The mediators released by damaged cells, chemicals released by invading microorganisms, products of the complement system and reaction products of the blood-clotting system also trigger the process of inflammation. Many of these substances strongly activate macrophages and other cells of the phagocytic system.

Attack by Tissue Macrophages

Macrophages are already present in the tissues—the alveolar macrophages (in lungs), histocytes (in subcutaneous tissue), Kupffer cells (in liver), microglia (in brain). However, tissue macrophages are present in small numbers initially. In the case of inflammation, these macrophages immediately proceed towards the injury site to begin their phagocytic actions. They serve as the first line of defence against infection.

The next step in inflammation is the redirection of blood phagocytes towards the lesion site. This is facilitated by **vasodilation** of blood vessels and capillaries at the site of injury. The increase in the diameter of blood vessels is brought about by chemical mediators like histamine and brady-kinins. These mediators bind to the receptors on nearby capillaries and vessels, causing vasodilation. The engorged capillaries are responsible for increased blood accumulation and the redness of the inflamed tissue (**erythema**).

The increased permeability of the capillaries permits the influx of fluid from the engorged capillaries into the site of irritation or injury. The accumulation of fluid at the site of irritation results in tissue swelling (**oedema**).

Margination and Diapedesis

The products released from the inflamed area also cause phagocytes (now mainly neutrophils) to move towards the inflamed area, a process facilitated by vasodilation and increased capillary permeability. The emigration of phagocytes involves the adherence of the cells to the capillary walls, a process called **margination**. The adhered phagocytes then pass from the blood vessels into tissue spaces through the spaces between capillary endothelial cells. This process is termed **diapedesis**. Once in the tissue spaces, the phagocytes migrate towards the injured tissue.

Once phagocytes engulf the invading bacteria and the necrotic tissue, many phagocytes eventually die. However, some enzymes may leak out into the extracellular environment from the neutrophils before the phagosome closes. This process, termed sloppy eating, damages healthy cells. After a few days, a cavity containing necrotic tissue, dead bacteria and dead phagocytes is formed at the site of inflammation. This fluid mixture is often called **pus**. Ordinarily, pus formation continues until all infection is suppressed. The blood-clotting system is also activated in the inflamed region which results in the deposition of insoluble strands of fibrin. This seals off the injured area from the rest of the body, thereby preventing the spread of infection. This pus, enclosed in a wall of fibrin, usually forms an opening on the surface of the body, from where it empties out. The fibrin clot dissolves, tissue repair occurs and scar tissue is formed.

Acute-Phase Proteins

Acute-phase proteins are a group of heterogeneous plasma proteins that are important in the innate defence against microorganisms (mostly bacteria and protozoa) and in limiting the damage caused by infection, trauma, malignancy and other diseases to tissues. In response to tissue insult, cells circulating in the blood, such as macrophages and neutrophils, secrete a variety of cytokines that stimulate the liver to produce acute-phase proteins. Some important acute-phase proteins include C-reactive complement protein (CRP) components, mannose-binding metal protein (MBP), binding proteins and protease inhibitors Acute-phase proteins function by stimulating phagocytosis (acting as **opsonins**), activating the complement system and clumping the invading microbes.

Table. Acute-phase proteins and their functions.

Protein	Function
C-reactive protein phosphatidyl-choline of microbes, damaged tissue	Binds C polysaccharide of *S. pneumoniae*; binds and activates complement (C1) pathway/cascade
Serum amyloid protein	Binds DNA, activates complement (C1) pathway/cascade
Complement components C2,C3, C4, C5,C9, Factor B	Chemotaxis and cell lysis
Haptoglobin/ Haemopexin	Binds iron and makes it unavailable for bacterial growth
Ceruloplasmin	Binds copper and renders it unavailable for bacteria
Mannose-binding protein; Lipopolysaccharide binding protein	Binds mannose and lipopolysaccharide on bacterial surface(respectively), acts as opsonin, activates complement pathway/cascade
Fibrinogen	Blood coagulation
vonWillebrand factor	
Antitrypsin. Antichymotrypsin	Proteinase inhibitor

Adaptive Immunity

Adaptive immunity is a more evolved and specific defence mechanism. The characteristics of adaptive immunity are exquisite antigenic specificity and the ability to "remember" different types of antigens. Adaptive immunity is activated only against invading foreign material and never against its own molecules (except in autoimmune diseases). Thus, it has the ability to distinguish between self and non-self. Since it can differentiate between a variety of different antigens, invading pathogens and self-antigens, and induce different levels/types of immune response, adaptive immunity is often referred to as specific immunity.

Cells of The Immune System

An effective immune response is mediated by a variety of cells including neutrophils, lympho-cytes, natural killer (NK) cells, eosinophils, basophils and antigen-presenting cells.

Lymphocytes are mainly responsible for initiating adaptive immune response in the human body. All lymphocytes are produced in bone marrow stem cells by a process known as **haematopoeisis**.

(A) B– Lymphocytes

B lymphocytes originate in the bone marrow where they continue to differentiate and mature. Each B cell is genetically programmed to encode a unique antigen-binding receptor on its membrane. This B-cell receptor is actually a membrane-bound antibody molecule. When a naive B cell (which has not previously encountered antigen) comes in contact with an antigen via its membrane-bound antibody, it multiplies and differentiates into two types of cells—plasma cells and memory cells.

B–cell **T–cell**

Plasma cells lack B-cell receptors. However, plasma cells produce large numbers of soluble molecules called antibodies. B-cell receptors and antibodies differ only at their c terminal. B-cell receptors have an additional transme mabrane segment at their c terminal that anchors them to the cell membrane. The antibody molecule is a large, polyfunctional glycoprotein found in blood and other tissue fluids. One antibody molecule consists of two identical heavy chains and two identical light chains. Each light chain is linked to one heavy chain to form a heterodimer. Two such heterodimers are linked by disulphide bonds to form a tetrameric Y-shaped antibody molecule. The tips of the two arms of the Y-shaped molecule form the antigen-binding site. Thus, both the heavy chain and the light chain form a part of the antigen-binding site.

Unlike plasma cells which last for a few days, memory B cells have a longer life (sometimes up to 20 years) than naive B cells. In humans, B cells last for several months. Memory cells express some membrane bound antibody molecules like their parent naive cells. Memory cells are functionally inactive unless they are stimulated by the same antigen again.

(B) T– Lymphocytes

Like B lymphocytes, T lymphocytes too are produced in the bone marrow but they migrate to the thymus to mature. During maturation, each T cell acquires a specific receptor on its membrane termed as the T-cell receptor. A T-cell receptor does not recognize soluble antigens. It recognizes antigens only when they are associated with a protein complex called the major histocompatibility complex (MHC).

MHCs are diverse transmembrane glycoproteins present on a variety of cells. These molecules present (display) the antigens to the cells of the immune system. MHC molecules are so named because they were first identified as "antigens" responsible for the acceptance or rejection of tissue grafts (**histocompatibility**). There are three classes of MHC proteins. Class I MHC molecules are expressed on nearly all nucleated cells. They present antigenic determinants to a specific class of effector T cells. Class II MHC molecules are

expressed only by antigen-presenting cells. They present antigens to the other class of effector T cells. Class III MHC proteins are of diverse types and are indirectly involved in the immune response.

There are several different groups of T cells. One group interacts with antigen-class II MHC molecule complex, gets activated and starts secreting cytokines. This group of T cells is called T helper (T_H) cells. T_H cells activate B cells via cytokines and help them to divide, differentiate and secrete antibody molecules. Another group of T cells is responsible for the destruction of virus-infected host cells, intracellular pathogens harbouring host cells, tumour cells and cells of a foreign tissue graft. These T cells are called cytotoxic T cells (T_{CL}). They recognize an antigen when it is associated with class I MHC molecules. It should be emphasized here that T cells recognize antigens only when they are presented in association with MHC molecules. The receptor that recognizes an antigen-MHC complex is present on the cell membrane of T cells and is termed T-cell receptor (TCR).

(C)Antigen-Presenting Cells

Any cell that is capable of processing and presenting antigens to the cells of the immune system may be termed an antigen-presenting cell. However, this term is reserved for only those cells that display antigens associated with class II MHC molecules. These antigen-presenting cells present antigens to T_H cells. Antigen-presenting cells include dendritic cells (interdigitating), macrophages and B lymphocytes.

Class II MHC molecule and processed peptide

Antigen-presenting cells take up antigens either by endocytosis or phagocytosis. The intern- alized antigen is then degraded and processed. Small peptides derived from antigen processing are then displayed on the surface of the antigen-presenting cells in association with class II MHC molecules. This antigen-class II MHC complex is recognized by T_H cells. This binding with T_H cell also activates antigen-presenting cells, which produces signal molecules (cytokines) that leads to the activation of the T_H cell which further augments the immune response.

(D)Natural Killer Cells

Natural killer (NK) cells play an important role in immune surveillance and innate immunity. These large granular cells, that have a natural "instinct" to kill tumour or virus-infected cells with-out prior immunization, are called natural killer cells. It is believed that NK cells recognize cell that have lost class I MHC molecules from their surface. Many surface changes occur on tumour cells and some virus-infected cells, including loss of class I MHC molecules. NK cells recognize an damage these class-I-MHC-negative cells very effectively.

(E)Eosinophils

Eosinophils are a specialized group of leukocytes that are weak phagocytes exhibiting chemotaxi Eosinophils are produced in large numbers in individuals with parasitic infections. Sometime when a parasite is too big to be phagocytosed, (such as schistosomes or flatworms), they attack themselves to these parasites and release their intracellular granular contents that contain sever toxic proteins to kill the parasite.

(F)Basophils and Mast Cells

Basophils circulating in the blood are very similar, though not identical, to the large mast cells located outside many blood capillaries (mainly in tissues). These cells (basophils and mast cells) have stored intracellular granules that contain a variety of molecules capable of triggering inflammation upon their release. These mediators are released when the cells are activated. Basophils and mast cells have bound antibody (IgE) molecules on their surface. The binding and cross-linking of these antibody molecules by an allergen stimulates

the release of pharmacologically active granules that are stored in basophils/mast cells.These granules, when released inside the host body, elicit symptoms of an allergic reaction.

(G)Mononuclear Phagocytes

The mononuclear phagocyte system comprises blood monocytes and tissue macrophages (histocytes). Monocytes are incompletely differentiated cells that circulate in blood. These monocytes are referred to as macrophages when they migrate through blood vessels and are sequestered into the surrounding organs and connective tissues. Monocytes are derived from the myeloid lineage of bone marrow stem cells. Their major functions involve phagocytosis and destruction of invading pathogens. Moreover, these cells are also very effective at presenting antigens to T lymphocytes.

(H) Neutrophils

A neutrophil is another important phagocyte that has a multilobed nucleus, and is often referred to as polymorphonuclear neutrophil (PMN). Neutrophils express F_c receptors for antibodies (receptors that bind the F_c **region of the antibody**) and receptors for complement proteins that help to phagocytose the opsonized bacteria.

These cells are also involved in inflammation, and migrate from blood vessels into inflamed tissue in response to chemical signals received from the lesion site. However, after phagocytosing 5-20 bacteria, neutrophils themselves die, probably by the action of a variety of lytic enzymes released near the neutrophils during phagocytosis. Thus, neutrophils are short-lived cells.

Organs of Immune System and Their Functions

A number of morphologically and functionally diverse organs and tissues have various functions in the development of immune response. This can be divided on the basis of function into the primary and secondary lymphoid organs. The thymus and bone marrow constitute the primary lymphoid organs where maturation of lymphocytes occurs. The lymph nodes, spleen and various mucosal associated tissues (MALT) compose the secondary (or pheripheral) lymphoid organs, which trap antigen and provide site for mature lymphocytes to interact with that antigen. In addition, tertiary lymphoid tissues, which normally contain few lymphoid cells, can import lymphoid cells during an inflammatory response. Most prominent of these are cutaneous associated lymphoid tissue. Once mature lymphocytes are generated in the primary lymphoid organs, they circulate in the blood and lymphatic system.

(A) Primary Lymphoid Organs.

Immature lymphocytes generated during hematopoiesis mature and become committed to a particular antigenic specificity within the primary lymphoid organs. Only after a lymphocyte has matured within a primary lymphoid organ is the cell immuno competent. In mammals, B cell maturation occurs in the bone marrow and T cell maturation occurs in the thymus.

(*i*) **Thymus :** The thymus is a flat, bilobed organ situated above the heart. Each lobule is organized into two compartments : the outer compartment or cortex, is densely packed with immature T cells, called thymocytes, whereas the inner compartment or medulla is sparsely populated with thymocyte.

After the progenitor T cells formed during hemotopoiesis enter the thymus, they are thought to multiply rapidly within the cortex. The small subset of more mature thymocytes are then thought to migrate from the cortex to the medullar where they continues to mature and finally leave the thymus via post capillary venules. Before leaving the thymus, when the developing thymocytes begin to express antigen binding receptors, they are subjected to a 2 step selection process, so that only T cells recognizing antigenic peptides in the content of self MHC molecules are released from the thymus.

(*ii*) **Bone Marrow :** In birds a lymphoid organ called the bursa of fabricius is the primary site of B cell maturation. There is no bursa in mammals and no single counterpart to it as a primary lymphoid organ. Instead, regions of the bone marrow and possibly of other lymphoid tissues serve as the 'bursal equivalent' where B cell maturation occurs.

Stromal cell within the bone marrow interact directly with the B cells and secrete various cytokins that are required for B cell developmental process. Similar to thymic selection process within the bone marrow eliminates B cells with self reactive antibody receptors.

(B) Secondary Lymphoid Organs : Lymph nodes and spleen are the most highly, organized of the secondary lymphoid organs, in addition to lymphoid follicles, they possess distinct regions of T cell and B cell activity and are surrounded by a fibrous capsule. Less organized lymphoid tissue collectively called mucosal associated lymphoid tissue (MALT) occurs in various body sites.

(i) **Lymph Nodes :** Morphologically, a lymph nodes can be divided into 3 rougly concentric regions, the cortex, paracortex and medulla each of which provides a distinct micro environment. The outer most layer, the cortex contains lymphocytes (mostly B cells), macrophages and follicular dendritic cells arranged in primary follicles. Intense B cell activation and differentiation into plasma and memory B cells occurs in the germinal centres of lymph nodes. Beneath the cortex is the paracortex which is populated largely with T lymphocytes and also contains interdigitating dendritic cells thought to have migrated from tissues to the node. These interdigitating dendritic cells express high levels of class II MHC molecules, which is necessary for antigen presentation to T_H cells.

(ii) **Spleen :** The spleen is a large, ovoid secondary lymphoid organs. Unlike lymph nodes, which are specialized to trap localized antigen from regional tissues spaces, the spleen is adapted to filtering blood and trapping blood borne antigens, and thus can respond to systemic infections. Spleen is compartmentalized into red pulp and white pulp, which are separated by a diffuse marginal zone. The splenic red pulp consists of a network of sinusoids populated with macrophages and numerous red blood cells, it is the site where old and defective red blood cells are removed. The splenic white pulp surrounds the arteries forming a periarteriolar lymphoid sheath (PALS) populated mainly by T lymphocytes. The marginal zone, located peripheral to PALS, is rich in B cells organized into primary lymphoid follicles. The initial activation of B and T cells takes place in the T cell rich PALS.

(iii) **Mucosal Associated Lymphoid Tissue :** The mucosa lining the alimentary, respiratory, genitourinary and other lumina and surfaces are constantly exposed to numerous antigens. These areas are endowed with a rich collection of lymphoid cells, either specialized aggregates like the Payer's patches or scattered isolated lymphoid follicles collectively called the mucosa associated lymphoid tissue. Those lymphoid tissue present in gut are GALT and in bronchus are BALT. MALT contains lymphoid as well as phagocyte cells. Both B and T cells are present. While the predominant immunoglobulin produced in the mucosa is secretory IgA, other immunoglobulin classes, IgG, IgM, IgE are also formed.

(C) Tertiary Lymphoid Tissues : The skin provides an important barrier to the enternal environment and its large surface area makes this tissue important in nonspecific innate defences. The outer epidermal layer of the skin is composed largely of specialized epithelial cells called keratinocytes.

Antigens

Classically, an antigen is defined as a substance that has the

(a) ability to induce the formation of specific antibodies,

(b) ability to react specifically with the generated antibodies, and not with other antibodies that are also present in the system.

A more recent term used is **immunogen**. An immunogen may be defined as any chemical substance that either elicits the production of specific antibodies or binds to T-cell receptors. Thus, an immunogen stimulates both B cells that form antibodies and T cells that mount an effective immune response. This may be represented as :

Immunogen + B cells $\longrightarrow$ Antibodies

Immunogen + T$\longrightarrow$cells $\longrightarrow$ Activated effector T-cells (for example, T_{cyt}, T_H, etc.)

The complete antigen molecule is not immunogenic. Instead small regions of the antigen are antigenic or immunogenic. These regions are termed **antigenic determinants or epitopes** (Greek: *epi*—non-self and *topos*—map or place).

Antigenic determinants, are regions of an antigen that can bind an antibody or an antigen-specific membrane receptor on a lymphocyte. The size of an antigenic determinant usually ranges from 6 to 8 amino acids in protein antigens and 6 to 8 monosaccharide units (molecular weight: 750 Da) in the case of polysaccharide antigens.

Antigenic determinants are normally well-separated spatially and two separate antibody molecules can bind to the same antigen molecule without influencing the binding of the other. Such determinants are called **non-overlapping** determinants. In other instances, antigenic determinants may be located so close that the binding of one antibody to a determinant may sterically interfere with the binding of another antibody molecule. In such cases, determinants are said to be **overlapping**.

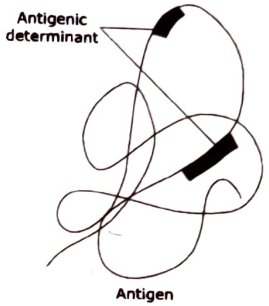


Fig: Schematic representation showing protein antigen and antigenic determinant.

General Properties of Antigen

Molecular

A relative molecular mass of about 1,000 is the lowest limit below which molecules are found to be non-immunogenic. Molecules between 5,000-10,000 Da are weakly immunogenic though better than 1000 Da molecule. Immunogens of relative molecular mass above 10,000 can generally induce a sufficiently strong immune response, and molecules above 100,000 are most effective in triggering an immune response.

Self or Foreign

Generally, the body's own antigens are recognized as self and hence cannot act as antigen /immunogen. However, homologous antigens from another species will be recognized as an immunogen and will be dealt with accordingly. Generally, the greater the **phylogenetic distance** between two species, the greater the structural (and hence antigenic) disparity between them; and hence, stronger the immune response.

Chemical Complexity

Synthetic homopolymers (polymer of one amino acid) fail to elicit an immune response. Heteropolymers (polymers of different amino acids) are the best immunogens as they are structurally complex. A complex of lipids (that are themselves non-antigenic) associated with polysaccharides is called a lipopolysaccharide. This is a chemically complex molecule that is strongly immunogenic.

Route of Entry

The route, or portal of entry (Latin: *porta*—gate) as it is commonly called, refers to the route taken by the pathogen (or antigen) to enter the body of the host. The immune response to an antigen varies according to the portal of entry of that antigen. Protein antigens that enter subcutaneously or intradermally are usually immunogenic as these antigens are taken up by the antigen-presenting cells (Langerhans cells) and transported to lymph nodes, where the immune response occurs. In contrast, large doses of protein immunogen, administered intravenously, often induce specific irresponsiveness. Such irresponsiveness could be due to **tolerance** induction in lymphocytes or suppression of reactive T-cell clones.

B-cell and T-cell Epitopes

The mechanism of recognition of antigens by T-cells and B-cells is basically different. Studies have revealed that B and T-cells recognize different antigenic determinants on the same antigen molecule.

Properties of B-cell Epitopes

The properties of B-cell epitopes are as under :

- generally composed of hydrophilic amino acids;
- located on the surface of the native protein so that it is topographically accessible to the antibody or antibody-like B-cell receptor;
- can contain linear or conformational epitopes;
- tend to be located on the flexible region of the immunogen, thus maximizing its easy binding with the antibody that might not otherwise react with it, if it was rigid; and
- may contain overlapping and non-overlapping determinants. Bovine serum antigen (BSA) has 25 overlapping antigens on its surface.

Some determinants induce a more pronounced immune response than other epitopes in a given animal. Such epitopes are called immunodominant. These are usually those epitopes that project distally from the central mass of the immunogen.

Properties of T-Cell Epitopes

The properties of T-cell epitopes are as follows:

- Processing of an antigen is required before it can be presented to a T cell. This processing yields peptides which bind to class I or II MHC molecules and this complex is then presented to T cells.
- T-cell epitopes are always presented together with MHC. T-cell receptors do not recognize any epitopes that are presented alone.
- Antigens recognized by T cells have two regions—epitope, that interacts with T-cell receptor, and agretope, that interacts with the MHC molecule. Interactions between the epitope and T-cell receptor, and the agretope and MHC are purely non-covalent.
- Epitopes recognized by T cells are often internal and usually sequential.

Superantigens

Recently, a class of protein antigens with "super" antigenic properties has been characterized and named **superantigens**. Superantigens bind simultaneously to T-cell receptors and class II MHC molecules. This non-specific binding activates into the lymphoproliferative phase in T cells and induces pronounced cytokine production by T cells *in vivo*, thereby causing a variety of pathological consequences such as fever, malaise, diarrhoea, etc. Unlike conventional antigens, superantigens are not internalized and degraded by antigen-presenting cells. Instead, they bind directly to class II MHC molecules *outside* of the antigen-binding cleft.

Haptens

The term hapten (Greek; *aptein*—to grasp or fasten) was introduced by Ehrlich who considered the antigen (he was studying the diphtheria toxin then) to be a distinct molecular entity, containing two active groups—haptophore, the group for binding and toxophore, the group for toxicity.

A hapten can be defined as any substance of molecular weight less than 1,000 Da, that, though *incapable* of stimulating antibody formation by itself, is able to react with pre-formed antibody molecules. Examples of haptens include 2,4 dinitriphenol p-aminobenzenearsonate, monosaccharides, amino acids, etc. Many biologically important substances such as sugars, amino acids, drugs, peptide hormones, etc. function as haptens. A large number of chemicals such as dinitrophenol (DNP), and drugs such as penicillin, are classified as hapten.

A hapten becomes immunogenic when it is attached to an immunogenic **carrier** molecule such as a protein, yielding an immunogenic hapten-carrier conjugate.

Consider the case of DNP. DNP is a hapten and is unable to induce antibody formation when injected in test animal, alone. However, when DNP is linked to an immunogenic carrier molecule such as bovine serum albumin (BSA), it results in the formation of anti-DNP antibodies.

Although anti-hapten antibodies are usually predominant, animals immunized with such a conjugate produce antibodies, not only against hapten, but also against epitopes of the carrier molecule. Moreover, antibodies are also formed against neoantigenic determinants formed by parts of both carrier (BSA) and hapten molecules.

Adjuvants

Adjuvants (Latin: *adjuvare*—to help) are substances which when mixed with antigen and injected with it, accelerate, enhance and prolong the immunogenicity of the antigen without altering the chemical composition of the antigen. Adjuvants are agents that enhance the immune response and are usually used when either the concentration of the antigen or its immunogenicity is low. In both cases, the addition of adjuvant to the antigen augments the immune response manifold. Chemicals commonly used as adjuvants are **alum** (aluminium potassium sulphate), **Freund's complete** and **incomplete adjuvants**.

Freund's incomplete adjuvant is an emulsion of oil in water (aqueous solution of mineral oil and mannide monooleate which acts as emulsifier). Freund's complete adjuvant contains heat-killed myco-bacterium in oil-and-water emulsion. Freund's complete adjuvant, though a highly effective adjuvant, is never used in human vaccines.

For the vaccination of human beings, the most commonly used adjuvants are still common ones such as aluminium hydroxide, aluminium phosphate or alum precipitate

Major Histocompatibility Complex and Antigen Processing and Presentation

Although molecules coded for by the MHC were originally identified based on their role in transplant rejection, they actually evolved to present foreign antigens to T cells. Two classes (class I and II) of MHC genes, closely linked on chromosome 6 in humans, code for human leukocyte antigen (HLA) which are the molecules critical to antigen presentation. MHC molecules are important in determing the ability of an animal to mount an immune response. Because there are atleast 20 MHC genes and atleast 50 alleles for each genes, it is virtually impossible for any two people to have matching sets of MHC markers on their cells, thus the MHC is a biochemical finger print unique to each individual. The MHC contains 3 groups of genes known as class I, II, III.

Class I and II gene code for integral membrane proteins, whereas class III gene codes for serum proteins T lymphocytes recognize surface antigen only when associated with class I, II MHC molecules. Two main classes of MHC molecules mark cells as 'self. Class I MHC molecules are located on all nucleated cells, almost every cell of the body. Class II MHC molecules are restricted to macrophages and B lymphocytes. They play a important role in interaction between cells of immune system. Class I region is encoded by DP, DQ and DR. Since there is a pair of chromosome 6 in diploid cell, an individual has two a genes and two b genes. These alleles are codominant which means product of both the alleles are expressed in cells.

Class I Gene Products

These are glycoproteins found at the surface of all nucleated cells. Class I protein is a single polypeptide chain associated with smaller proteins called β_2-microglobulin which is invariant and is not coded by MHC. The proteins specified by different alleles differ in their amino acid composition at amino terminal ends of molecule. Class I proteins restrict the activity of cytotoxic T cells. For example T cytotoxic lymphocytes is specific for measles virus, will kill cells infected with that virus only. If the T cell and the virus infected cells have same. Class I antigens it also acts as variability factor what the means is that the T cells must recognize self MHC proteins inorder to react against the non self (viral proteins)

Class II Gene Products

These are also glycoproteins found on restricted range of cells, such as antigen presenting cells, B cells. Class II proteins consist of 2 dissimilar polypeptides called α and 3 subunits both of which are encoded within the MHC. Most of the differences between class II proteins coded by a different alleles lie in the amino acid composition of the β chain.

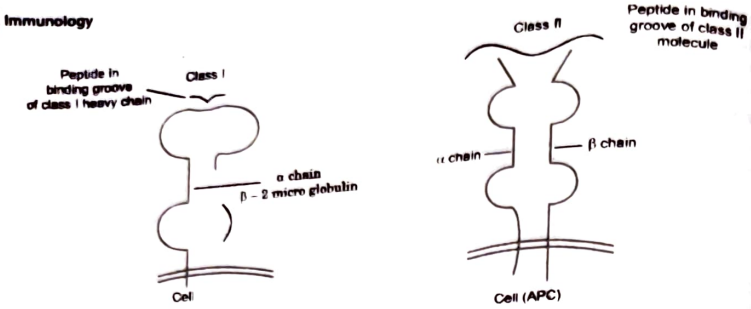


Fig. MHC class I and class II molecules binding peptide.

The polymorphic regions of MHC class I and II are the peptide binding domains of these molecules and bind peptides ranging from 9 -10 (for MHC class I) and 12 - 20 amino acid residues (for MHC class II) MHC class II molecules are expressed on B cells, dendritic cells and macrophages, efficient APCs for the activation of CD^{4+} helper T cells.

Anting Processing and Presentation

In order for a foreign protein antigen to be recognized by a T cell, it must be degraded into small antigenic peptides that form complexes with class I or class II MHC molecules. This conversion of proteins into MHC associated peptide fragments is called antigen processing and presentation. Whether a particular antigen will be processed and presented together with class I or class II MHC molecules appears to be determined by the route that the antigen takes to enter a cell.

1. **Endogenous antigens :** It is produced within the host cell itself. Two common examples are viral protein sythesized with virus infected host cells and unique proteins sythesized by cancerous cells. Endogenous antigens are degraded into peptide fragments that bind to class I MHC molecules within the endoplasmic reticulum. The peptide class I MHC complex is then transported to the cell membrane. Since all nucleated cells express class I MHC molecules, all cells producing endogenous antigen use this route. T cells displaying CD8 recognize antigen associated with class I MHC molecules and thus are class I MHC restricted. These cytotoxic T cells attack and kill cells displaying the antigen - MHC class I complex for which their receptors are specific.
2. **Exogenous antigen :** It is produced outside of the host cell and enters the cell by endocytosis or phagocytosis. APCs cells (macrophages, dendritic cells and B cells) degrade ingested exogenous antigen into peptide fragments within the endocytic processing pathway. Experiment suggest that class II MHC molecules are expressed within the endocytic processing pathway and that peptides produced by degradation of antigen in this pathway bind to the cleft within the class II MHC molecules. The MHC molecules bearing the peptide are then exported to the cell surface. Since expression of MHC class II molecules is limited to APCs, presentation of enogenous peptide class II MHC complexes is limited-to these cells. T cells displaying CD4 recognize antigen combined with class II MHC molecules and thus are said to be class II MHC restricted. These cells generally function as T helper cells.

Antibody Structure, Function And Diversity

Antibodies - Immunoglobulins

Antibodies or immunoglobulins (Ig) are produced by B lymphocytes (plasma cells) in response to the presence of antigens. Immunoglobulins constitute 20-25 percent of the total serum proteins. Based on physiological and antigenic difference five classes of immunoglobulins have been recognized

IgG, IgA, IgM, IgD and IgE.

Structure of Antibody:

An antibody molecule consists of two identical 'light' protein chains and two identical 'heavy' protein chains, all held together by both hydrogen bonding and precisely localized disulfide linkages. The N terminal regions of the light (L) and heavy (H) chains together form the antiogen recognition site of each antibody what is called F_{ab} fragment. The sites that recognize and bind antigens consist of three complementary determining regions (CDRs) that lie within the variable (V_H and V_L) regions at the N terminal ends of the two heavy and two light chains. There are five different kinds of H chains (referred to as μ, δ, γ, ε and α chains), which determine the class of antibody (IgM, IgD, IgG, IgE and IgA, respectively).

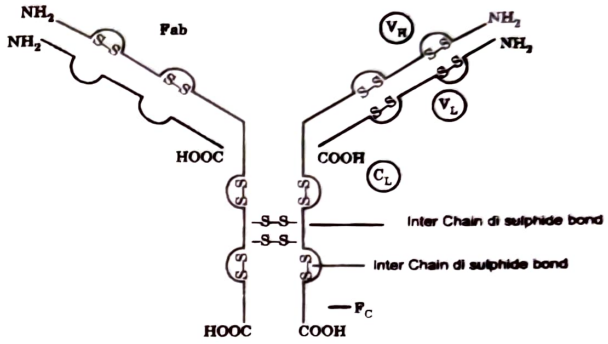


Fig. Structure of Immunoglobulin

There are also two different kinds of L chains - K and λ, each with a MW of 23 KD. Each antibody unit can have only K or λ, L chains but not both. In addition to the variable regions (V_H, V_L), each chain contains one constant region, or domain (C_L) and each H chain has three constant regions, or domains (C_{H1}, C_{H2}, C_{H3}). Most of the antibody molecule (the C terminal three quarters of the H chain and the C terminal half of the L chain) are constant regions of the antibody molecule and are the same for all antibodies of the same class and subclass. The C terminal half of the H chain the F_c region (fragment that crystallized) serves other functions i.e., combines with complement, is cytophilic. Carbohydrates are also present on antibodies on the F_c portion of H chain when antibodies are digested with the protolytic enzyme papain, three fragments are released :

Two identical (F_{ab}) fragments, and one F_c fragment.

Antibody Classes

Different antibody classes with different biological activities have evolved to deal with antigens (e.g. microbes) with different properties and which enter the body at different sites - through the skin, the gastrointestinal or the genitourinary tract.

IgG

They have a MW of 150 KD are found in vascular and entravascular spaces as well as in secretions. IgG is the most abundant immunoglobulin in the blood and is the only antibody class to cross the placenta to provide passive humoral immunity to the developing fetus. There are different subclasses of IgGl designated

IgG, IgG2, IgG3, IgG4.

IgG3 is most effective complement activator.

IgA

This immunoglobulin is present in the serum as a 170 kD, four polypeptide (two L and two H) chain protein. It is the major immunoglobulin present in external secretions such as colostrum, milk and saliva where it exists as a 420 kD dimer. In addition to k or λ chain and the IgA heavy chain (α) it also contains two other polypeptide chains secretory component (SC) and J chain (joining chain).

Most IgA is synthesized locally by plasma cell in mammary and salivary glands, and along the respiratory, gastrointestinal and genitourinary tracts. It is then transported through epithelial cells to the lumen. This antibody is a first line of defense against microbial invaders at mucusal surfaces.

IgM

It accounts for 5-10% of the total serum immunoglobulin. IgM is the first antibody produced by, and expressed on the surface of a B cell. It acts as an antigen receptor for these cells, and is also present as a soluble molecule in the blood. On the B cell surface this molecule is expressed as a four chain unit - two μ H chain and two L chain. In the blood, IgM is composed of five four chain units held together by disulfide bridges at the carboxy terminal ends of the μ chain J chain is also associated with IgM in the blood. Because of its size (900 kD), IgM is found primarily in the intravascular space. As IgM is the first antibody produced in an immune response. Although IgM antibodies usually have low affinity binding sites for antigen, they have 10 combining sites per molecule which can synergize with each other on the same molecule when it binds to a microbe. Thus, the overall tightness of binding of an IgM molecule to a microbe (avidity) is quite high, making antibodies of this class very effective in removal of the microbe.

IgD

IgD is present in low quantities in the circulation (0.3 mg/ml in adult serum). Its primary function is that of an antigen receptor on B lymphocytes. B cells thus can express both IgM and IgD and both are specific for the same antigen when IgM and IgD expressed on a B cell interact with an antigen for which they are specific, the antigen is internalized, and processed and presented to helper T cells which trigger the B cells to proliferate and differentiate into plasma cells, thus initiating the development of a humoral immune response.

IgE

IgE is present in serum at very low levels (nanograms per milliliter), but plays a significant role in enhancing acute inflammation, in protection from injection by worms, and in allergic reactions. Antibody mediated allergy is predominantly associated with IgE. Antigen reintroduced into a perviously sensitized individual binds to antigen specific IgE on 'armed' most cells and triggers release of the pharmacologically active agents (e.g. histamine). It is involved in immediate hypersensitivity syndromes such as hay fever and asthma.

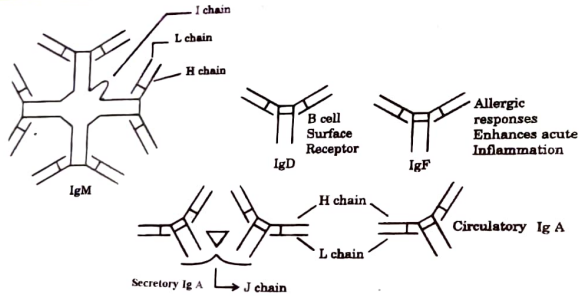


Fig. IgA Immunoglobulim

Affinity : Strength of binding of antigen to antibody. The term affinity alludes to the strength of the bond between antigen determination and monovalent antibodies.

Avidity : Overall strength of interaction between a large antigen with multiple epitopes and polyvalent antibody molecule such as IgM.

Allotypes

These are genetic markers on immunoglobulin (Ig) that segregate within the species. If Ig expressing a particular allotype is injected into an individual whose Igs do not express that allotype, an immune response with develop against the allotype. Like blood types, they are inherited in mendelion fashion but are usually of no functional consequence.

Idiotypes

These are unique antigenic determinants associated with antigen binding sites of antibodies and are the result of the different amino acid sequences which determine their specificities.

Functions of Antibodies

Antibody alone can neutralize viruses and toxins if it binds tightly to, and blocks, a part of the toxin or virus critical to its biological activity. Similarly, antibodies can inhibit microbes from colonizing mucosal areas and in some cases may induce programmed cell death (apoptosis).

IgG or IgM antibodies can, on binding to antigen, activate the classical pathway of comple-ment leading to lysis of the cell on which the antigen is located or to attraction of immune cells (chemotoxis) which phagocytose the antigen expressing cells.

A variety of effector cells have receptor for the F_c region of antibodies. Phagocytes (PMNs, macrophages and cosinophils) utilize their F_c receptors (F_cR) for IgG ($F_c\gamma R$) or IgA ($F_c\alpha R$) to enhance phagocytosis of antibody opsonized microbes. In addition these F_cR can mediate killing of cells through antibody dependent cellular cytotoxicity (ADCC). Binding of antigens to IgE already bound to F_c receptors for IgE on mast cells results in degranulation and subsequent enhancement of the acute inflammatory response.

T- Cells : Receptors and Activation

T lymphocytes are produced in the thymus from cirulating T cell precursors derived from HSC (Henoepoietic stem cell) in the bone marrow. These T coll presursors differentiate within the thymus into mature T cells expressing molecules important to T cell function, such as the T cell receptor, CD4 and CD8. The thymus thus generates large numbers of functionally mature antigen specific T cells which then migrate to the peripheral lymphoid tissues to mediate protection against invading microbes.

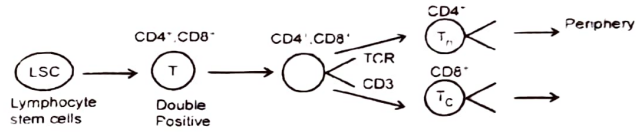


Fig. Development of CD4 and CDS' T cell in the thymus.

Generation of T cell diversity

Each of the very large number of T cells produced in the thymus has only one specificity, coded for by its antigen receptor. Millions of T cells, each with receptor specific for different antigens, are generated by gene rearrangement from multiple (inherited) germ line genes. The T cell receptor consists of two polypeptide chains :

α and β or γ and δ.

Each chain is a member of the Ig superfamily and thus has a uniform domain structure produced by intra chain disulfide bonding. Unlike most proteins produced in the body, each polypeptide chain of the T cell receptor is coded for by several different genes.

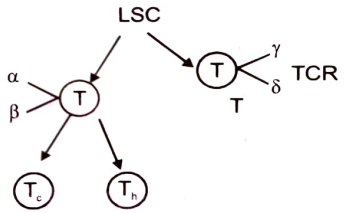


Fig. Development of αβ and γδ T cells.

The genes for the TCR α, β and γ chains are coded by three unlinked gene groups, whereas the δ chain genes lie within the α chain genes. Each gene group consists of multiplegene segments which code for V, D and J gene segments (β and δ genes) or V and J gene segments (α and γ genes) as well as constant region genes. In the germ line the V and J (α chain) or V, D and J (β chain) segments are separated by non coding DNA. During T cell differentiation these V, D and J segments are rearranged to form a complete V gene. As in the case of Igs rearrangements, the expression of a V-C (constant region) and the T cell encludes further rearrangement (allelic exclusion) and the T cell thus becomes committed to the expression of a single V-C α chain combination and a single V-C β chain combination, which together constitute an antigenic binding site and determine the specificity of the T cell.

T Cell Receptor (TCR) for Antigen :

The TCR for antigen is only found on the T cell membrane and is composed of two polypeptide chains, α and β (50 and 39 kDa). Each of these glycoproteins is made up of constant and variable regions like those of Ig and together the α and β variable regions compose a T cell antigen binding site. Some T cells, whose function is not clear, express a TCR consisting of γ and δ chains.

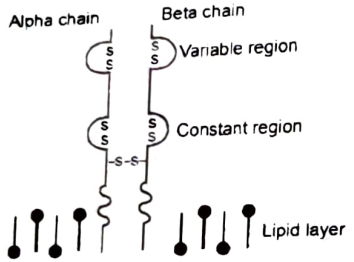


Fig. T cell antigen receptor αβ dimer.

T Cell Receptor Complex

The T cell receptor complex consists of the antigen receptor, the αβ or γ δ dimer associated with several other polypeptides important in T cell signalling and recognition. In particular, the TCR is associated with CD3, a signaling complex which is itself composed of several polypeptides including γ,δ and ε chains. Together these molecules form an ,α β - CD3 complex. T cells expressing the γ δ antigen receptor form a γ δ-CD3 complex. Two other molecules on T cells also play a role in T cell recognition of antigen. CD4 is not just a marker on Th cells; but also a molecule that can bind to the non polymorphic region of MHC class II. This molecule restricts these Th cells to recognizing only peptides presented on MHC class II molecules. Similarly, CD8 on cytolytic T cells binds the nonpolymorphic region of MHC class I, restricting these killer T cells to recognizing only cells presenting peptide in MHC class I molecules.

Selection of the T Cell

In the thymus, those T cells that express a TCR that binds weakly to self MHC are positively selected. Those that express a TCR that binds strongly to self MHC are negatively selected. Thus, the T cells that survive and mature are those that recognize modified self MHC molecules-self MHC molecules plus foreign peptides.

T Suppressor Cells (TS)

There are Th cells which promotes immune response. It is clear that some T cells help to suppress the humoral and the cell mediated branches of the immune system but there is also a regulating set (TS) of T cells whose job is to prevent helpers from getting out of hand.

Two Types of T Helper Cells

CD4 cells can differentiate down at least two pathways to become inflammatory T cells (Thl) or Th2 cells which help B cells. Thl cells are involved in mediating inflammatory responses (through the activation of macrophages), while Th2 are involved in the induction of humoral immunity. Thl cells produce cytokines that act on macrophages

(IFNY and TNFα) whilet Th2 cells are involved mainly in B cell differentiation and maturation (via IL-4, IL-5, IL-6).

T Cell Activation

Initial recognition of processed antigen by T cells is via the T cell antigen receptor. Accessory molecules further link the APC and the T cell leading to a stronger cell interaction. For e.g. CD4 binds to the constant region domain of class II MHC molecules whilst CD8 binds to class I MHC molecules. Other ligand-receptor pairs such as intracellular adhesion molecules (ICAMS) and lymphocyte function associated antigens (LFA) are also important.

Co Stimulatory Molecules

Full activation of antigen specific T cells requires two signals-one signal coming via the TCR and the other signal through engagement of co-stimulatory molecules. T cells receiving one signal via their TCR are turned off (become anergic), whilst those receiving two signals e.g. via T cell CD28 binding to B7 on the APC induce T cell lymphokine production and T cell proliferation.

Super-Antigens and Th Cell Activation

Some protein products of bacteria and viruses produce proteins known as super-antigens that bind to lateral surfaces of the MHC class II molecules and the V region of the β submit of the TCR. These antigens are not processed into peptides but are able to bind to a specific family of TCR. Among the bacterial super antigens are the staphylococcal enterotoxins (SE) that cause common food poisoning and the toxic shock syndrome toxin (TSST).

Activation of Cytotoxic T Cells

$CD8^+$ Tc cells also require activation to produce functional CTLs containing granules and perforsin. This is mediated through attachment of their TCR to MHC class I peptide complexes on APCs. In this case, however, it appears that, the second signal can only be supplied by APCs that have been conditioned by Th cells. As with Th activation, cytokines produced by Th cells and APCs are important for Tc activation.

T Lymphocyte Activation

In order for a T cell to fulfill its function it has to convey a signal from the cell surface via the cytosol into the nucleus to give rise to a specific gene transcription. This signal transduction is brought about by several molecules at the cell surface. The key players in this amplification process are enzymes which phosphorylate (kinases) and desphosphorylate (phosphatases) particular amino acids thus activating them in a sequential fashion leading eventually to activation of specific transcription factors in the nucleus and production of functional proteins. Initiation of the process is through activation of a CD4 associated kinase (lymphocyte kinase - Ick) by CD45 (a phosphatase), one of its functions being to phosphorylate tyrosines of the immuno stimulatory tyrosine - based activation motifs (ITAMs) on the zeta chain of CD3. This, in turn activates another protein tyrosine kinase ZAP 70 which activates an enzyme responsible for activating the phosphatidyl inositol pathway and further down stream events leading to the activation of transcription factors regulating production of IL-2 and other important proteins.

B – Cells : Maturation and Activation

The generation of mature B cell first occurs in the embryo and continues throughout life. Before birth, the yolk sac, fetal liver, and fetal bone marrow are the major, sites of B-cell maturation, generation of mature B cells occurs in the bone marrow.

Progenitor B Cells Proliferate in Bone Marrow

B cell development begins as lymphoid stem cells differtiate into the earliest distinctive B-lineage cell-the progenitor B cell (pro B cell) which expresses a transmembrane tyrosine phosphatose called CD45R. Pro B cells proliferate within the bone marrow. Proliferation and differentiation of pro B cells into precursor B cells (pre-B cells) requires the micro environment provided by the bone marrow stromal cells. The stromal cells play two important roles : they interact directly with pro B and pre B cells, and they secrete various cytokines (IL-7) that support the developmental process.

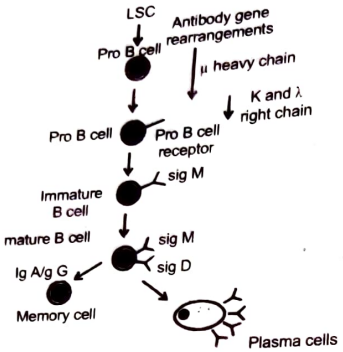


Fig: Life History of B-Cell.

B cell maturation depends on rearrangement of the Ig DNA in the lymphoid stem cells. First to occur in the pro B cell stage is a heavy chain D_H to J_H gene rearrangement this is followed by a V_H to D_H J_H rearrangement. If the first heavy chain rearrangement is not productive, then $V_H - D_H - J_H$ rearrangement continues on the other chromosoms upon completion of heavy chain rearrangement, the cell is classified as a pre B cell. Continued development of a pre B cell into an immature B cell requires a productive light chain gene arrangement. Because of alletic exclusion, only one light chain isotype is expressed on the membrane of a B cell. Immature B cells express mIgM on the cell surface.

B Cell Receptor Complex (BCR)

The BCR complex consist of the antigen receptor Ig, in association with two other polypeptides, Igα and Igβ. Igα and Igβ are signaling molecules for the BCR and are also required for assembly and expression of Ig. B cell coreceptors, including CD21, CD32, CD19, and CD81, associate with the BCR complex especially when both the BCR and one or more of the coreceptors are linked through an antigen-complement / antibody complex. Depending on which molecules are ligated, signaling by the Ig-Igα / Igβ complex is enhanced or inhibited.

B Cell Activation and Proliferation

After export of B cells from the bone marrow, activation, proliferation and differentiation occur in the periphery and require antigen. This type of response mediated by antibodies is called humoral immune response. Depending on the nature of the antigen, B cell activation proceeds by two different routes, one dependent upon T_H cells, the other not.

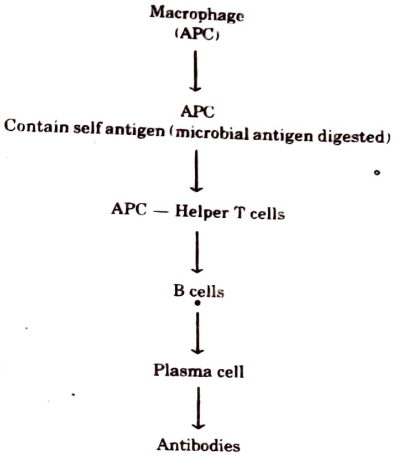
Thymus Independent (T-I) Antigens

The T independent antigens, without the interferance of T-cells, can also induce a response from B cells. The T-independent antigens contain the repeating subunits of polysaccharide of protein. Multiple bonds are formed with B cells by these antigens.

Thymus Dependent (T-D) Antigens

The T dependent antigen consists of bacteria, some of proteins, RBC etc. The B cells need the help of antigen presenting cells (APC) and specialized helper T cells to produce antibodies against T dependent antigens.

The cells recognize antigenic peptides on the surface of antigen specific B cells because the B cell can capture antigen specifically via membrane Ig and associate it with class II MHC molecules once triggered via the TCR, Th cells express CD40L which triggers B cell activation via its CD40 surface receptor. Activated B cells reciprocally co-stimulate T cells via CD28, which produce IL- 2, IL- 4 and IL- 5. As a result, both the T cells and B cells clonally expand and differentiate.



CD79 a/b molecules transmit the first signals following B cell interaction with antigen through ITAM of their intracytoplasmic tails. Co receptors of B cell receptor complex (CD21, CD19 and C8SI) modulate these initial signals CD21 binds to C3d if it is bound to the specific antigen and provides an additional positive signal in B cell activation. Cross linking of membrane receptor antibodies on B cells by T-I antigens induces clustering of co-receptors leading to multiple signals enhancing activation of kinase networks and of IgM producing B cells. B cells primed by a T - D antigen receive a second signal when the Th cell CD40L binds to CD40 on the B cell. Together with cytokines such as 1 L-2, 1 L- 4 etc, this signaling induces proliferation and differentiation of the B cells into plasma cells or memory cells.

Polyclonal Antibody

Antibodies derived from different B cell-lines. They are a mixture of immunoglobulin molecules secreted against a specific antigen each recognizing a different epitope.

Production:

These antibodies are typically produced by immunization of suitable mammal, such as a mouse, rabbit or goat. Larger mammals as often preferred as the amount of serum that can be collected is greater. An antigen is injected into the mammal. This induces the B lymphocyte to produce IgG immunoglobulins specific for the antigen. This polyclonal IgG is polyclonal purified from mammal's serum.

Monoclonal Antibodies

If a specific lymphocyte, after isolation and culture in *vitro*, becomes capable of producing a single type of antibody which bears specificity against specific antigen, it is known as 'monoclonal antibody'. Due to the presence of desired immunity, monoclonal antibodies are used in the diagnosis of diseases. Major difficulties with antibody secreting cells are that they cannot be maintained in culture. But myeloma cells (bone marrow tumour cells due to cancer) grow indefinitely to produce a huge quantity of identical cells (clones) and also produce immunoglobulins in the same amount. The immunoglobulins are infact monoclonal antibodies.

(*i*) **Hybridoma :** Cesar Milstein of Argentina tried to culture myeloma cells. In 1973, in collaboration with Cotton, he succeeded to fuse rat and mouse myeloma cells with the result of production of hybrid cells. The hybrid cells secreted the immunoglobulins which consisted of several types of polypeptides. In 1974, George Kohler and Milstein successfully isolated clones of cells from the fusion of two parental cell lines *i.e.* between lymphocyte from spleen of mice immunized with red blood cells from sheep and myeloma cells. These cells were maintained in *vitro* and produced antibodies. The antibodies for specific antigens immunized the

myeloma cells (Kohler and Milstein, 1975). Moreover, the hybrid cells maintained the character of lymphocyte to secrete antibodies, and of myeloma cells to multiply in culture. These hybrid cell lines (cell clones) are known as hybridomas which are capable of producing unlimited supply of antibodies. For the discovery of production of monoclonal antibodies Kohler and Milstein along with Neils Jerne were awarded Nobel Prize in 1984, in physiology and medicine.

The cells growing on culture and producing sufficient amount of monoclonal antibodies can be frozen. The cells are collected through centrifugation and resuspended at a density of 10^7 cells per ml freezing medium in cold at 0°C. Samples (0.7-1.0 ml) poured in sterile glass vials are sealed and stored at 80°C

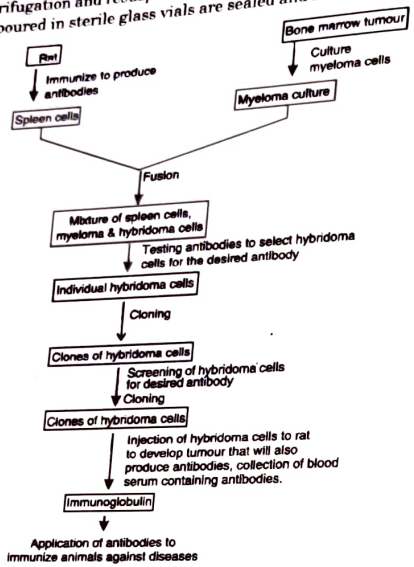


Fig. Outline of production of monoclonal antibodies.

Uses: Monoclonal antibodies are useful in the following cases:

(i) In diagnosis of ABO blood groups, cancer, pregnancy, allergies, viral diseases and certain hormones.

Antibody Diversity/Antibody Genes

There are three unlinked gene groups encoding immunoglobulins-one for κ chains, one for λ chains and one for H-chains, each on a different chromosome. Within each of these gene groups on the chromosome there are multiple coding regions (exons) which recombine at the level of DNA to yield a binding site. In a mature B cell or plasma cell, the DNA encoding the V region for the H-chain of a specific antibody, consists of a continuous uninterrupted nucleotide sequence. In contrast, the DNA in a germ line cell (or non B cell) for this V region exists in distinct DNA segments, exons, separated from each other by regions of noncoding DNA . The exons found in the area encoding the V region of the H-chain are: *V* segment (encoding approximately the first 102 amino acids), *D* segment (encoding 2-4 amino acids), and *J* segment (encoding the remaining 14 or so amino acids in the V region). For L-chains there are only V (encoding the first approximately 95 amino acids) and J segment (encoding the remaining 13 or so amino acids) exons. In each gene group, there are from 30-65 functional V segment genes. The D and J regions are between the V and C regions on the chromosome and there are multiple different genes for each. Thus, DNA segments that ultimately encode the binding site of antibodies have to be moved over distances (translocated) on the chromosome to form a DNA sequence encoding the V region.

The DNA sequences encoding the C region of the L-and H-chains are 3′ to the V genes, but separated from them by unused *J* segment genes and noncoding DNA. Furthermore, each gene group usually has one functional gene segment for each class and subclass. Thus, the H-chain gene group has nine functional C region genes, one each encoding μ, δ, γ1, γ2, γ3, γ4, ε, α1, α2. For the L-chain gene groups, there is one gene segment encoding the C region of κ L-chains, but four encoding λ L-chain C regions.

Table. Genes for human immunoglobulins

Ig polypeptide	Chromosome
H-chain	14
κ-chain	2
λ-chain	22

Gene rearrangement

During its development, a single B cell randomly selects one V, one D and one J (for H-chains), and one V and one J (for L-chains) for rearrangement (translocation). Gene segments encoding a portion of the V region are moved adjacent to other gene segments encoding the rest of the V region to create a gene segment encoding the entire V region, with the intervening DNA removed. More specifically, the H-chain gene group is the first to rearrange, initially moving one of several D segment genes adjacent to one of several J segment genes. This creates a DJ combination, which encodes the C terminal part of the H-chain V region. A V segment gene then rearranges to become contiguous with the DJ segment, creating a DNA sequence (VDJ) encoding a complete H-chain V region (Fig). This VDJ combination is 5′ to the group of H-chain C region genes, of which the closest one encodes the μ chain. A primary mRNA transcript is then made from VDJ through the μC region gene, after which the intervening message between VDJ and the μC region gene is spliced out to create an mRNA for a complete μ H-chain.

After the H-chain has successfully completed its rearrangement, one of the V region gene segments in either the λ or κ gene groups (but not both) translocates next to a J segment gene to create a gene (VJ) encoding a complete L-chain V region (Fig). For κ chains, the DNA sequences encoding the C region of the L-chains are 3′ to the V genes, but separated from them by unused J segment genes and noncoding DNA (Fig.), For λ, chains, since the J segment genes are each associated with a different Cλ gene, translocation of a V gene segment to a J gene , segment results in a V region next to a particular Cλ gene (e.g. Cλ2). It is important to repeat that in each B cell, only one of two L-chain gene groups will be used. A primary mRNA transcript is then made from VJ.

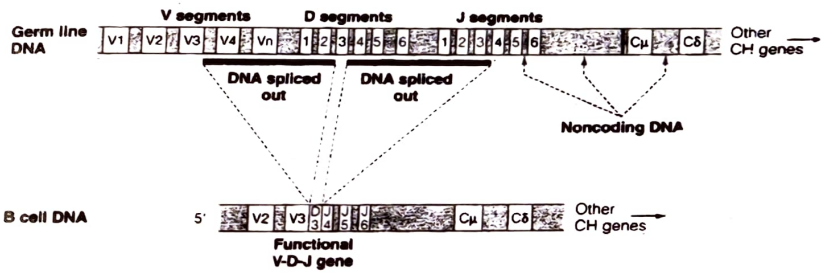


Fig. H-chain genes and translocation. In the germ line, and therefore in a cell destined to become a B cell, the H-chain gene loci contains many V segment genes. In a developing B cell, one of these V segments recombines with one of many D segments, which has already recombined with one of several J segments, to produce a functional DJ gene. In each B cell, the rearranged gene is transcribed, spliced and translated into an H-chain protein.

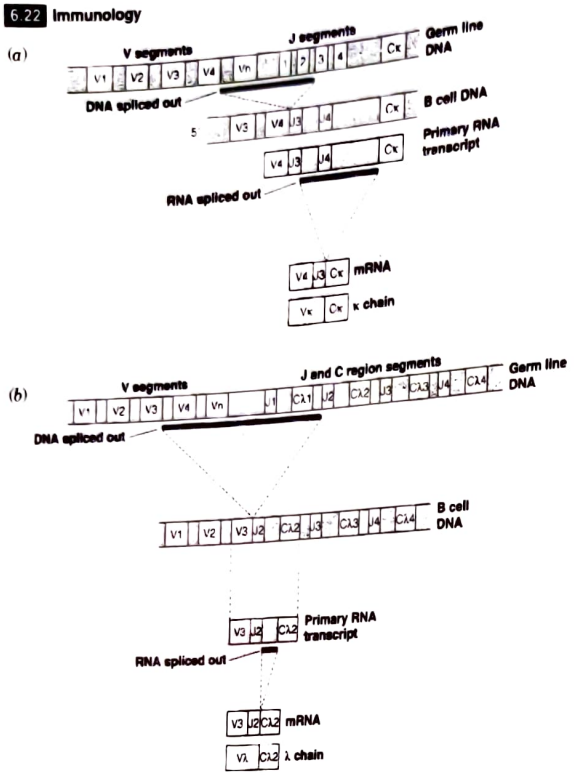


Fig. *L-chain genes and translocation. During differentiation of a B cell, and after rearrangement of the H-chain genes, one of the two L-chain groups rearrange. In particular, either* ***(a)*** *a germ line Vκ gene combines with a J segment gene to form a VJ combination; or* ***(b)*** *a germ line Vλ gene combines with one of the J segment Cλ gene combinations to form a VJ Cλ combination. The rearranged gene is then transcribed into a primary RNA transcript which then has the intervening noncoding sequences spliced out to form mRNA. This is then translated into light chain porotein.*

through the L-chain C region gene, after which the intervening message between VJ and the C region gene is spliced out to create an mRNA for a complete L-chain.

Gene rearrangement in B cells requires the products of two recombination-activating genes, RAG-1 and RAG-2, which appear to be only expressed together in developing lymphocytes. These enzymes break and rejoin the DNA during translocation and are thus critical to the generation of diversity.

Allelic exclusion

After successful rearrangement of the Ig DNA segments, the cell is committed to the expression of a particular V region for its H-chain and a particular V region for its L-chain and excludes other H- and L-chain V region rearrangements. This process is referred to as allelic exclusion and is unique to B and T cell antigen receptors. If an aberrant rearrangement occurs on the first chromosome the process will continue, *i.e.*, the process does not stop if the cell does not get it right the first time. The process stops, however, if the cell gets it right or runs out of chromosomes to rearrange. In fact, following successful VH gene rearrangement on one chromosome there is active suppression of further rearrangement of the other VH gene segments. Similarly, following successful VL gene rearrangement there is active suppression of further rearrangement of other V and J gene segments.

Thus, each B cell makes L-chains all of which contain a V region encoded by the same VJ region sequence and H-chains all of which contain a V region encoded by the same VDJ sequence. Each B cell will therefore express antibodies on its surface, all of which have exactly the same specificity. This cell and all of its progeny are committed to express and produce antibodies with these V regions.

Synthesis and assembly of H and L chains

After successful rearrangement of both L- and H-chain DNA, L- and H-chain mRNA is produced and translated into L- and H-polypeptide chains that combine in the ER to form an antibody molecule, which is transported to the plasma membrane as the antigen specific receptor for that B cell. Since the gene encoding the H-chain also contains coding sequences for a transmembrane domain, the H-chain produced contains a C terminal amino acid sequence which anchors the antibody in the plasma membrane. In plasma cells, the part of the mRNA encoding the H-chain transmembrane domain important for its membrane expression on B cells is spliced out. Thus, the antibody produced by a plasma cell does not become associated with the membrane, but rather is secreted.

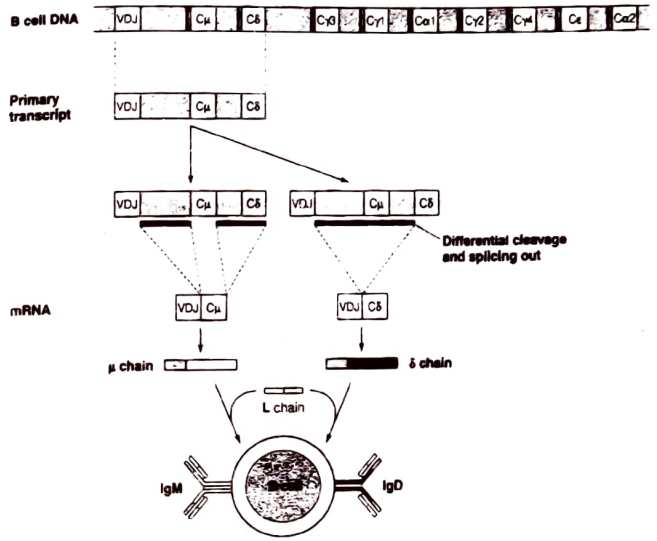


Fig. Expression of IgM and IgD on a mature B cell.

Differential splicing and class switching

As indicated above, the first antibody produced by a B cell is of the IgM class. Soon thereafter the B cell produces both an IgM and an IgD antibody, each having the same V regions and thus the same specificity. This is the result of the differential cleavage and splicing of the primary transcript. In particular, a primary transcript is made which includes information from the VDJ region through the Cλ region (Fig). This transcript is differentially spliced to yield two mRNAs - one for an IgM H-chain and the other for an IgD H-chain. In a mature B cell both are translated and expressed on the B cell surface with L-chain.

B cells expressing IgM and IgD on their surface are capable of switching to other H-chain classes. This isotype (class) switching requires stimulation of the B cell by T cells and cytokines. These signals induce translocation of VDJ and its insertion 5′ to another constant region gene (Fig). Class switch is guided by repetitive DNA sequences 5′ to the C region genes and occurs when these switch regions recombine. The intervening DNA is cut out and the resulting DNA in the B cell which has class switched, and in plasma cells derived from this B cell, no longer contains Cμ, Cδ or other intervening H-chain C region genes. A primary transcript is made and the RNA between the VDJ coding region and the new H-chain coding region is spliced out to give an mRNA for the new H-chain.

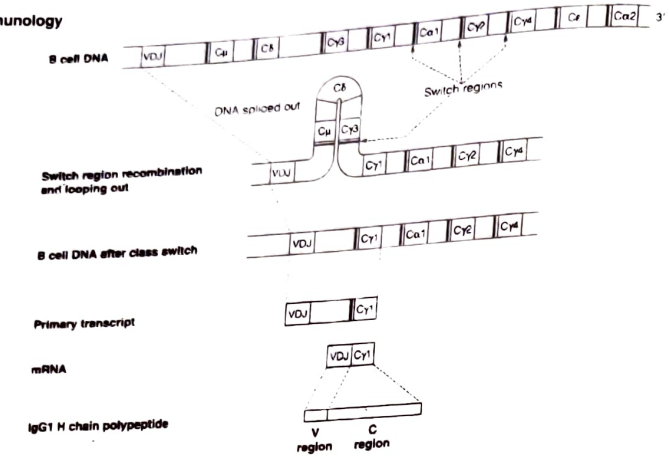


Fig. Class switching.

Ways of creating diversity

Ig diversity (the generation of antibodies with different specificities) is created by several antigen independent mechanisms. In addition, in B cells that have been stimulated by antigen and received T cell help, Ig genes undergo mutational events that may increase the affinity of the antibody produced by the B cell Overall, diversity is generated by:

Antigen independent events

- at the DNA level as a result of multiple germ line V, D and J and V and J gen.
- at the DNA level as a result of random combination of V, D and J segments C, V and J segments,
- at the DNA level as a result of imprecise joining of V, D and J segments,
- at the protein level as a result of random selection and pairing of different combinations of L- and H-chain V regions in different B cells.

Antigen dependent events

- at the DNA level as a result of somatic mutation in the V region, which may create higher affinity antibody binding sites.

Although rearrangement of the gene segments that will make up the V region genes occurs in an ordered fashion, they are chosen at random in each B cell. As these events occur in a vast number of cells, the result is that millions of B cells, each with a different antigen specificity, are generated. Additional diversity is created during recombination of V and J (L-chain) and V, D and J (H-chain) gene segments due to imprecise joining of the different gene segments making up the V region. That is for example, although translocation of a V gene segment to a J gene segment could occur with all three nucleotides of the last codon of the V segment joining with all three nucleotides of the first codon of the J segment, it is also possible that one or two nucleotides at the 3' end of the V segment could replace the first one or two nucleotides of the J segment. Such a difference in the position at which recombination occurs can change the amino acid sequence in the antigen binding area of the resulting V region of the antibody, and thus change its specificity.

Furthermore, after antigen stimulation of the B cell, the DNA of its L- and H-chain V regions becomes particularly susceptible to somatic mutation. This results in changes in the nucleotides of the DNA and thus corresponding changes in the amino acid sequence of the V regions of the antibody expressed by the B cell. As a result, the B cell may have a different specificity and not bind to or be stimulated by the original antigen. However, it often happens that at least some mutations result in amino acid changes which increase the tightness of binding of the antibody on the B cell to its antigen. These B cells will compete more efficiently for antigen than the original B cell, and will differentiate into plasma cells producing a higher affinity antibody **(affinity maturation)**.

Diversity is also generated as a result of the fact that any L-chain can interact with any H-chain to create a unique binding site. Thus, for example, an L-chain with a particular VJ combination for its binding site could be produced by many different B cells and interact with the different H-chains (i.e. different in their VH region) generated in each of these B cells to create many different specificities.

In sum, almost unlimited diversity is created from a limited number of V regions. The diversity almost certainly exceeds the amount of diversity needed to bind the immunogens of microbes. Moreover, the vast majority of the different B cells generated will never encounter antigen to which they can bind, and thus will not be stimulated to further development. And yet, such apparent wastefulness is justified by the fact that this mechanism of creation of diversity ensures that there are B cells and thus antibodies reactive with virtually all antigens that will be encountered. When an antigen to which this antibody binds is encountered, the B cell is triggered to divide and to give rise to a clone of cells, each one of which makes, at least initially, the originally displayed antibody molecule.

Antigen selection of B cells for stimulation

During an antibody response to an antigen, the overall affinity of the antibodies produced increases with time. This is partly because, as the B cell numbers increase during an immune response, antigen becomes limiting and there is competition for antigen among B cells. Those expressing higher affinity antibody compete more successfully for antigen and clonally expand, contributing a higher proportion to the antibody pool. Thus, the affinity of antibody produced in the secondary immune response is higher and thus more efficient at effector functions than that produced in the primary response.

In addition and of particular importance, B cells already stimulated by antigen and T cells undergo somatic mutation in the V regions of their H and L genes, which results in affinity maturation the generation of B cells expressing antibody with higher affinity for the antigen. These B cells contribute significantly to the pool of antibodies in the circulation. Typical antibodies have binding constants of $10^{6-7}\ M^{-1}$. After successive immunization with limiting antigen they are usually $10^{8-9}\ M^{-1}$ but may be as high as $10^{12}\ M^{-1}$.

Cytokines

Cytokines are small molecules, secreted by cells in response to a stimulus. They may have an effect on the cell that produces them and are critical to signaling between cells, with each cytokine often inducing several different biological effects. Many different cells release cytokines, but each cell type releases only certain of these molecules. As a group, cytokines induce growth, differentiation, chemotaxis, activation, and/or enhanced cytotoxicity. Moreover, it is not uncommon for some cytokines to have similar activities and for many cytokines, some with opposing activities, to be released by a particular stimulus. Thus, the resulting biological effect is a factor of the sum of all of these activities.

Cytokine nomenclature

To some extent cytokines can be grouped by the cell populations that secrete them. **Monokines** are cytokines secreted by cells of the myeloid series (monocytes, macrophages) and **lymphokines** are cytokines secreted primarily by lymphocytes, although some cytokines are produced by both lymphocytes and myeloid cells. The term **interleukin** (IL) is often used to describe cytokines produced by leukocytes although some cytokines labeled as IL are also produced by other cell populations. A group of small heparin binding cytokines, **chemokines,** has recently been recognized which direct cell migration, and may also activate cells in response to infectious agents or tissue damage. **Interferons** are produced by a variety of cells in response to viral infection. In addition, a variety of other less easily categorized cytokines exist which have critical immunologic activities.

Interferons

These molecules can be divided into two groups, type I IFNs (IFNα and IFNβ) anc type II IFN (IFNγ) also called immune IFN. IFNα and IFNβ are produced in response to viral infection (and probably to double-stranded viral RNA) by many different cells. They inhibit viral replication in uninfected cells, inhibit cell proliferation, increase the lytic activity of NK cells and modulate the cellular expression of MHC molecules. The receptor for both interferon α and interferon β is the same and found on most nucleated cells. Binding of interferons α or β to this receptor inhibits viral replication in that cell as a result of blocking translation of viral proteins and induction of the synthesis of inhibitory proteins. In addition these interferons induce increased expression of MHC class I and other components of the class I processing and presentation pathway which leads to induction of antigen specific $CD8^+$ CTL responses against virally infected cells. Induction of MHC class I is also important for protection of uninfected cells from killing by NK cells.

Lymphokines

A variety of cytokines are produced by lymphocytes and lymphocyte subsets many of which are growth factors for lymphocytes or which influence the nature of the immune response. IL-2 is made by T cells as a critical autocrine growth factor that is required for proliferation of T cells, especially Th0 and Th1 cells and CTL. On activation of these T cells as a result of the interaction. of their antigen receptor complexes with antigenic peptide in MHC molecules on APCs, the T cells make IL-2 for secretion and at the same time IL-2 receptors with which to bind and be stimulated by the released IL-2. In the absence of IL-2 and/or its receptor, many antigen specific T cells do not expand, severely compromising immune responses.

IL-3 is a cytokine that appears to be involved in the growth and differentiaton of a variety of cell types as a result of its synergistic activity with other cytokines in hematopoiesis. IL-4 is produced by Th2 cells and mast cells. It is a growth and differentiation factor for T (primarily Th2 cells) and B cells, and in particular for B cell class switch to the production of IgE antibodies. Closely related to this activity, IL-4 has an important role in influencing the nature of the immune response as it can induce the development of Th2 cells from Th0 cells and can inhibit the development of Th1 responses. Thus, IL-4 not only is involved in B cell growth, but it can also influence the B cell and its subsequent plasma cells to produce IgE antibody. Like IL-4, IL-5 is also produced by Th2 cells and mast cells and is important to B cell activation and production of IgA antibody. It also has a role in eosinophil growth and differentiation. IL-10, which is produced by Th2 cells and MØ, is involved in B cell activation. It induces Th2 responses and inhibits Th1 responses, perhaps by enhancing IL-4 production and/or by suppressing MØ activity and production of IL-12, a Thl-stimulatory cytokine.

Monokines

This group of cytokines has many different local and systemic activities that are critical to immune defense. In addition, these molecules, also called pro-inflammatory cytokines, are important mediators of inflammation. In particular, as a result of an appropriate stimulus, including ingestion of gram negative bacteria and subsequent activation by LPS, MØ secrete IL-1, IL-6, IL-8, IL-12 and TNFα. IL-1, TNFα and IL-6 have activities which include: *(a)* increasing body temperature and lymphocyte activation, which decrease pathogen replication and increase specific immune responses; *(b)* mobilization of neutrophils for phagocytosis; *(c)* induction of release of acute phase proteins (CRP, MBP) and thus complement activation and opsonization.

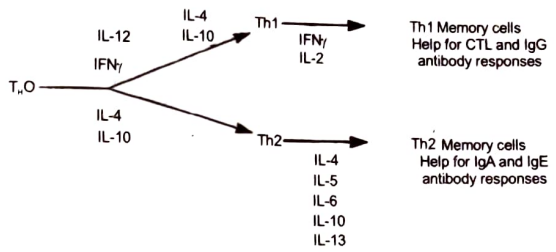
Table : Representative lymphokines and monokines

Cytokine	Produced by	Activity
IL-1	MØ, epithelial cells	Activates vascular endothelium; tissue destruction; increased effector cell access: fever; lymphocyte activation; mobilization of PMNs; induction of acute phase proteins (CRP, MBP)
IL-2	T cells	Proliferation of T and NK cells
IL-3	T cells, thymic cells	Proliferation, differentiation of hemat-opoietic cells
IL-4	Th2 cells, mast cells	B cell activation, proliferation, IgE response; induces Th2 and inhibits Th1 responses
IL-5	Th2 cells, mast cells	Eosinophil growth, differentiation; B cell activation, IgA response
IL-6	T cells, MØ	Lymphocyte activation; fever
IL-8	MØ, MØ, Fb, Kr	Increases tissue access for, and chemotaxis of PMNs
IL-10	Th2 celis, MØ	B cell activation; suppression of MØ activity; induces Th2 and inhibits Th1 responses
IL-12	B cells MØ	Induces Th1 and inhibits Th2 responses; activates NK cells
IFN-γ	T cells, NK cells	MØ and PMN activation; induces Th1 and inhibits Th2 responses
TNFα	MØ, T cells	Activates vascular endothelium; fever; shock; increases vascular permeability; mobilization of metabolites

Monocytes (Mo), macrophages (MØ), endothelial cells (En), fibroblasts (Fb), keratinocytes (Kr), neutrophils (PMNs), chondrocytes (Co)

Table: Cytokine effects on Th1 and Th2 immune responses

Enhance	Inhibit	Cytokines produced



IL-1 also activates vascular endothelium (in preparation for neutrophil chemo taxis) and induces systemic production of IL-6. IL-8 increases access for, and chemotaxis of, neutrophils. It also activates binding by integrins, which facilitates neutrophil binding to endothelial cells and migration into tissues. Like IL-1, TNFα also activates vascular endothelium and is able to increase vascular permeability. It activates MØ and induces their production of nitric oxide (NO). Although produced by monocytes and MØ, TNFα is also produced by some T cells. Finally, IL-12, which is also produced by B cells, activates NK cells which then produce IFNγ, a cytokine important to inducing differentiation of Th0 cells to Thl cells (Table).

Chemokines

This group of more than 50 small, closely related cytokines (mol. wt 8-10 kDa) are primarily involved in chemoattraction of leukocytes (lymphocytes, monocytes and neutrophils) (Table). They are made by monocytes/macrophages, but also by other cells including endothelial cells, platelets, neutrophils, T cells, keratinocytes and fibroblasts. Chemokines can be divided into four different groups based on unique aspects of their amino acid sequence, and in particular the position of conserved cysteine residues. One group has two adjacent cysteines (CC), a second has two cysteines separated by another amino acid (CXC), another has one cysteine, and the last has two cysteines separated by three other amino acids. For the most part, CC chemokines such as monocyte chemotactic protein (MCP-1) are chemotactic for monocytes, inducing them to migrate into tissues and become macrophages, whereas CXC chemokines such as IL-8 are chemotactic for neutrophils inducing them to leave the blood and migrate into tissues. Some of these chemokines also are chemotactic for T cells. Chemokines can be produced by many cells in response to an infectious process or to physical damage. They not only direct a cell to the source of infection /damage, but may also activate the cell for dealing with the infectious agent or damage.

Table. Representative chemokinte

Class1	Name	Source	Chemoattractant for	Activation of
CXC (α)	IL-8	Mo, MØ, Fb, Kr	Naive T cells, PMNs	PMNs
	NAP-2	Platelets	Neutrophils	
	MIP-1β	Mo, MØ, En, PMNs	CD8 T cells	
CC(β)	MCP-1	Mo, MØ. Fb, Kr	Memory T cells, Mo	Monocytes
	Rantes	T cells	Memory Th cells	
C (γ)	Lymphotactin		Lymphocytes	
CX_3C (δ)	Fractalkine		Lymphocytes, monocytes, NK cells	

Receptors for chemokines are all integral membrane proteins with the characteristic feature that they span the membrane seven times (seven transmembrane glycoproteins). These molecules are coupled to G (guanine nucleoside binding) proteins which act as the signaling moiety of the receptor. Although most of these receptors can bind more than one type of chemokine, they are usually distributed only on particular cell populations, permitting different chemokines to have selective activity.

Some of these chemokines, for example IL-8 and MCP-1, have been shown to work by first binding to proteoglycan molecules on endothelial cells or on the extracellular matrix. On this solid surface they then bind blood neutrophils or monocytes, slowing their passage and directing them to migrate down a chemokine concentration gradient toward the source of the chemokine. Although the role that each plays in immune defense and pathology is still being clarified, it is evident that these molecules are potent agents for activating and directing effector cell populations to the site of tissue damage as well as for controlling leukocyte migration in tissues.

Other Cytokines

Of the many other cytokines which are important to immune defense, several are particularly noteworthy (*Table*). A group of CSFs, including granulocyte-monocyte CSF (GM-CSF), granulocyte CSF (G-CSF) and monocyte CSF (M-CSF) are cytokines which drive the development, differentiation and expansion of cells of the myeloid series. GM-CSF induces expansion of myeloid progenitor cells and their commitment to the monocyte/MØ and granulocyte lineage, after which G-CSF and M-CSF induce specific commitment to the granulocyte or monocyte lineage, respectively, and then their subsequent expansion. These factors, and especially G-CSF, are important clinical tools in a number of disease situations as they are used to expand myeloid effector cell populations critical to defense against pathogens (Topic G3).

TGFβ is produced by a variety of cells including monocytes, MØ, T cells and chondrocytes, and plays an important role in suppressing immune responses, as it can inhibit activation of MØ and growth of B and T cells. TNFβ (lymphotoxin) is a molecule which is cytotoxic to a variety of cell types, including ineffectual chronically infected MØ. The released pathogen is then killed by less compromised MØ.

Table. Other cytokines

Cytokine	Produced by	Activity
GM-CSF	MØ, T cells	Stimulates growth, differentiation & activation of granulocytes, Mo, MØ
G-CSF	Mo, Fb, En	Stimulates PMN development
M-CSF	Fb	Stimulates Mo, MØ development
TGF-β	Mb, T cells, Co	inhibits cell growth and inflammation
TNFβ (lymphotoxin)	T cells	cytotoxic to T, B and other cells

Cytokine treatment

Several cytokines which enhance immune responses, including IL-2, IFNγ and IFNα have been approved for treatment of certain tumors. .G-CSF is extremely useful in treating cancer patients with low PMN counts resulting from chemotherapy or irradiation.

Table. Cytokines and their inhibitors as tools in the clinic.

Cytokine treatment	Patient group
For infection	
IFNα	Some infections
IFNγ	Treatment of patients with CGD
G-CSF	Treatment of patients with low granulocyte counts (*e.g.* resulting from chemotherapy or irradiation)
For tumors	
IL-2	Renal cell carcinoma (20% of patients have partial responses)has been used
IFNα	together with tumor infiltrating lymphocytes for some tumors
Cytokine inhibitors (anti-inflammatory)	Hairy cell leukemia
Anti-TNFα	Some success in treating RA
Soluble cytokine receptors e.g. IL1, TNFα	Some success in treating RA

Cytokine receptor targeting

Antibodies to cytokine receptors or soluble cytokine receptors themselves used to treat autoimmune diseases where pro-inflammatory cytokines are involved in maintenance of chronic inflammation e.g. TNFα in RA.

Molecules of The Innate Immunity

Complement System

The complement system (C.S.) is the primary numeral (plasma) mediator of antigen (Ag) and antibody (Ab) reaction. The system consist of 20 chemically and immunologically distinct plasma proteins found in blood plasma along with blood clotting fibrinolysis and kinin formation, complement forms one of the triggered enzyme system. Synthesized by hepatocytes and monocytes, these proteins .help (complement) antibody responses and have a wide spectrum of activities including a pivotal role in innate defense mechanism. Complement may be activated by either the classical pathway (through antibody) or the alternative pathway (innate). The two pathways converge at a point from which a final common pathway ensues, the terminate at the end point of complement activation, which is cytolysis or cytotoxicity. There are other non complement enzymes of serum or cellular origin which can activate the complement system in midway, e.g. trypsin like enzymes such as plasmin. The most abundant and pivotal component of complement system is C3. This is present at a concentration of 1.2 mg/ml in plasma and has mol.wt of 195 kD.

Classical C.P.

C.C.P. is activated by antigen antibody binding. Among the IgG subclasses IgG3 has been known to become most active, followed by IgG1 and IgG2. IgM is the most efficient immunoglobulin activator of the C.C.P. Activation occurs by binding of first complement Cl to a site located in CH2 region of immunoglobulin.

The formation of an Ag-Ab complex induces conformational changes in the F_c portion of IgM molecule that expose a binding site for the C1 component of the C.S. C1 in serum is a complex consisting of C1q and two molecules each of the C1r and C1s, held together in a complex ($C1qr_2s_2$) stabilized by Ca^{2+} ions. C1 is normally present as a complete entity. Individual proteins of Cl are found only in the diseased state. Clq is unique as its structure is similar to collagen. C1q consists of 18 polypeptide chains of 3 distinct type that are organized into structure consisting of 6 globular proteins connected by fibrinular strand to a central structure. C1 is able to bind 6 IgG molecules. Once C1q binds to Ig molecule C1r, α, β globulin acquires the ability to enzymatically activate C1s. C1s once activated acquires proteolytic activity, and the initial phase of complement pathway is activated. The earlier reactants including Ag, Ab and C1q or C1r are not necessary for the progression of complement reaction. Activated C1s mediates the next phase of complement cascade. C1s cleaves C4 yielding C4a and C4b. It also cleaves C2, initiate the formation of the key enzyme C4b2a on the activator surface. C4b contains a labile binding site which binds with activation in a transient manner. C2a also has a labile binding site which allows it to bind to C4b. Mg^{2+} ions required for the formation of C4b2a complex. Activated C4b2a is a proteolytic enzyme that assumes the role of continuing the classical complement reaction. Once it has been formed, earlier reactant are no longer required. C4b2a is termed as C3 convertase thus activates C3. The fragment from C4 cleavage, C4a, is an anaphylatonin, or mediator of inflammation.

Cleavage of C3 results in formation of C3a and C3b. C3a is biologically coating peptide, the attachment of C3b to membrane in the incinity of C4b2a molecule lead to generation of last enzyme of classical pathway C4b2a3b. Called C5 convertase. The C3b component of this complex binds C5 and alters its conformation, so that the C4b2a component can leave C5 into C5a which diffuses away, and C5b which attaches to C6 and initiates formation of the membrane attack complex described later.

Alternative Pathway

The alternate pathway was originally described as properdin system, consist of a group proteins involved in resistance to infection. There were thought to be similar to but distinct from complement. The properdin system was found to be involved in destruction of certain bacteria, neutralizing of virus and lysis of erythrocytes. This system does not seem to require specific Ab. Several factors involved in the system has now been isolated and identified. These includes properdin, the γ_2 globulin, factor α has been identified as C3. Factor B, a β_2 protein is similar to but distinct from C2. Factor D and α globulin and Factor I and H β globulin. The properdin system is now known as alternative C.P., becaus of its major role in resistance to infection it is considered to be the

pathway of most, important to man. Initial requirement for activation in the presence of C3b, is continuously generated. This occurs following cleavage of the trioester bond in C3 thus forming C3* (active) which reacts with factor B and D to generate an enzyme able to cleave C3a and C3b. It is possible that C3 in circulation is also cleaved by enzymes of coagulation system, the enzyme zymogen a yeast polysaccharide also cause C3 activation in the presence of Mg^{2+} ions. The most of newly generated C3b remains in the fluid phase, some binds to cellular surfaces. In either cases this C3b is rapidly inactivated by controlled proteins factor I and H. This surface bound, protected C3b interact with factor B in presence of Mg^{2+} ions to form C3bB with the addition of factor D, which acts on bound factor B to cleave it yielding Bb, and enzyme C3bBb is generated. This surface bound enzyme is also called the amplifying C3 convertase and is able to cleave very large amount of C3 this results in accumulation of C3b which undergoes lyclic amplification of critical C3b component. This is also called as a positive feeback mechanism which enhances C3b formation.

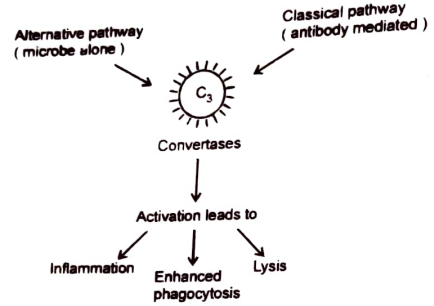


Fig. Complement system

Lectin Pathway:

It orginates with host proteins binding microbial surfaces. Lectin are proteins that recognize and bind to specific carbohydrates. The lactin pathway like alternative pathway does not depend on antibody for its activation. However, the mechanism is more like the classical pathway, because after initiation it proceeds, through the action of C4 and C2, to produce a C5 convertase.

Three Complement Pathways Converge at the Membrane Attack Complex :

The terminal sequence of complement activation involves C5b, C6, C7, C8 and C9, which interact sequentially to form a macromolecular structure called the membrane attack complex (MAC). This complex forms a large channel through the membrane of the target cell, enabling ions and small molecules to diffuse freely across the membrane.

Functions : The major function of complement system are :

(*i*) Initiation of inflammation as a result of the stimulation of degranulation of most cells by C3a and C5a (anaphylectoxins).

(*ii*) Attraction, by C5a, of neutrophils to the site of microbial attack fchemotoxis). (Hi)

(*iii*) Enhancement of the attachment of the microbe to the phagocyte - PMNs attracted to a site of complement activation by C5a bind to C3b enhancing their internalization of the microbe (opsonization).

(*iv*) Killing of the microbe by the 'membrane attack complex' which through C9 produces 'pores' in the target cell membrane leading to lysis of the microbe.

Regulation : Complement components rapidly lose binding capacity after activation, limiting their membrane damaging ability to the immediate vicinity of the activation site. The complement system is also tightly regulated by inhibitor / regulatory proteins which inclue Cl inhibitor, Factor I, C4b binding protein, Factor H, decay accelerating factor (DAF), membrane cofactor protein (MCP) and CD59 (protecting These molecules protect host cells from destruction or damage at different stages of the complement cascade.

Acute Phase Proteins

These are a heterogeneous group of plasma proteins important in innate defense against microbes and in limiting tissue damage caused by infection, trauma, malignancy and other diseases. They include C reactive protein (CRP), serum amyloid protein A (SAA) and mannose binding protein (MBP).

Acute phase proteins are mainly produced in the liver, usually as the result of a microbial stimulus. Production is at a low level but on stimulation increases rapidly. These proteins are produced in response to the cytokines IL-1, IL-6, TNFα, IFNγ that are released by activated macrophages and NK cells.

Interferons

Interferons (IFNs) are proteins involved in protection against viral infections. The two kinds of interferons type I and type II, have different cellular origins and mediate a range of different activities. These molecules interfere with viral replication but also are signaling molecules between cells.

Type I: IFNs consist of IFN-alpha (IFNα) and beta (IFNβ) are produced by many different cells in the body and interfere with viral replication. Both IFNα and IFNβ inhibit protein synthesis in virally infected cells by preventing mRNAs translation and DNA replication. IFNα and β form the part of innate immune response.

Type II : immune interferon IFNγ (gamma), has no structural homology with type I is mainly produced by the Thl (T-helper 1) cells and NK cells and functions to regulate Thl responses, increase phagocytic mechanisms and increase antigen presentation.

Table : Interferons

	IFN-α/β	IFN-γ
Chromosomal location	9	12
Origin	All nucleated cell, especially fibroblasts, macrophages and dendritic cells	NK cells and Thl, γ S and CD 8 T cells
Induced by	Viruses, cytokines some intracellular bacteria and protozoans	Antigen stimulated T cells
Functions	Antiviral, increases MHC class I expression, inhibites cells proliferation	Antiviral, increases MHC I and II expression, activates macrophages

Hypersensitive Reactions

An immune response mobilizes a battery of effector molecules that act to remove antigen by various mechanisms. Generally, these effector molecules induce a localized inflammatory response that eliminates antigen without extensively damaging the host's tissue. Under certain circumstances, however, this inflammatory response can have deleterious effects, resulting in significant tissue damage or even death. This inappropriate immune response is termed **hypersensitivity** or **allergy**.

Classification

Hypersensitivity reactions occur at different times after coming into contact with the offending antigens, within a few minutes (i e. immediate), minute to hours (intermediate) or after many hours (delayed). Generally, the delayed responses are mediated by the cellular components of the immune system (i.e. T cells) whilst the former are the result of the humoral arm of the immune response which incldces antibodies and the complement system. The original classification by Gell and Coombs was into four main types, a fifth has since been added. *Table1* summarizes the main immune system components which contribute to tissue damage. It should be stressed that more than one of these mechanisms can contribute to any one particular disease process.

Table 1. Classification of hypersensitivities

Time of appearance	Type	Immune mechanism
2-30 min (immediate)	I	IgE antibodies (enhancement of acute inflammatory response)
5-8 h (cytotoxic)	II	Antibody and complement
2-8 h (immune complex)	III	Antibody/antigen complexes
24-72 h (delayed)	IV	T cell mediated (can be granulomatous)
(Stimulatory)	V	Antibody mediated

IgG-mediated Type I Hypersensitivity : Allergy

This most common type of hypersensitivity is mediated by IgE and causes mild (hayfever) to life threatening (bee sting) clinical situations. Some individuals (atopic) have a genetic predisposition to make high levels of IgE. Skin tests can be used to test for sensitivity to allergens.

Sensitization Phase

Sensitization to a particular antigen is dependent on stimulation of IgE antibody production. This requires CD4+ Th2 cells to induce class switching of antigen specific B cells and to secrete IL-4 for B cell growth and differentiation.

Effector phase-IgE -mediated mast cell degranulation

IgE antibodies produced following initial contact with the specific antigen, bind to IgE receptors on mast cells and basophils. Cross-linking by antigen of the IgE and the receptors with which it is associated results in rapid degranulation and release of pharmacological mediators (e.g. histamine) causing local inflammation (anaphylaxis). In the case of systemic release, systemic anaphylactic reactions are produced requiring treatment with adrenaline (epinephrine) to restore blood pressure.

Common antigens causing type I Hypersensitivity

These include grass and tree pollens, insect venoms, nuts, drugs and animal dander. Fungal and worm antigens also induce this type of hypersensitivity.

Drugs and immunotherapy (desensitization)

Drugs used to counteract Type I hypersensitivity inhibit production or release of inflammatory mediators (nonsteroidal anti-inflammatory drugs (NSAIDs), such as aspirin and indomethacin, glucocorticoids and cromolyn) or Inhibit the action of inflammatory mediators which then relieve symptoms (benadryl, dramamine, glucocorticoids). Epinephrine is used to counteract mediator effects such as low blood pressure and bronchospasm. The aim of desensitization is to induce an IgG immune response and/or divert the immune response away from production of IgE. This approach has been used successfully for only a few allergens (e.g. bee venom).

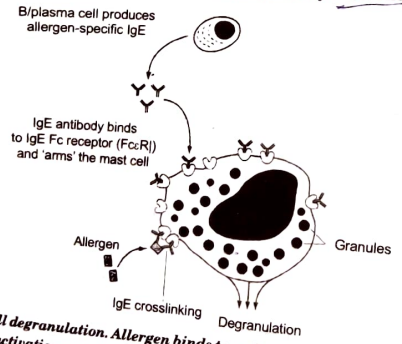


Fig.IgE-mediated mast cell degranulation. Allergen binds to and crosslinks cytophiltc (cell oound) IgE, signaling FceH to trigger mast cell activation and degranulation with release of histamine, leukotrienes, etc.

IgG and Igm-mediated Type II Hypersensitivity

Antibody (IgM or IgG) directed mainly to cellular antigens (e.g. on erythrocytes) or surface autoantigens can cause damage through opsonization, lysis or antibody dependent cellular cytotoxicity. Also called cytotoxic hypersensitivity.

Rhesus incompatibility

Pregnant mothers who are rhesus D (RhD) antigen positive can respond to RhD antigen inherited from the father. Sensitization occurs either through prior blood transfusion with RhD+ erythrocytes or at parturition when fetal erythrocytes pass into the maternal circulation. During subsequent pregnancies, small numbers of fetal erythrocytes stimulate a memory response with the result that IgG antibodies to RhD antigen pass across the placenta and destroy the fetal erythrocytes (hemolytic disease of the new-born).

Transfusion reactions

Natural antibodies (isohemagglutinins) to major blood group antigens (A, B) bind to transfused erythrocytes carrying the target antigens resulting in massive hemolysis. This is now rare due to blood group typing.

Autoantigens

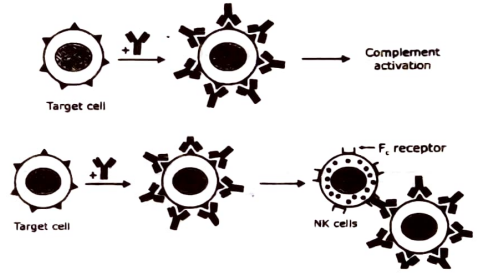
Antibodies to a variety of self antigens such as basement membranes of lung and kidney (Goodpasture's syndrome), the acetylcholine receptor (Myasthenia gravis) and erythrocytes (hemolytic anemia) can result in tissue damaging reactions.

Drugs

Drugs such as penicillin can attach to erythrocytes and cause IgG-mediated damage to erythrocytes.

Stimulatory hypersensitivity

A variant of type II hypersensitivity (sometimes called type V), this results in binding to a receptor and acting as the natural ligand e.g., in Graves disease antibodies are present which react with the thyroid stimulating receptor, stimulating hyperthy



Immune-complex Mediated Type III Hypersensitivity

Immune complexes can form to serum products as well as microbial and self antigens, either in local sites or systemically, leading to phagocytic and complement mediated damage.

Mechanisms of type III hypersensitivity

Tissue damage is caused mainly by complement activation and release of lytic enzymes from neutrophils. Local damage (Arthus reaction) can be seen in pulmonary disease resulting from inhaled antigen. Systemic antibody complexes with microbial or autoantigens result in immune complex deposition in blood vessels (vasculitis) or in the renal vessels (glomeruli) of the kidneys leading to glomerulonephritis.

Diseases associated with type III hypersensitivity

Pulmonary diseases result from inhalation of bacterial spores (Farmer's lung) or avian serum/fecal proteins (bird fancier's disease). Systemic disease can occur from streptococcal infections (streptococcal nephritis), autoimmune complexes (e.g. systemic lupus erythematosus (SLE)) or drugs (e.g. penicillin) or antisera made in animals.

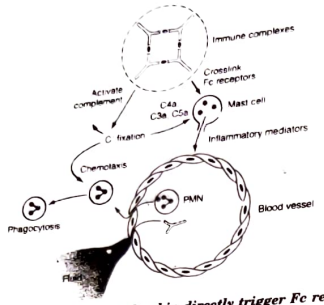


Fig. The Arthus reaction. Small immune complexes in the skin directly trigger Fc receptors and activate complement resulting essentially in an acute inflammatory response mediated through mast cells. Small immune complexes can also lodge in blood vessels and induce vasculitis or glomerulonephritis in the kidney.

Delayed-Type Type IV Hypersensitivity

This occurs from 24 h after contact with an antigen and is mediated by T cells together with dendritic cells, macrophages and cytokines. The persistent presence of the antigen *e.g.* chronic mycobacterial infections, results in granulomas. Skin contact with a number of small molecules (chemicals and plant molecules) can also result in delayed hypersensitivity.

The tuberculin reaction

This is a 'recall' response to purified mycobacterial antigens and is used as the basis of a skin test for an immune response (not necessarily curative) to TB.

The production of granulomas

The inability to kill all mycobacteria in macrophages by T cells often results in a chronic stimulation of the mycobacterial specific T-cells. The cytokines produced are responsible for 'walling off' the macrophages containing the persistent antigens and thus the production of granulomas. This also occurs as a response to shistosomula worms and is seen in some clinical conditions with, as yet, undefined antigens.

Contact sensitivity

Contact with a number of small molecular weight chemicals (e.g. nickel in a watch strap buckle) and molecules from some plants (poison ivy) can penetrate trie skin, bind to self proteins and induce a specific CD4+ T cell response. The resulting cytokines induce a local redness and swelling which usually disappears on removal of the antigen.

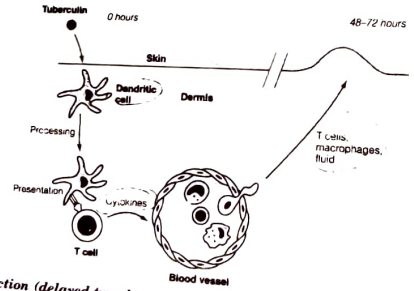


Fig. 1. The tuberculinr reaction (delayed-type hypersensitivity). Tuberculin protein introduced into the dermis is processed and presented by dendritic cells to T cells via MHC class II molecules. Cytokines produced by the T cells alter local endothelial cell adhesion molecules allowing monocytes to enter the site of injection and develop into macrophages. T cells and macrophage products result in edema (fluid) and swelling. A positive skin test shows up as a firm red swelling which is maximal at 48-72 h after injection.

Autoimmunity

Human beings suffer from several autoimmune diseases, Some of these diseases are caused by autoantibodies, while others are due to T cells that cause tissue destruction. Autoimmune diseases are either organ-specific or systemic. In case of **organ-specific diseases**, the immune response is directed to a target antigen unique to a single gland or organ. As a result, the manifestations are largely limited to that organ; the manifestation may be tissue damage or blockage/overstimulation of the normal function of the organ. Direct cellular damage occurs when lymphocytes or antibodies bind to cell-membrane antigens causing cellular lysis and/or an inflammatory response in the affected organ. Gradually, the damaged cellular structure is replaced by connective tissue leading to a decrease in function of that organ. Examples of such diseases are autoimmune anaemias, insulin-dependent diabetes mellitus, etc. In some diseases, *e.g.*, Grave's disease, the antibodies bind to hormone receptors and activate them, while in some others, *e.g.*, myasthenia gravis, antibody binding blocks the activation of the receptors.

In case of systemic autoimmune diseases, the immune response is directed towards a broad range of target antigens and, as a result, involves a number of organs and tissues. These diseases reflect a general defect in immune regulation that results in hyperactive T cells and B cells. Tissue damage is widespread, is caused by both T cells, e.g., multiple sclerosis and autoantibodies, e.g., systemic lupus erythematus, or by accumulation of immune complexes, e.g., rheumatoid arthritis.

Causes of autoimmune disease.

Autoimmune diseases could arise in many possible ways, some of which are as follows.

(1) A tissue antigen that is sequestered from circulation will not be employed in the thymus for clonal deletion of T cells reactive to this antigen. When such an antigen is released into circulation due to trauma to tissues caused by an accident or a viral or bacterial infection, it may induce autoantibody formation.

(2) A number of bacteria and viruses have antigenic determinants that are identical or similar to normal host cell components. Infection by such pathogens may initiate autoimmunity.

(3) A number of viruses and bacteria can induce nonspecific polyclonal B cells that produce IgM in the absence of T_H cells. Such infections could activate B cells reactive to self-antigens and, thereby, cause autoimmunity.

(4) Expression of MHC II in cells that normally do not express them could lead to autoimmunity. Such an inappropriate expression of MHC II is induced by certain agents like IFN-γ.

Table Some autoimmune diseases of human beings

Autoimmune disease	Self-antigen	Immune response
	Organ-specific autoimmune diseases	
Autoimmune haemolytic anaemia	RBC membrane proteins	Autoantibodies
Grave's disease	Thyroid-stimulating hormone receptor	Autoantibody (stimulation of the receptor)
Insulin-dependent diabetes mellitus	Pancreatic beta cells	T_{DTH} cells*, autoantibody
Myasthenia gravis	Acetylcholine receptors	Autoantibody (blockage of the receptor)
Spontaneous infertility	Sperm	Autoantibodies
	Systemic autoimmune diseases	
Multiple sclerosis	Brain	T_{DTH} and T_c cells, autoantibodies
Rheumatoid arthritis	Connective tissue,IgG	Autoantibodies, immune complexes
Scleroderma	Nuclei, heart, lungs, kidney, gastrointestinal tract	Autoantibodies
Systemic lupus erythematus	DNA, nuclear protein, and platelet membranes	RBC Autoantibodies, immune complexes

*T_{DTH} = *T cells involved in delayed type hypersensitivity.*

Treatment of Autoimmune Diseases

The current therapies provide relief by nonspecific suppression of the immune system. Some agents like cyclosporin A are somewhat selective in that they inhibit only antigen-activated T cells. More directed approaches are still at experimental level; some of these are briefly summarise below.

1. T cells specific for the concerned antigen are injected into individuals to immunize them against these T cells (***T cell vaccination***).
2. A synthetic peptide differing by only one (or few) amino acid from the autoantigenic peptide may be used as therapy. The synthetic peptide competes with the autoantigenic peptide for MHC molecules (***peptide blockade of MHC molecules***).
3. ***Monoclonal antibodies*** may be directed against the following: (1) CD4 (depletes all T_H cells), (2) the α subunit of IL-2 receptor, which is expressed only by antigen-activated T_H cells (blocks antigen-activated T_H cells), (3) specific T cell receptor, and (4) the specific allelic variant of the MHC molecule that is associated with the autoimmune disease.
4. Oral administration of the concerned autoantigen way induce tolerance to the antigen.

Transplantation Immunology

Introduction

Transplantation is the process of taking tissues or organs (or even cells) and placing them into the same or different individual. The tissues, organs or cells that are transferred from one individual to another are called **grafts.** The individual who donates the graft is called **donor** and individual who receives the graft is referred to as **recipient**. Clinical transplantation is usually performed to overcome a functional or anatomical deficit in an individual. The transplantation of kidney, heart, liver, cornea, lungs, pancreas and bone marrow is now performed worldwide.

Autologous graft (autograft): These are graft transplants from one region to another on the same individual.

Syngenic graft (syngraft)/Isograft: Isograft involves the transfer of graft between *genetically identical* (syngenic) individuals of the *same species*. This is possible between genetically identical twins or genetically identical mice.

Allogeneic graft (or allograft): A graft transplanted between two genetically non-identical individuals of the *same species* is called allograft.

Xenogeneic graft (or xenograft): These are graft transplants between individuals of different species.

Vaccines

A vaccine can be defined as a preparation of bacterial, viral or other pathogenic agents or of their isolated antigens which is administered with the objective of stimulating a recipient's protective immunity. After primary exposure of antigen to immunocompetent lymphocytes, there occurs an initial but slightly delayed immune response called **primary immune response**. Primary immune response peaks on approximately the 14th day of antigen exposure. There are two main outcomes of primary response : first, specific immuno-competent cells (B and T cells) are activated; and second, and the more important one, memory cells are formed. Subsequent exposure to the same antigen stimulates these memory cells which results in a rapid and more heightened immune response. Secondary response usually occurs within two to three days. It must also be mentioned that a primary immune response elicits the formation of IgM antibodies while a secondary response protects the host with higher-affinity IgG antibodies. It is the rapidity with which secondary response occurs upon secondary exposure to antigen that protects the host against any potential threat by repeated exposure to pathogen.

Thus, a vaccine is basically an antigen or its component that can induce secondary or adap-tive immunity in the host. It aims to prevent severe complications of infections by reinforcing or broadening the defences by introducing immunological memory.

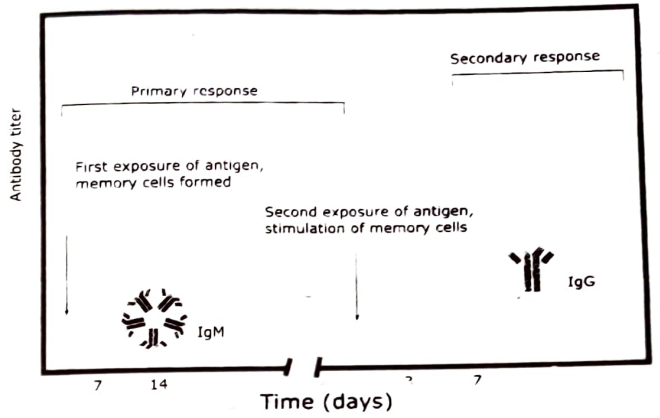


Fig. Primary and secondary immune responses.

Types of Vaccines

There are several types of vaccines that are currently and conventionally used. These include natural live vaccine, live attenuated vaccine, inactivated vaccine, toxoid vaccine, polysaccharide vaccine, recombinant antigen vaccine, live vector vaccine and a more recent DNA vaccine.

Table Types of common vaccines for humans.

Type	Examples of Vaccines used or Explored
Natural live vaccines	Cowpox (vaccinia) virus used for immunization against smallpox (historic importance, now redundant); simian rotavirus is used in vaccinations against gastroenteritis.
Live attenuated vaccines	BCG; Sabin polio vaccine; MMR*; varicella- zoster vaccine (for chickenpox), yellow fever vaccine
Inactivated vaccines	Salk polio vaccine, rabies vaccine, pertussis vaccine
Toxoid vaccines	Diptheria vaecine; tetanus vaccine
Polysaccharide vaccines	Hib[+] vaccine; Vaccine for pneumococcal pneumonia; meningococcal meningitis
Recombinant antigen vaccines	HBs Ag[^] vaccine
Viral vector vaccines	An HIV vaccine containing *env. gag* and, *pol* genes in canary pox virus - vector (in clinical trials); animal model is experimented with hepatitis B antigen; influenza antigens inserted in vaccinia virus vector
Bacterial vecto vaccines	Attenuated strain of *Salmonella typhimurium* (Ty2la) is explored as vector for cholera vaccine
DNA vaccines	Rabies vaccine; Influenza vaccine; HIV vaccine-are currently being explored

Not : *[*Mumps-Measels-Rubella Vaccine; [+]Haemophilus influenza type b; [^]Recombinant hapatitis B V-irus*

Natural Live Vaccines

These preparations include natural non-pathogenic organisms, but which still induce specific immunily. The problems of these vaccines reside in their ability (albeit hidden) to mutate and convert into forms that could be pathogenic to human hosts.

Live Attenuated Vaccines

Attenuation (Latin: *attenuare*-to weaken) refers to the weakening of the pathogenic bacteria or virus by making it less virulent without altering its immunogenicity. Microorganisms are attenuated or weakened so that they do not cause any disease. Attenuation can be achieved by growing pathogenic microorganisms (bacteria or virus) for a long period of time in a foreign host such as embryonated eggs or tissue culture cells. The second and more famous attenuation was successfully carried out by two scientists Albert Calmette and I. Camille Guerin in 1921 (the first being carried out by Louis Pasteur himself in 1880 on bacteria of fowl cholera). Calmette and Guerin grew the bovine strain of *Mycobacterium tuberculosis* (called *M. bovis*) for several years on a medium containing increasing concentration of bile. This *in vitro* cell culture of pathogenic *M. bovis* changed it into a less virulent and more suitable form of bacilli known as BCG (Bacillus Calmette-Guerin) a commonly used vaccine against tuberculosis. Attenuated viruses are also used as vaccine, the vaccines of polio, yellow fever and measles viruses being the most successful.

The process of attenuation involves growing microbes (bacteria or viruses) under abnormal *in vitro* conditions, be it high bile concentration or passage through foreign cell/tissue such as embryonated eggs. These abnormal environmental conditions select those mutant cells that are able to survive and multiply under these conditions. These microbes are then harvested and used as vaccine.

Inactivated Vaccines

Another commonly used method for vaccine production is to inactivate the whole pathogen and then use it for vaccination. The inactivation could be achieved by modifying the antigen (pathogen) chemically by formaldehyde treatment or physically by heat treatment. The killed bacteria or viruses are then used in vaccines.

Toxoid Vaccines

The virulence or some pathogenic bacteria depends primarily on the production ot **exotoxin**. Exotoxins are cytotoxic microbial poisons (proteinic in nature) produced by living cells. Some bacterial pathogens such as diphtheria and tetanus bacilli produce exotoxins that induce several characteristic symptoms associated with these infections.

In other words, the harmful part of these bacteria is their exotoxin, so if these exotoxins can be neutralized by the body the disease will not occur. These exotoxins are isolated and chemically modified (usually with formaldehyde) so that their toxicity (but not immunogenicity) is lost. These non-toxic yet immunogenic derivatives of exotoxin, or **toxoids**, are commonly used in vaccines.

Polysaccharide Vaccines

The surface of bacteria is the first thing that is exposed to the immune system. Therefore, the surface antigens can serve as an excellent vaccine. Moreover, some bacteria have resistance to phagocytic invasion because these cells have capsular polysaccharides that resist phagocytosis (for example, the polysaccharide capsules of *S. pneumoniae, Haemophilus influenzae and Klebsiella pneumoniae*). The coating of these capsular polysaccharides with antibodies and/or complement components marks them for destruction by phagocytosis. These observations have paved the way for synthesizing vaccines against bacterial-purified polysaccharides.

Recombinant Antigen Vaccines

With the advent of recombinant DNA technology, virtually any gene-encoding immunogenic protein can be introduced and expressed in yeast, bacterial or even mammalian cells, using recombinant DNA technology. These cells are then cultured in the laboratory and the protein produced endogenously is harvested. The genes that are selected for making recombinant antigen vaccine are usually surface antigens (mostly glycoproteins). A number of genes coding for various surface antigens have been successfully cloned in bacterial, yeast and mammalian cells cultures. Yeasts have emerged as a better choice for making these surface antigens as they add and process carbohydrate molecules on the protein in their Golgi bodies in a manner more similar to mammals, and less care is needed in comparison with mammalian cells. One such vaccine approved for human use is hepatitis B vaccine.

Live Vector Vaccines

In live vector vaccines, the desired gene coding for target antigens of the virulent pathogen is put into a vector (attenuated bacteria or virus) and then this vector is infected (or administered orally) to the vaccinee. This vector slowly replicates inside the inoculated individual and it serves as a source of the antigen, delivering a large amount

of antigen into the system and provoking a strong immune response. A number of organisms have been used as vectors. The most commonly used viral vectors are the vaccinia virus (the smallpox vaccine virus), adenovirus and canary pox virus (that infects cells but does not replicate in humans).

Viral Vector Vaccines

The most commonly used viral vector is the vaccinia virus, though genetically engineered recombinant viral vectors such as pox viruses or adenovirus are also deliberated upon.

Genes that have the potential to induce protective immunity (such as coat antigens) are inserted into attenuated live virus. Vaccinia, a commonly used virus has a large, double-stranded genome (about 187,000 kB and approximately 200 genes). Because of its large size, a large number of genes can be replaced with foreign (antigen) genes without causing any loss of their capacity to infect host cells. The disadvantage is, because of the large size, it becomes difficult to manipulate or insert the gene *in vitro*. So a simple but long method is used to prepare the vaccinia virus containing a foreign gene. The foreign gene (that encodes the antigen) is first inserted into a plasmid vector. The plasmid vector is then transfected in tissue culture cells.

These tissue culture cells are then infected with the vaccinia virus. Inside the tissue culture cells, the vaccinia virus replicates in the cytosol where it undergo homologous recombination with plasmid containing antigen gene. This recombination event causes the antigen gene (which had the vaccinia virus promoter associated with it) to get transferred to the vaccinia virus genome. This foreign gene is usually inserted into the viral **tk** (thymidine kinase) gene inactivating it. The viruses that show tk-negative phenotype are selected.

This genetically manipulated vaccinia virus expresses a high level of gene product (foreign antigen) and this serves as a potent source of immunogen inside the host. Effective live vaccines based on this method have been developed for rabies in animals. Experimental viral vector vaccines generated by this method include those that express influenza virus haemaglutinin, malaria protein, hepatitis B surface antigen, herpes simplex virus proteins, among others. An HIV vaccine that carries HIV *env*, *gag*, protease, part of *pol* genes in canary pox are in clinical trials.

Bacterial Vector Vaccine

Like the live viral vectors, some attenuated bacterial strains have been engineered to carry genes of virulent pathogens. The DNA encoding the antigenic determinants is inserted into the attenuated bacterial genome. The bacteria then express the antigen along with its own protein. The production and expression of antigen by the bacterial vector inside the host body stimulates the immune system. An attenuated strain of *Salmonella typhimurium* (Ty2la), the causative agent of food poisoning, is being explored as a vector and is currently in human trials.

DNA Vaccine

The DNA vaccine (or more properly DNA-based vaccine) represents a new class of vaccines in which there is a deliberate introduction of a DNA plasmid usually into the muscle cell of the recipient. The plasmid contains a protein-coding gene (of antigen) that gets expressed in the cell, leading to both humoral and cell-mediated immune responses. The plasmid DNA can be introduced into the muscle cell either by infection or by bombarding the skin with DNA-coated gold particles with a fine airgun (gene-gun). Currently, attempts are also underway to introduce DNA into the nasal tissue via nasal drops. It should be noted that once inside the cells of the recipient, the plasmid does not replicate, but only expresses itself, that is, protein is produced. Usually bacterial plasmid is used and a gene coding the antigen is inserted into the control of a mammalian promoter and this chimeric plasmid is then introduced into the recipient. The recipient cell then expresses the foreign antigenic protein coded by the introduced DNA in the host body. The immune system then responds to the antigen as to any other antigen entering the body.

Immuno Deficiency Diseases

Immunodeficiencies occur when one or more components of the immune system is defective. Most of immunodeficiency diseases are inherited.

Acquired immunodeficiencies are caused due to malnutrition or by agents, such as drugs or irradiation that damage lymphocytes, or infection with measles or HIV. Immunodeficiencies may be *primary* and *secondary*.

Primary immuno deficiency diseases are disorders caused by a defect in one or more genes involved in the development or function of the immune system. By contrast, a *secondary immune deficiency* disease is one in which the immune system is compromised by factors from the outside, such as viruses, radiation treatments or chemotherapy.

Example of primary immuno deficiency diseases:

Severe combined immunodeficiency (SCID)

SCID is a genetic disorder which is characterized by a very low number of circulating lymphocytes. Both arms (B cells and T cells) of the adaptive immune system become non-functional. As such patients make neither specific T-cell dependent antibody responses nor cell-mediated immune responses, and thus cannot develop immunological memory. Several different defects can lead to the SCID phenotype. In X-linked SCID, which is the commonest form of SCID, T cells fail to develop because of defect in the genes code for several cytokine receptors, including those for the interleukins IL-2, IL-4, IL-7, IL-9, and IL-15. The autosomally inherited SCID occurs due to *adenosine deaminase deficiency*. Adenosine deaminase catalyzes conversion of adenosine to inosine, and its deficiency results in accumulation of adenosine, which interferes with purine metabolism which result in an accumulation of nucleotide metabolites that are particularly toxic to developing T cells.

Chediak-Higashi syndrome

Chediak-Higashi syndrome is an autosomal recessive disease. It is characterized by recurrent bacterial infections, lack of skin and eye pigment. Phagocytes from patients with this immune defect contain giant granules but do not have the ability to kill bacteria. With time, patients develop massive infiltrates of lymphocytes and macrophages in the liver, spleen, and lymph nodes. The molecular basis of the defect is a mutation in a protein involved in the regulation of intracellular trafficking. The mutation impairs the targeting of proteins to lysosomes.

DiGeorge syndrome

DiGeorge syndrome, or *congenital thymic aplasia,* is not hereditary but occurs sporadically and is result of a deletion in chromosome 22. The syndrome is caused by defective migration of fetal neural crest cells into the third and fourth *pharyngeal pouch*. DiGeorge syndrome in its most severe form is the complete absence of a thymus. This developmental defect causes immunodeficiency along with hypoparathyroidism, and congenital heart disease. The immune defect includes a profound depression of T-cell numbers and absence of T-cell responses. Although B cells are present in normal numbers, affected individuals do not produce antibody in response to immunization with specific antigens.

Failures of host defense mechanisms

The propagation of a pathogen depends on its ability to multiply in a host. Hence pathogens must therefore grow without activating an immune response. The most successful pathogens persist either because they do not elicit an immune response, or because they evade the response once it has occurred. Pathogens have developed various strategies for avoiding destruction by the immune system. Some of strategies are mentioned below:

1. One way in which a pathogenic agent can evade immune response is by altering its antigens. There are three ways in which antigenic variation can occur. First, many pathogenic agents exist in a wide variety, of antigenic types. For example, there are many types of *Streptococcus pneumoniae.* Each type differs from the others in the structure of its polysaccharide capsule. The different types are distinguished by serological tests and so are often known as serotypes. Infection with one serotype of *Streptococcus pneumoniae* can lead to type-specific immunity, which protects against reinfection with that type but not with a different serotype.
2. A second, more dynamic mechanism of antigenic variation arises due to **antigenic drift** and **antigenic shift**. *Antigenic drift* is caused by point mutations in the genes encoding epitopes. For example in influenza virus new variant arises with mutations in genes encoding the major surface proteins *hemagglutinin* and *neuraminidase*. Thus individuals who were previously infected with, and hence are immune to, the old variant are thus susceptible to the new variant. Periodically influenza viruses also show an *antigenic shift* through reassortment of their segmented genome with another influenza virus, changing their surface antigens radically. Such antigenic shift variants are not recognized by individuals immune to influenza.

3. The third mechanism of antigenic variation involves programmed rearrangements in the DNA of the pathogen. For example, African trypanosomes which cause sleeping sickness in humans changes the major surface antigen repeatedly within a single infected host. The trypanosome is coated with a single type of glycoprotein, the *variant-specific glycoprotein* (VSG), which elicits an antibody response. The trypanosome genome, however, contains about 1000 VSG genes, each encoding a protein with distinct antigenic properties. Only one of these is expressed at any one time. The VSG gene expressed can be changed by gene rearrangement. So, by having their own system of gene rearrangement that can change the VSG protein produced, trypanosomes keep one step ahead of an immune system capable of generating many distinct antibodies by gene rearrangement.
4. Some pathogens have evolved specialized mechanisms to resist destruction by host defense mechanisms. For instance, some bacteria that are engulfed in the normal way by macrophages have evolved means of avoiding destruction by these phagocytes. For example, *Mycobacterium tuberculosis* is taken up by macrophages but prevents the fusion of the phagosome with the lysosome, protecting itself from the bactericidal actions of the lysosomal contents.

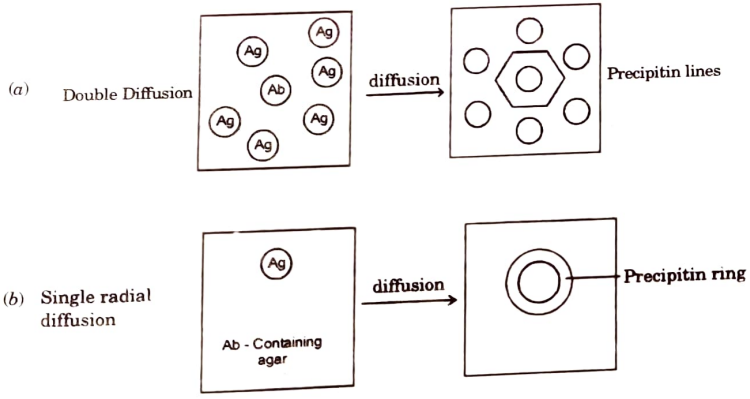
Immunological Techniques

When there is both sufficient Ag and Ab, the combination of Ag and Ab proceeds until large aggregates are formed which are insoluble in water and precipitate. The extent to which a lattice forms depends on the relative amount of Ag and Ab present.

Lattice formation, and precipitation are the basis for several qualitative and quantitative assays for Ag or Ab. These assays are done in semisolid gels into which holes are cut for Ag /or Ab and diffusion occurs until Ag and Ab are at equivalence and precipitate.

Immunodiffusion

Two types of immunodiffusion reactions can be used to determine relative concentrations of Abs or Ag, to compare antigens, or to determine the relative purity of an Ag preparation. They are radial immunodiffusion (the mancini method) and double immunodiffusion (the ouchterlong method); both are carried out in a semisold medium such as agar. In radial immunodiffusion, an antigen sample is placed in a well and allowed to diffuse into agar containing a suitable dilution of an antiserum. As the antigen diffuses into agar, the region of equivalence is established and a ring of precipitation, percipitin ring forms around the well. The area of the precipitin ring is proportional to the concentration of antigen. By comparing the area of the precipitin ring with a standard curve, the concentration of the antigen sample can be obtained. In the ouchterlong, method both Ag and Ab diffuse radially from wells toward each other, thereby establishing a concentration gradient. As equivalence is reached, a visible line of precipitation, a precipitin line forms.



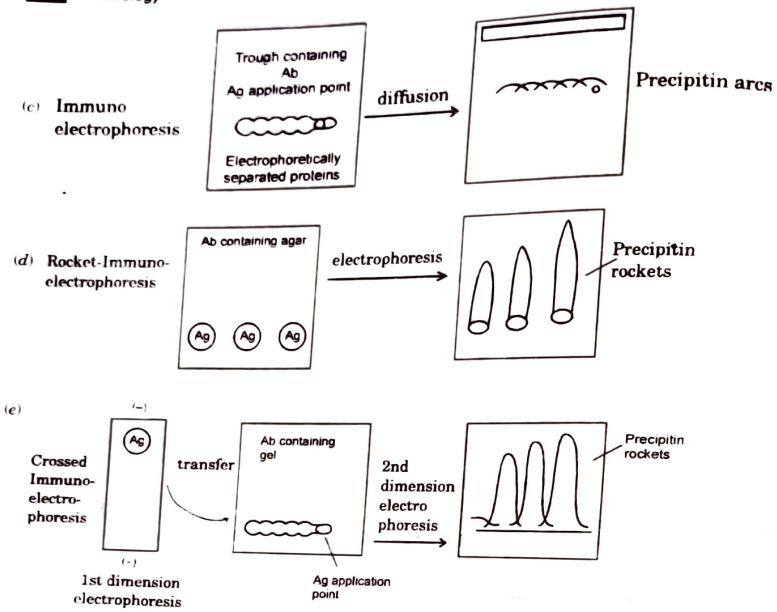


Fig. Diagrammatic representation of immunoprecipitation techniques in agar

Immunoelectrophoresis Combines Electrophoresis and Double Immunodiffusion

In immunoelectrophoresis, a mixture of proteins containing the antigen are first separated by agar gel electrophoresis. Antibody is then allowed to diffuse through the gel from a through cut in the gel parallel to the direction of electrophoresis. Precipitin arcs are then formed where antibody meets antigen at equivalence. A disadvantage of this technique is the relatively poor resolution of antigen mixtures using agar gel electrophoresis, but it can be useful for detecting precipitating antibodies. Clinically, the technique is carried out on samples of patient's serum, concentrated urine or spinal fluid to detect abnormalities in concentrations of antigens/or the presence of abnormal proteins relative to normal control samples analysed at the same time.

Rocket immunoelectrophoresis is an adaption of single radial immunodiffusion and involves electrophoretic migration of antigen from wells cut in antibody containing gel. At equivalence, rocket-shaped precipitin lines are formed, the area under which is proportional to the antigen concentration. The technique can be used to determine antigen concentrations in unknown samples (*e.g.* serum protein levels) by reference to a standard curve as in RID, or to investigate immunochemical relationships between different samples if these are placed in adjacent wells close together.

In crossed immunoelectrophoresis, proteins are first separated by agar gel electrophoresis, after which they are electrophoresed into an antibody containing gel at right angles to the direction of the first electrophoresis. The technique can be used for analysis of serum proteins.

RIA and ELISA :

The presence of Ab to a particular Ag in the serum of a patient can be determined using very sensitive radioimmunoassay (RIA) or enzyme linked immuno-absorbent assays (ELISA). Such assays are of particular value in demonstrating Ab to Ags to infectious agents, e.g. virus, bacteria etc.

Radioimmunoassay

At the present time, one of the most common uses of solid scintillation counting is in certain radioimmunoassay (RIA) procedures. The RIA technique, developed in 1950s by S.A. Berson and R.S. Yalow, is a tracer procedure used to measure trace amounts of any substance with antigenic properties. Depending on the specific activity of the labeled compound, quantities of antigen that are as low as 10 picograms (i.e., 10 millionths of a microgram) may be detected. Because many different kinds of chemical substances are antigenic (i.e. they can elicit an immune response), they can be reliably quantitated using the RIA procedure. The sensitivity of the RIA technique equals or surpasses all chromatographic and spectrophotometric methods currently in use. The technique has been especially useful for assaying hormones.

In the RIA technique, a sample containing the Ag to be quantified is mixed with a second sample containing radioisotopically labelled Ag of very high specific activity. To the mixture is added antiserum containing antibody previously prepared against the antigen. The amount of Ab added must, be limiting so that antigen is always in excess. The resulting Ag-Ab reaction produces a complex in which the amount of radioisotope present is inversly proportional to the original quantity of unlabeled antigen in the sample being assayed. In the final stage of RIA, the Ag-Ab complex is separated from free Ag by precipitation. The radioisotopes used are ^{3}H, ^{125}I. The first RIA described was for insulin.

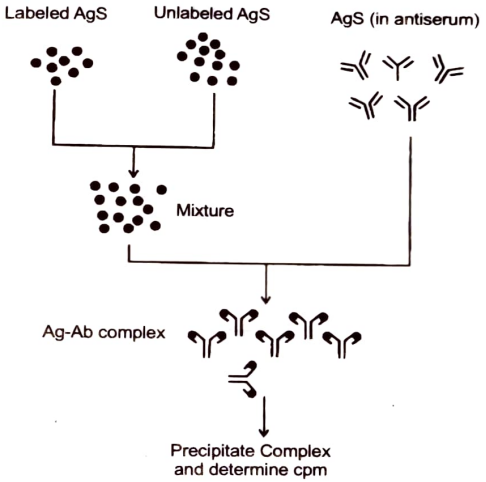


Fig. Radioimmunoassay

Enzyme-Linked Immunosorbent Assay

Enzyme-linked immunosorbent assay, commonly known as ELISA (or EIA), is similar in principle to RIA but depends on an enzyme rather than a radioactive label. An enzyme conjugated with an antibody reacts with a colorless substrate to generate a colored reaction product. Such a substrate is called a chromogenic substrate. A number of enzymes have been employed for ELISA, including alkaline phosphatase, horseradish peroxidase, and β-galactosidase. These assays approach the sensitivity of RIAs and have the advantage of being safer and less costly.

Variants of ELISA : A number of variations of ELISA have been developed, allowing qualitative detection or quantitative measurement of either antigen or antibody. Each type of ELISA can be used qualitatively to detect the presence of antibody or antigen. Alternatively, a standard curve based on known concentrations of antibody or antigen is prepared, from which the unknown concentration of a sample can be determined.

Indirect ELISA: Antibody can be detected or quantitatively determined with an indirect ELISA. Serum or some other sample containing primary antibody (Ab_1) is added to an antigen-coated microtiter well and allowed to react with the antigen attached to the well. After any free Ab_1 is washed away, the presence of antibody bound to the antigen is detected by adding an enzyme conjugated secondary anti-isotype antibody (Ab_2), which binds to the primary antibody. Any free Ab_2 then is washed away, and a substrate for the enzyme is added. The amount of colored reaction product that forms is measured by specialized spectrophotometric plate readers, which can measure the absorbance of all the wells of a 96-well plate in seconds.

Indirect ELISA is the method of choice to detect the presence of serum antibodies against human immunodeficiency virus (HIV), the causative agent of AIDS. In this assay, recombinant envelope and core proteins of HIV are adsorbed as solid-phase antigens to microtiter wells. Individuals infected with HIV will produce serum antibodies to epitopes on these viral proteins. Generally, serum antibodies to HIV can be detected by indirect ELISA within 6 weeks of infection.

Sandwich ELISA: Antigen can be detected or measured by a sandwich ELISA. In this technique, the antibody (rather than the antigen) is immobilized on a microtiter well. A sample containing antigen is added and allowed to react with the immobilized antibody. After the well is washed, a second enzyme-linked antibody specific for a different epitope on the antigen is added and allowed to react with the bound antigen. After any free second antibody is removed by washing, substrate is added, and the colored reaction product is measured.

Competitive ELISA: Another variation of measuring amounts of antigen is competitive ELISA. In this technique, antibody is first incubated in solution with a sample containing antigen. The antigen-antibody mixture is then added to an antigen-coated microtiter well. The more antigen present in the sample, the less free antibody will be available to bind to the antigen-coated well. Addition of an enzyme-conjugated secondary antibody (Ab_2) specific for the isotype of the primary antibody can be used to determine the amount of primary antibody bound to the well as in an indirect ELISA.

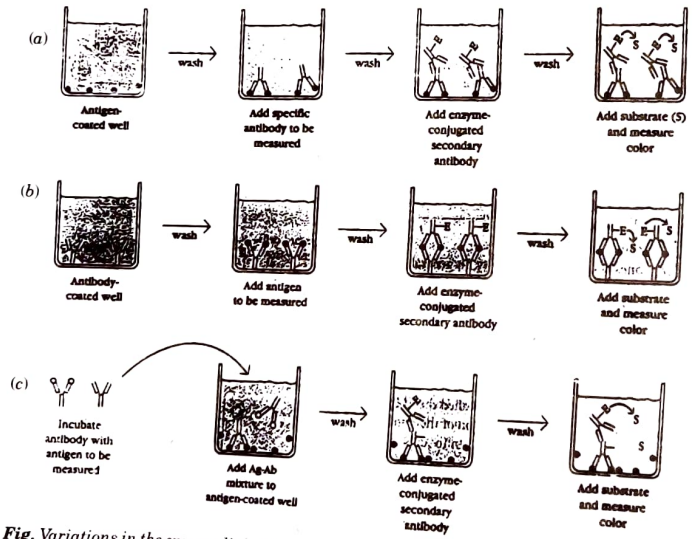


Fig. *Variations in the enzyme-linked immunosorbent assay (ELISA) technique allow determination of antibody or antigen. Each asaay can be used qualitatively, or quantitatively by comparison with standard curves prepared with known concentration of antibody or antigen. (a) whereas antigen can be determined with a sandwich ELISA (b) competitive ELISA (c) In the competitive ELISA, which is an inhibition-type assay, the concentration of antigen is inversely proportional to the color produced. Antibody can be determined with an indirect ELISA.*

EXERCISE – I

MCQ TYPE QUESTIONS

1. Which of the following cells secretes IL-1?
(a) Monocytes (b) Leucocyte
(c) Stromal cells (d) Both A & B

2. The characteristic features of multiple myeloma is
(a) Paraproteinemia (b) Proteinuria
(c) Lytic bone lesions (d) All of these

3. Immunoglobulin's are secreted by which cells
(a) plasma cells (b) T-lymphocytes
(c) T-cytotoxic cells (d) None

4. Which method is used to detect the HIV ANTIBODY?
(a) ELISA (b) RIA
(c) Immunofluorence (d) A and C

5. The features of neutrophils is\are
(a) Active glycolysis
(b) Contain certain unique enzymes
(c) Fight against, viruses
(d) A and B

6. By humoral immunity, the antibodies can destroy the target cell by which of the following mechanism?
(a) Classical complement pathway
(b) Agglutination
(c) Opsinization
(d) All of these

7. Which of the following Ig fixes most cells and basophils? .
(a) IgM (b) IgA
(c) IgE (d) EgG_1

8. Which of the following additional unit is present in the IgA molecule?
(a) 'S' piece (b) 'J' piece
(c) 'S' and 'J' piece (d) None

9. The producer of the cells lymphokines
(a) T-cytotoxic cells
(b) B-cell
(c) T-cells when stimulated by antigens
(d) Both A& B

10. Which type of variation is individually specific to each Ig molecule?
(a) Isotopic
(b) Idiotypic
(c) Allotypic
(d) None

11. The immunity which shows delayed type of hypersensitivity is
(a) Cell mediated immunity
(b) Humoral immunity
(c) Antibody immunity
(d) ADCC

12. In which condition IgM level in blood is increased considerably with a monoclonal peak?
(a) Waldenstroms macroglobulinemia
(b) Heavy chain disease
(c) Bence-Jones proteinuria
(d) Both A and B

13. On which of the following loci the MHC genes are present?
(a) A, C, B, D and DR (b) A, C, B, D and DR
(c) B, A, C, B and DR (d) A& B

14. The functions of IFN's is
(a) T and B cell proliferation
(b) Proliferation of macrophages
(c) Antiviral
(d) Both B & C

15. IgG molecule is divided into which sequence of the heavy chain constant region?
(a) Amino acid sequence
(b) Globin sequence
(c) Protein sequence
(d) Heavy chain sequence

16. Indirect ELISA technique is used to detect
(a) Antibody (b) DNA
(c) RNA (d) Both A& B

17. T-helper cells carry which determinants on the cell surface?
(a) CD8 (b) CD12
(c) CD4 (d) A, B

18. In which type of electrophoresis the precipitation arcs are formed?
Where the Ag-Ab molecules are 1:1% .
(a) PAGE
(b) SDS-PAGE
(c) Immunoelectrophoresis
(d) All of these

19. The use of Western blot technique is used to find the
(a) Size of specific proteins
(b) Quantitative specific protein molecules
(c) Shape of the specific proteins
(d) Both A & B

20. Which proteins are helpful to determine the shape and flexibility of the red blood cells?
(a) Spectrin (b) Ankyrin
(c) Actin (d) A and B

21. Universal donor blood group is
(a) A (b) AB
(c) B (d) O

22. Ga/NAc is a
(a) Immunodominant sugar
(b) Immunosupress sugar
(c) Tumor marken
(d) None of these

23. The receptor used by the B-Cell is
(a) Immunoglobulin (b) Protein
(c) Nucleic acids (d) Lipids

24. Antigen processing and presentation are processes that occur
(a) Outside the cell
(b) Within the cell
(c) Within receptors
(d) Within plasmamembrane

25. Antigen processing and presentation are processes result in
(a) Proteolysis
(b) Association of the fragments with MHC molecules
(c) Expression of the peptide-MHC molecules
(d) All of these

26. Dendritic cells, ingest antigens by
(a) Pinocytosis
(b) Phagocytosis
(c) Both i and ii
(d) Doesnot ingest antigens

27. The antigens that can polyclonally activate T cells
(a) Polyantigens (b) Polyclonal antigens
(c) Supreantigens (d) Superantigens

28. An individual does not want functional T cells in the periphery because of
(a) Damages of healthy, normal tissues
(b) Damage of healthy, normal organs
(c) Damage of healthy, normal receptors
(d) Damage of healthy, normal MHCs

29. If a self reactive B cell does get to the periphery, it will not be activated due to
(a) Lack of T cell help
(b) Lack of cytotoxic cells help
(c) Lack of macropnages help
(d) None of these

30. A given T cell is restricted to recognize a peptide antigen only when it is bound to
(a) Self-MHC (b) Anti-MHC
(c) Auto-MHC (d) None of these

31. CD4 binding to Class II MCH , which ensures that
(a) Tc cells can interact with target cells
(b) CD2binding to LFA-3
(c) LFA-1 binding to ICAM-1
(d) The cells only interact with APCs

32. B cells are an essential component of
(a) Innate immunity (b) Adaptive immunity
(c) Both A and B (d) None of these

33. I. The path leading to the association of protein fragments with MHC molecules differs for class I and class II MHC.
II. MHC class II molecules present degradation products derived from intracellular (endogenous) proteins in the cytosol.
III. MHC class I molecules present fragments derived from extracellular (exogenous) proteins that are located in an intracellular compartment.
Which of the above statements are true about MHC ?
(a) I, II (b) II, III
(c) I (d) I, II, III

34. I. Binding of peptide within the MHC is not covalent, there are many factors while help stabilize the immunological synapse.
II. Class I MHC molecules are composed of two polypeptide chains, a long α chain and a short β chain called β1-microglobulin.
III. The α chain of MHC class I has four regions.
Which of the above statements are true?
(a) I, II (b) II, III
(c) I, III (d) I, II, III

35. The CD3 complex is composed of
(a) One ɣ, one δ, two ε and 2 ξ chains
(b) Two ɣ, one δ, two ε and 2 ξ chains
(c) One ɣ, two δ, two ε and 2 ξ chains
(d) One ɣ, one δ, one ε and 2 ξ chains

36. Immature B cells are produced in the bone marrow with an exception of
(a) Horse (b) Rabbit
(c) Drosophila (d) Pig

37. When the B-cell fails in the maturation process, it will die by a mechanism called
(a) Senescence (b) Phagocytosis
(c) Pinocytosis (d) Apoptosis

38. Immunoglobulin which is expressed on B1 cell is
(a) IgG (b) IgA
(c) IgM (d) IgD

39. I. B cells recognize their cognate antigen in its processed form
II. T cells recognize their cognate antigen in a native form
III. A critical difference between B cells and T cells is how each lymphocyte recognizes its antigen.
IV. Among the above statements how many of them are true about B and T- cell ?
(a) I,II (b) II,III
(c) I (d) III

40. Lipids and nucleic acids are antigenic when combined with
(a) ATP and NADPH
(b) Magnesium and calcium
(c) Proteins and polysaccharides
(d) Enzymes

41. A class of antigens which cause non-specific activation of T-cells is
(a) Superantigen (b) Allergen
(c) Epitope (d) Tolerogen

42. Medulla is surrounded by cortex except
(a) Spleen (b) Adenoids
(c) Hilum (d) Peyer's patches

43. The Humoral Immune Response is the aspect of immunity that is mediated by
(a) Antibodies (b) Antigen
(c) Dendritic cells (d) Macrophages

44. A woman with infertility receives an ovary transplant from her sister who is an identical twin. What type of graft is
(a) Xenograft (b) Autograft
(c) Allograft (d) Isograft

45. Avidity is influenced by
(a) Valence of the antibody
(b) Valence of the antigen
(c) Both (a) and (b)
(d) None of these

46. Which of the following is false
(a) Theobald – Smith phenomenon is a type-I hyper sensitivity
(b) Serum sickness is a type-II hypersensitivity reaction.
(c) Allograft reaction is a type-IV hypersensitivity reaction
(d) Transfusion reaction is a type-II hypersensitivity reaction

47. The most widely complement fixing Ab is
(a) IgA (b) IgG
(c) IgM (d) IgE

48. Skin test based on neutralization reaction is/are
(a) Casoni test (b) Lepronin test
(c) Tuberculin test (d) Schick test

49. The use of monoclonal Ab's
(a) Immunotherapy
(b) Immunological identification of the cells and tissues
(c) Radioimmunoimaging
(d) All of these

50. Most sensitive test for Ag detection is
(a) RIA
(b) ELISA
(c) Immunofluoresence
(d) Passive hemagglutination

51. Reaction of soluble Ag with Ab is known as
(a) Aggulutination (b) Precipitation
(c) Flocculation (d) Complement fixation

52. Monoclonal Ab's are associated with name of
(a) Medawar (b) Oven
(c) Burner (d) Pasteur

53. The prototype of type-II hypersensitivity reaction is
(a) Arthus reaction
(b) SLE
(c) Auto immune hemolytic anemia
(d) Contact dermatitis

54. Which is an example of type-III hypersensitivity
(a) Contact dermatitis
(b) Hemolytic anemia
(c) Serum sickness
(d) Good pasture's syndrome

55. An example of type-I hypersensitivity reaction is
(a) Schick test (b) Mantoux test
(c) Lepronin test (d) Casoni's test

56. The main aim of adjuvant is to increase
(a) Distribution (b) Absorption
(c) Antigenicty (d) Metabolism

57. Delayed hypersensitivity involves
(a) Neutrophills (b) Monocytes
(c) Eosinophils (d) Lymohocytes

58. Rosette formation with sheep RBC's indicate functioning
(a) T-cells (b) B-cells
(c) Neutrophils (d) Monocytes

59. Prozone phenomenon is due to
(a) Disproportionate Ag-Ab levels
(b) Excess Ab
(c) Excess Ag
(d) Hyper immune reaction

60. T-cell multiplication is stimulated by
(a) Macrolin (b) Heat
(c) Bovine serum (d) Phytohemagglutin

61. T-cells are identified by
(a) Rosette formation with Sheep RBC
(b) Immunoglobulin's their surface
(c) EAC Rosetter with sheep erythrocytes
(d) Have filamentous projections on their surface

62. Contact dermatitis is
(a) Type-I hypersensitivity
(b) Type-II hypersensitivity
(c) Type-III hypersensitivity
(d) Type-IV hypersensitivity

63. Humoral graft rejection is characterized by
(a) Intense mono-nuclear infiltration
(b) Neovascularisation of grafted tissue
(c) Thrombosis of the blood vessels and Ischemic necrosis
(d) All of these

64. In situ hybridization is used for
(a) Diagnosis of specific genetic disease
(b) Diagnosis of enzyme deficiency disease
(c) Treatment of genetic diseases
(d) All of these

65. VDRL is a
(a) Slide flocculation test
(b) Tube flocculation test
(c) Gel precipitation
(d) All of these

66. Hep-2 cells are a type of
(a) Primary cell culture
(b) Diploid cell lines
(c) Continuous cell lines
(d) Explant-culture

67. M-band is present in
(a) IgG Myeloma (b) H-chain disease
(c) L-chain disease (d) All of these

68. Phagocytosis enhanced by coating the surface of Ag is called
(a) Opsonization (b) Chemotaxis
(c) Decoding (d) CFT

69. Agglutination test is
(a) ABO incompatibility (b) VDRL
(c) Weil Felix (d) FTA AB

70. Rapid-Serological diagnosis test include all except
(a) Latex agglutination (b) Spectrophotometry
(c) Gel electrophoresis (d) RIA

71. Delayed tuberculin test response is due to
(a) B-lymphocytes (b) T-lymphocytes
(c) Monocytes (d) Histocytes

72. Ab's are more responsible to
(a) Recipient tissue (b) Donor tissue
(c) Isograft (d) Allograft

73. In the precipitin test, reaction takes place between
(a) Ag and its heterologous Ab
(b) Soluble Ag and solution of its homologous Ab
(c) Soluble Ag and solution of its heterologous Ab
(d) (b) and (c)

74. A tube precipitin test formed in a fluid phenomenon is called as
(a) Agglutination test
(b) Ring test
(c) Radial immunodiffusion
(d) Both (b) and (c)

75. Agglutination occurs when bacterial cell Ag's are exposed to
(a) Homologous Ab's (b) Heterologous Ab's
(c) Homologous Ag's (d) Both (a) and (c)

76. In hemagglutination test the intensity of the agglutination depends on
(a) Concentration
(b) Infective titer of the virus
(c) Temperature and concentration
(d) Both (a) & (b)

77. Hemagglutination test is used to diagnose
(a) Mumps (b) Influenza virus
(c) ECHO viruses (d) All of these

78. Influenza causing hemagglutination, is inhibited by
(a) Decrease in Ab
(b) Increase in Ab
(c) Excluding false reactions
(d) Increase in the Ag

79. Complement fixation tests are based on
(a) Presence of complement fixing Ag's in serum
(b) Presence of complement fixing Ab's in serum
(c) Presence of complement fixing Ab's in plasma
(d) None of these

80. How complement fixing Ab's are produced in an animal body
(a) Stimulation of complement
(b) Stimulation of Bacterial Ag's
(c) Stimulation of bacterial macrophages
(d) All of these

81. If complement is fixed in the complement fixing tests, then Ag and Ab's are
(a) Separate
(b) Union
(c) Aggregates
(d) Agglutinins/precipitates

82. Best known applications of the complement fixation test
(a) Wassermann test (b) Widal test
(c) Weil-Felix (d) None of these

83. Wassermann test for
(a) ECHO viruses (b) Influenza virus
(c) Syphilis (d) Mumps

84. Which precipitates at 50°C-60°C but disappears on heating
(a) H-chain (b) L-chain
(c) Both (a) and (b) (d) None of these

85. Prozone phenomenon is due to
(a) Ag excess (b) Ab excess
(c) False –ve reaction (d) Both (b) and (c)

86. The following methods of diagnosis utilizes labeled Abs except
(a) ELISA
(b) Hemagglutination inhibition test
(c) RIA
(d) Immunofluorescence

87. Which of the following tests denotes classical component pathway activation in immunoinflammatory condition ?
(a) C_2, C_4 and C_3 decreased
(b) C_3 and C_4 normal
(c) C_3 is decreased
(d) C_2, C_4 and C_3 all elevated

88. Which of the following is true
(a) Rose wallet test is a complement-fixation test
(b) Indirect hemagglutination test is less sensitive than gel diffusion test
(c) Ag, Ab reaction cannot cross in absence of electrolytes
(d) (a) and (c)

89. The combining site of an antibody is located in the Fab portion of the molecule and is constructed from the
(a) Hypervariable regions of heavy chain
(b) Hypervariable regions of light chain
(c) Both (a) and (b)
(d) None of these

90. Classic complement is activated by
(a) IgG (b) IgA
(c) IgM (d) (a) and (c)

91. The secretory component of molecule is
(a) Formed by epithelial cell and plasma cell
(b) Formed by plasma cells
(c) Formed by epithelial cell of lining mucosa
(d) Secrated by bone marrow

92. Which of these sentences are true about anaphylaxis
(a) Type I reaction
(b) Large amount of histamine released
(c) Cytokines like IL-4, IL-5, IL-6 and GMCSF are released
(d) All of these

93. Type-I hypersensitivity does not belong to
(a) Autoimmune hemolytic anemia
(b) Anaphylaxis
(c) Extrinsic Anemia
(d) Hay fever anemia

94. Which of the following doesn't belong to heterophile agglutination test ?
(a) Weil-Felix test (b) Widal test
(c) Paul-Bannet test (d) All of these

95. Match the following

1. ELISA	P. MCA
2. RIA	Q. MIT
3. Hybridoma technology	R. Picograin
4. T-cell function	S. HIV

(a) 1-P; 2-Q; 3-R; 4-S (b) 1-S; 2-R; 3-P; 4-Q
(c) 1-Q; 2-S; 3-P; 4-R (d) 1-Q; 2-S; 3-R; 4-P

96. Match the following

1. Autograft	P. Self
2. Isograft	Q. Twins
3. Allograft	R. Skin graft
4. Orthotopic graft	S. Unrelated members of same species

(a) 1-P; 2-Q; 3-R; 4-S (b) 1-P; 2-Q; 3-S; 4-R
(c) 1-S; 2-R; 3-P; 4-Q (d) 1-R; 2-S; 3-P; 4-Q

97. Highest avidity immunoglobuline
(a) IgG (b) IgM
(c) IgD (d) IgE

98. Combined immuno-deficiencies are due to
(a) B-cell defect (b) N-cell defect
(c) NK cell defect (d) Both (a) and (b)

99. Immune granulomas are not seen in
(a) Tuberculosis (b) Syphills
(c) Sillicones (d) Berryliosis

100. Animal used to demonstrate anaphylaxis in the lab is
(a) Rabbit (b) Adult mice
(c) Monkey (d) Gulnea pig

101. Which of the following cells is known as large granular lymphocyte ?
(a) Plasma cells (b) NK cells
(c) T-cells (d) K-cells

102. Which leukotriene is the adhesion factor for the neutrophill on the cell surface to attack to endothelium ?
(a) B4 (b) C4
(c) D4 (d) E4

103. All of the following are immune complex disease except
(a) Serum sickness (b) Farmer's lungs
(c) SLE (d) Graft rejection

104. Phagocytic cells elaborate
(a) Prostaglandins (b) Thromboxane
(c) Leukotriene (d) Both (a) and (c)

105. Runt disease is
(a) Graft rejection
(b) Graft Vs. host rejection
(c) Deficient T-cell function
(d) Complement deficiency

106. Which complement component shows thermo-stability at 56°C for 30 min
(a) C_{1q} (b) C_9
(c) C_3 (d) C_2

107. α-2 globulin can inhibit which enzymes
(a) Kallikerin (b) C_1 esterase
(c) Hageman's factor (d) All of these

108. IL-1 produces
(a) T-Lymphocyte activation
(b) Delayed wound healing
(c) Increased pain perceptin
(d) decreased PMN from bone marrow

109. Which of following is false statement regarding NK cells?
(a) It is activated by IL-2
(b) Express CD3 receptor
(c) It is a variant of large lymphocyte
(d) There is Ab induced proliferation of NK cells

110. All of the following are glycol-proteins except
(a) Blood Ag (b) Albumin
(c) IgD (d) HCG

111. Diagnosis of ABO incompatibility cannot be done by
(a) Sweat (b) Saliva
(c) Semed (d) CSF

112. The chain which precipitates at 50° - 60° but disappears on heating is
(a) Heavy chain (b) Light chain
(c) Both (a) & (b) (d) None of these

113. Bence-Jones proteins are best described as
(a) μ-chain (b) γ-chain
(c) k & λ chains (d) Fibrin split products

114. NK cells kill the viral infected cells due to
(a) Increased expression of MHC class-I molecules
(b) Decreased expression of MHC class-I molecules
(c) Increased expression of MHC class-II molecules
(d) Decreased expression of MHC class-II molecules

115. MHC class-III genes encode
(a) Complement component
(b) Tumor necrosis factor-α
(c) Tumor necrosis factor-β
(d) Both (b) & (c)

116. Perforins are produced by
(a) Cytotoxic T-cells (b) Suppressor T-cells
(c) Plasma cells (d) NK cells

117. Which of these sentences are true regarding immunoglobulin?
(a) Half life of IgG is 23 days
(b) IgD is heat stable
(c) IgA cross placenta
(d) Both (a) & (b)

118. IL-1 is produced by
(a) Macrophages
(b) Helper-T-lymphocytes
(c) B-cells
(d) Cytotoxic-T-cells

119. Which of these sentences are true about interferon?
(a) It is a synthetic antiviral agent
(b) Inhibits viral replication in cells
(c) Is specific for particular virus
(d) None of these

120. Which of these sentences are true about antibodies?
(a) Long latent period (b) Appear in short time
(c) Persists for long (d) Both (b) & (c)

121. Which of the following is very difficult to induce antibody?
(a) Polysaccharides (b) Protein
(c) Antigen's (d) Related interferon

122. Interferon is
(a) Species specific
(b) Reacts directly with virus particles to inactivate them
(c) Reacts with cells and the affected cell; then becomes resistant to a number of different virus
(d) Both (a) & (c)

123. Function of T-lymphocytes includes
(a) Production of Interferon
(b) Lymphokine production
(c) Rosette formation
(d) All of these

124. Secondary exposure to an Ag results in sudden increase in
(a) IgA (b) IgD
(c) IgG (d) IgM

125. Antigen recognition of the surface of Ag processing cells by
(a) T-cell recognition Ag
(b) Fc portion of Immunoglobulin
(c) Fab portion of Immunoglobulin
(d) Cell recognition Ag

126. NK cells are
(a) Activated macrophages
(b) Antibody-activated T-cells
(c) Null cells activated by complement
(d) Derived from plasma cells

127. T-cell mature in
(a) Peyers patch
(b) Lymph node
(c) Thymus
(d) All of these

128. T_4/T_8 ratio reversal is seen in
(a) T-cell Lymphoma (b) Hairy cell leukemia
(c) AIDS (d) All of these

129. Leukotriene is not secreted by
(a) Macrophages (b) T4 cells
(c) T8 cells (d) Platelets

130. HLA typing is useful in
(a) Blood grouping
(b) Assessing prognosis of diseases
(c) Cancer therapy
(d) Cases of paternal dispute

131. T-cell functions are assesed by
(a) Phagocytic index
(b) T-cell count
(c) Migration inhibition test
(d) Ig index

132. NK cell and cytotoxic cell are differentiated by
(a) Interferon's reduce NK cell activity
(b) Antibody speciticity
(c) Receptor for immunoglobulin
(d) Presence in spleen

133. Secretory piece of IgA is synthesized in
(a) T-cells (b) B-cells
(c) Lymph nodes (d) Mucosal epithelium

134. Life long immunity is seen in all of the following except
(a) Pertussis (b) Mumps
(c) Rubella (d) Plague

135. NK cells provides immunity against
(a) Virus (b) Bacteria
(c) Fungus (d) Chlamydia

136. Anti-Rh-Ab's belong predominantly to
(a) IgG_1 (b) IgG_2
(c) IgG_3 (d) IgG_4

137. Anti RA Ab's belong to
(a) IgA
(b) IgG
(c) IgE
(d) IgM

138. Type of receptors present in T-cells is
(a) IgA
(b) IgG
(c) Prostaglandins
(d) CD_4

139. Which of these sentences are true?

I. Antibodies to the Rh factor generally agglutinate red blood cells.

II. The agglutination test can be modified to be used for the measurement of soluble antigens

III. In direct immunofluorescence, the antibody specific to the antigen is directly tagged with the fluorochrome.

(a) I, II (b) II, III

(c) I, III (d) I, II, III

140. Active immunity is not acquired by

(a) Infection

(b) Vaccination

(c) Immunoglobulin tranfer

(d) Sub-clinical function

141. Opsonization is by

(a) IgA (b) IgE

(c) IgG (d) IgM

142. Lysozyme is present in the following secretions of the body except

(a) Lacrional secretions (b) CSF

(c) Saliva (d) Both (a) & (b)

143. The process increasing the ability for phagocytosis of foreigm bodies by body is called as

(a) Cross reactivity (b) Opsonization

(c) Immune tolerance (d) Immune surveliance

144. Virus infected cell is killed by

(a) Interferon's (b) Macrophages

(c) Neutrophil (d) Both (a) & (c)

145. The following are true for T-lymphocytes except

(a) Constitute 70-81% circulating pool of lymphocytes

(b) Release macrophage inhibition factor

(c) Secrete specific antibodies

(d) Both (b) & (c)

146. The reagin Ab is

(a) IgA (b) IgD

(c) IgE (d) IgB

147. The commonest IgG with maximum individual variation is

(a) IgG_1

(b) IgG_2

(c) IgG_3

(d) IgG_4

148. Blood transfusion mismatch in erythroblastosis fetalis is a type of

(a) Atopic hypersensitivity

(b) Cytotoxic type

(c) Arthus phenomenon

(d) Delayed hypersensitivity

149. Opsonins are

(a) C_3a (b) C_3b

(c) C_6a (d) C_5-C_9 complex

150. Which of the following cytokinin is secreted by T-helper cells except

(a) IL-1 (b) IL-2

(c) IL-3 (d) IL-5

151. IgA secretion is seen in

(a) Tears and saliva (b) CSF

(c) Hairs (d) Vaginal fluid

152. When an Ag is administered for the 1st time to animal / a human being who never been exposed to it, the 1st Ab to develop is

(a) IgG type (b) IgM type

(c) IgA type (d) IgE type

153. Ag combining site of Ab is

(a) Idiotype (b) Paratope

(c) Epitope (d) Hapten

154. Most important-inflammatory mediator is

(a) TNF (b) IL-2

(c) Interferon (d) PAF

155. Pentameric structure is

(a) IgM (b) IgG

(c) IgA (d) IgD

156. IL-11 is produced by which one of the following

(a) Macrophages (b) Dendritic cells

(c) Stromal cells (d) Both (a) & (c)

157. Group B-cell lymphocyte belongs to

(a) CD-19

(b) CD-69

(c) CD-59

(d) CD-68

158. Difference between NK and Cytotoxic cell is

(a) Interferon's decreased NK activity

(b) NK has CD_4

(c) NK contain azurophilic cytoplasmic granulocyte

(d) All of these

159. The exact part of the Ag that reacts with the immune system is called as

(a) Clone (b) Epitope

(c) Idlotype (d) Effector

160. Immunoglobulin found in B-lymphocytes is

(a) Ig A (b) Ig E

(c) Ig G (d) Ig D

161. Portion of immunoglobulin molecule with molecular weight 50,000 is

(a) Secretory piece (b) H-chain

(c) L-chain (d) J-piece

162. Function of IgA is

(a) Acts as mucosa barrier for infection

(b) Circulating Ab

(c) Kills virus infested cells

(d) Both (a) & (b)

163. Variable portion of Antibody molecule is

(a) C-terminal (b) N-terminal

(c) Intermediate region (d) Carbohydrate moiety

164. HLA-II genes codes in graft rejection

(a) Immunological reaction in graft rejection

(b) Complement

(c) Graft Vs Host system

(d) Immunoglobulin's

165. The structure of antibodies is composed of

(a) Single peptide chain

(b) 2-peptide chain

(c) 2-long and 2-short peptide chains

(d) None of these

166. CD_4 are associated with

(a) Helper T-cells

(b) Suppressor cells

(c) NK cells

(d) T-cells Ag receptor complex

167. Isoantigen is

(a) Found in some but all members of species

(b) Found in some but not all members of species

(c) Occurs in different biological species

(d) All of these

168. Co stimulatory molecule expressed by dendritic cells is

(a) B7 (b) B4

(c) B2 (d) B1

169. 2-basic-4-peptide units are present in which Ig?

(a) IgG (b) IgM

(c) IgE (d) IgA

170. S and additional units are present in which immunoglobulin?

(a) IgG (b) IgM

(c) IgE (d) IgD

171. Sedimentation coefficeint of IgG is

(a) 7s (b) 11s

(c) 19s (d) 8s

172. Electrophoretic mobility of IgE is

(a) γ (b) γ-β

(c) β,γ (d) None of these

173. Correct statement among the following is

(a) ADCC does not require antibody

(b) The cytolysis require complement activity

(c) The cytolysis does not require complement activity

(d) Both (a) & (c)

174. Myeloperoxidase is present in which cells?

(a) T-cells (b) Killer cells

(c) B-cells (d) Macrophages

175. Heavy and light chains of immunoglobulin molecule is combined

(a) Sulphide bridge (b) Sulphate

(c) Peptide (d) Disulfide

176. The major protective Ab in inborn/newborns

(a) IgM (b) IgG

(c) IgD (d) IgA

177. Natural immunoglobulin molecule is

(a) IgM (b) IgG

(c) IgD (d) IgA

178. Pepsin cleaves the immunoglobulin molecule in

(a) 1Fab+Fc (b) 2Fab+Fc

(c) $F(ab)^2$ (d) None of these

179. Active immunity is induced by

(a) Immunization with toxoid

(b) Killed / attenuated organisms

(c) Immunotherapy

(d) Both (a) & (b)

180. Genes involved in the recognition betwee and non-self Ag's

(a) MHC (b) HLA

(c) Transposons (d) None of these

181. Main transplanation antigen's is
(a) ABO Ag's (b) HLA Ag's
(c) Both (a) & (b) (d) None of these

182. Which immunoglobulin is responsible for Rh iso-immunization?
(a) IgG (b) IgM
(c) IgD (d) IgA

183. Functions of MIF is/are
(a) Activation of mobility of macrophages
(b) Inhibition of mobility of macrophages
(c) Both (a) & (b)
(d) Inflammation

184. IL-6 is secreted by
(a) T-cells (b) B-cells
(c) T and NK (d) Leukocytes

185. Which of the following is immunoglobulin molecule marker for B-cells?
(a) IgG (b) IgM
(c) IgG3 (d) IgD

186. Macrophages are transformed from which cells?
(a) Monocytes (b) Leukocytes
(c) Lymonocytes (d) All of these

187. Location of blood macrophages in the body is
(a) Peripheral blood (b) Lungs
(c) Lymphokines (d) None of these

188. Which macrophages are fixed to vascular endothelium?
(a) Alveolar macrophages
(b) Peritoneal macrophages
(c) Kuffer cell macrophages
(d) Both (a) & (c)

189. Cells which do not possess certain surface Ag's are
(a) Immunoglobulin's (b) Macrophages
(c) T-cells (d) Null cells

190. Susceptibility factors exceeding resistance factors means
(a) Good health (b) Disease
(c) Balance (d) All of these

191. Non-specific defense mechanisms includes
(a) Skin, mucous membranes
(b) Phagocytosis
(c) Propredin
(d) Both (b) & (c)

192. Primary characteristics of generalized immune response is
(a) Specificity (b) Anamnesis
(c) Discrimination (d) All of these

193. Which of the following are referred to as incomplete Ag's?
(a) High molecular weight Ag's
(b) Hapten
(c) Ag's which possess immunogenicity and specificity
(d) All of these

194. Hapten possesses
(a) Specificity (b) Immunogenicity
(c) Both (a) & (b) (d) None of these

195. Which antigen possesses both immunogenicity and specificity?
(a) High molecular weight Ag's
(b) Low molecular weight Ag's
(c) Incomplete Ag's
(d) Both (a) & (c)

196. Molecular weight of antigenic determinants is
(a) Less than 1000 daltons
(b) 1000 daltons
(c) Greater than 1000 daltons
(d) None of these

197. Which type of tissue-specific antigen is present in one individual of species but not in another?
(a) Allo Ag (b) Iso Ag
(c) Haptens (d) Both (a) & (b)

NUMERICAL TYPE QUESTIONS

1. How many heavy chain domains are present in the IgE molecule?
2. Half life of IgG molecule is
3. How many sub units does the component C_1 has?
4. What amount of T- Lymphocytes and B-cells are present in peripheral blood?
5. Effective valency of IgM
6. Molecular weight of IgM is
7. Half life of IgG is
8. Distribution of IgA (% intravascular)
9. In human beings, the MHC genes are present on which chromosome?
10. Molecular weight of IL-2 is
11. Number of amino acids included in variable region of heavy chains of antibodies

EXERCISE – II

(QUESTIONS FROM PREVIOUS GATE EXAMS)

MCQ TYPE QUESTIONS

2016

1. Junctional diversity of antibody molecules results from
(a) the addition of switch region nucleotides
(b) the addition of N and P nucleotides
(c) the joining of V, D and J segments
(d) mutations in complementarity-determining regions

2. Find the **INCORRECT** combination.
(a) Surface immunoglobulins – B cell antigen receptor
(b) Affinity maturation – isotype switching
(c) Fc region of antibodies – binding to complement proteins
(d) Spleen, the secondary lymphoid organ – no connection with the lymphatic system

3. Which of the following statement(s) is/are **CORRECT** for antigen activated effector T cells?
P. $CD4^+$ cells make contact with macrophages and stimulate their microbicidal activity
Q. $CD4^+$ cells make contact with B cells and stimulate them to differentiate into plasma cells
R. $CD8^+$ cells make contact with B cells and stimulate them to differentiate into plasma cells
S. $CD8^+$ cells make contact with virus infected cells and kill them
(a) Q only (b) Q and S only
(c) P, Q and S only (d) P, Q, R and S

2015

4. Anergy refers to
(a) mitochondrial dysfunction
(b) allergy to environmental antigens
(c) unresponsiveness to antigens
(d) a state of no energy

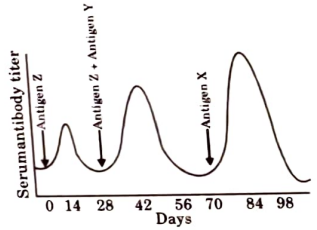
5. Based on the heavy chain, which one of the following antibodies has multiple subtypes?
(a) IgM (b) IgD
(c) IgE (d) IgG

6. Which one of the following complement proteins is the initiator of the membrane attack complex?
(a) C3a (b) C3b
(c) C5a (d) C5b

7. Production of monoclonal antibodies by hybridoma technology requires
(a) splenocytes (b) osteocytes
(c) hepatocytes (d) thymocytes

8. ABO blood group antigens in humans are differentiated from each other on the basis of
(a) sialic acid (b) lipids
(c) spectrin (d) glycoproteins

9. Three distinct antigens X, Y and Z were used to raise antibodies. Antigen Z was injected in a mouse on day zero followed by the administration of antigens X and Y on day 28. A second injection of antigen X was administered on day 70. The antibody titers were monitored in the serum every day and the results are shown below:



Which one of the following statements regarding the antibody titers in the serum is INCORRECT?
(a) Z-specific IgG will be high on day 14
(b) X-specific antibody titer will be high on day 84
(c) X-specific IgG will be high on day 42
(d) Y-specific IgG will be high on day 84

10. Match the cells in Group I with their corresponding entries in Group II.

Group I	Group II
P. Mast cells	1. Activation of the complement pathway
Q. Natural killer cells	2. Expression of CD 56
R. Neutrophils	3. Contains azurophilic granules
S. Dendritic cells	4. Defense against helminthic infection
	5. Production of antibodies specific to bacteria
	6. Contains long membranous projections

(a) P-4, Q-2, R-3, S-5 (b) P-4, Q-2, R-3, S-6
(c) P-3, Q-1, R-2, S-5 (d) P-3, Q-1, R-2, S-6

11. Match the antibiotics in Group I with their modes of action in Group II.

Group I	Group II
P. Chloramphenicol	1. Inhibits protein synth-esis by acting on 30S ribosomal subunit
Q. Rifampicin	2. Interferes with DNA replication by inhibiting DNA gyrase
R. Tetracycline	3. Inhibits protein synth-esis by acting on 50S ribosomal subunit
S. Quinolone	4. Interferes with RNA polymerase activity
	5. Inhibits β-lactamase activity

(a) P-1, Q-2, R-3, S-5 (b) P-3, Q-4, R-1, S-3
(c) P-3, Q-2, R-1, S-4 (d) P-1, Q-4, R-3, S-2

2014

12. Prior exposure of plants to pathogens is known to increase resistance to future pathogen attacks. This phenomenon is known as
(a) systemic acquired resistance
(b) hypersensitive response
(c) innate immunity
(d) antibody mediated response

13. Reactions between antibodies and antigens that are detected by precipitate formation in an agar gel are referred as
(a) immunoprecipitation assay
(b) immunodiffusion assay
(c) immunoaggregation assay
(d) immunofixation assay

14. Cholera toxin increases cAMP levels by
(a) modifying G_i protein
(b) modifying G_s protein
(c) binding to adenylate cyclase
(d) activating cAMP phosphodiesterase

15. Match the items in **Group I** with **Group II**

Group–I	Group–II
P. Receptor tyrosine kinase	1. Inactivation of G-proteins
Q. Cyclic GMP (cGMP)	2. Reception of insulin signal
R. GTPase activating protein (GAP)	3. Thyroid hormone
S. Nuclear receptor	4. Receptor guanylyl cyclase

	P	Q	R	S
(a)	1	3	4	2
(b)	2	4	3	1
(c)	3	1	4	2
(d)	2	4	1	3

16. Match the immunoglobulin class in Group I with its properties in Group II

Group–I	Group–II
P. IgG	1. Major antibody in external secretions such as bronchial mucus
Q. IgA	2. Protects against parasites
R. IgE	3. Antibody that appears first in serum after exposure to an antigen
S. IgM	4. Antibody present in highest concentration in serum

	P	Q	R	S
(a)	4	1	2	3
(b)	3	2	1	4
(c)	4	3	1	2
(d)	1	4	3	2

2013

17. Protein A, which has strong affinity to F_c region of immunoglobulin, is extracted from
(a) *Saccharomyces cerevisiae*
(b) *Staphylococcus aureus*
(c) *Streptococcus pyogenes*
(d) *Streptococcus sanguis*

18. The first humanized monoclonal antibody approved for the treatment of breast canceris
(a) Rituximab (b) Cetuximab
(c) Bevacizumab (d) Herceptin

19. The role of an adjuvant is to
(a) prolong the persistence of antigen
(b) cross link the antigen
(c) increase the size of antigen
(d) avoid inflammation

20. Endogenous antigens are presented on to the cell surface along with
(a) MHC-II (b) MHC-I
(c) $F_c\gamma$receptor (d) complement receptor

21. Which one of the following is an ABC transporter?
(a) multidrug resistance protein
(b) acetylcholine receptor
(c) bacteriorhodopsin
(d) ATP synthase

22. Match the antibiotics in **Group I** with the targets in **Group II.**

Group I	Group II
P. Sulfonamide	1. Peptidoglycan synthesis
Q. Quinolones	2. Peptide chain elongation
R. Erythromycin	3. Folic acid biosynthesis
S. Cephalosporin	4. Topoisomerase

(a) P-3, Q-4, R-1, S-2
(b) P-2, Q-4, R-3, S-1
(c) P-4, Q-1, R-2, S-3
(d) P-3, Q-4, R-2, S-1

2012

23. Idiotypic determinants of an antibody are associated with the
(a) constant region of the heavy chains
(b) constant region of the light chains
(c) variable region
(d) constant regions of light and heavy chains

24. Identification of blood groups involves
(a) precipitation
(b) neutralization
(c) opsonization
(d) agglutination

25. B-lymphocytes originate from the bone marrow whereas T-lymphocytes originate from
(a) thymus
(b) bone marrow
(c) spleen
(d) liver

26. A humanized antibody is one in which the
(a) heavy and light chains are from human
(b) heavy chain is from human and light chain is from mouse
(c) light chain is from human and heavy chain is from mouse
(d) CDRs are from mouse and the rest is from human

27. Nude mice refers to
(a) mice without skin
(b) mice without thymus
(c) knockout mice
(d) transgenic mice

28. Heat inactivation of serum is done to inactivate
(a) prions
(b) mycoplasma
(c) complement
(d) pathogenic bacteria

29. Match the products in Group I with the applications in Group II.

Group I	Group II
P. Digoxin	1. Muscle relaxant
Q. Stevioside	2. Anti-cancer agent
R. Atropine	3. Cardiovascular disorder
S. Vinblastine	4. Sweetener

Codes :

	P	Q	R	S
(a)	1	4	3	2
(b)	3	2	1	4
(c)	3	4	1	2
(d)	2	3	1	4

2011

30. Members of the antibody protein family that have common structural features are collectively known as
(a) haptens (b) allergens
(c) antigens (d) immunoglobulins

31. In ABO blood group system, antigenic determinants are
(a) nucleic acid
(b) carbohydrate
(c) lipid
(d) protein

32. The polymorphic domains for Class II MHC proteins are
(a) α_1 and β_2 domains only
(b) β_1 and α_2 domains only
(c) α_1 and β_1 domains only
(d) α_2 and β_2 domains only

33. Match the viruses in **Group I** with their h cell receptors in **Group II.**

Group I	Group II
P. Hepatitis A virus	1. Heparan sulphat
Q. Human immunodeficiency virus	2. Acetylcholine receptor
R. Rabies virus	3. CD4 protein
S. Herpes simplex virus type I	4. Alpha-2 macroglobulin

	P	Q	R	S
(a)	1	3	2	4
(b)	3	4	1	2
(c)	4	3	2	1
(d)	2	3	1	4

34. Determine the correctness or otherwise of the following **Assertion (A)** and the **Reason (R).**

Assertion : I_gM is found in serum as a pentameric protein consisting of five I_gM monomers.

Reason : The pentameric form of I_gM is due to cross-linking of I_gM monomers via peptide bond.

(a) both (A) and (R) are true and (R) is the correct reason for (A)

(b) both (A) and (R) are true but (R) is not the correct reason for (A)

(c) (A) is true but (R) is false

(d) (A) is false but (R) is true

35. Match items in **Group I** with **Group II**.

Group I	Group II
P. Alzheimer's disease	1. H1N1
Q. Mad cow disease	2. Hemoglobin
R. Sickle cell anaemia	3. Prions
S. Swine flu	4. Amyloid

	P	Q	R	S
(a)	4	3	2	1
(b)	3	4	1	2
(c)	2	1	4	3
(d)	1	2	3	4

2010

36. Hybridoma technology is used to produce

(a) monoclonal antibodies

(b) polyclonal antibodies

(c) both monoclonal and polyclonal antibodies

(d) B cells

37. Interferon-β is produced by

(a) bacteria infected cells

(b) virus infected cells

(c) both virus and bacteria infected cells

(d) fungi infected cells

38. A neonatally thymectomized mouse, immunized with protein antigen shows

(a) both primary and secondary responses to the antigen

(b) only primary response to the antigen

(c) delayed type hypersensitive reactions

(d) no response to the antigen

39. Lymphocytes interact with foreign antigens in

(a) Bone marrow (b) Peripheral blood

(c) Thymus (d) Lymph nodes

40. Dizygotic twins are connected to a single placenta during their embryonic development. These twins

(a) have identical MHC haplotypes

(b) have identical T_H cells

(c) have identical T cells

(d) can accept grafts from each other (both (a) and (b)

2007

41. Match the recombinant products in group 1 with their therapeutic applications in group 2

Group 1

P. Human growth hormone

Q. Platelet growth factor

R. Factor VIII

S. Erythropoietin

Group 2

1. Pituitary dwarfism
2. Chemotherapy inducred thrombocy-topenia
3. Haemophilia
4. Anaemia associated with chronic renal failure

(a) P-l, Q-2, R-3, S-4 (b) P-2, Q-1, R-3, S-4

(c) P-l, Q-4, R-3, S-2 (d) P-2, Q-4, R-3, S-l

42. Presence of $CX_{2-4}\ CX\phi X_8HX_3H$ sequence in a protein suggest that it is

(a) a protein kinase

(b) GTP binding protein

(c) zinc finger protein

(d) lipase

43. Oils rich in PUFA are NOT desirable for bio-diesel production because

(a) they form epoxides in presence of oxygen

(b) they do not form epoxides in presence of oxygen

(c) they have high ignition temperature

(d) they solidify at low temperature

2006

44. Which of these mice fail to develop a thymus ?

(a) Motheaten mice (b) Beige mice

(c) Knock out mice (d) Nude mice

NUMERICAL TYPE QUESTIONS

2014

1. In a mouse genome, the numbers of functional Vα, Jα, Vβ, Dβ, Jβ gene segments are 79, 38, 21, 2 and 11, respectively. The total number of possible combinations for αβ T cell receptors are ___$\times 10^6$.

ANSWERS

EXERCISE – I

MCQ Type Questions

1. (d)	2. (d)	3. (a)	4. (a)	5. (d)	6. (d)	7. (c)	8. (c)	9. (c)	10. (b)
11. (a)	12. (a)	13. (a)	14. (d)	15. (a)	16. (a)	17. (c)	18. (c)	19. (a)	20. (d)
21. (d)	22. (a)	23. (a)	24. (b)	25. (d)	26. (c)	27. (d)	28. (a)	29. (a)	30. (a)
31. (c)	32. (b)	33. (c)	34. (d)	35. (a)	36. (b)	37. (d)	38. (c)	39. (d)	40. (c)
41. (a)	42. (c)	43. (a)	44. (d)	45. (c)	46. (b)	47. (c)	48. (d)	49. (b)	50. (a)
51. (b)	52. (c)	53. (c)	54. (c)	55. (d)	56. (c)	57. (d)	58. (a)	59. (c)	60. (d)
61. (c)	62. (d)	63. (c)	64. (a)	65. (a)	66. (c)	67. (c)	68. (b)	69. (c)	70. (b)
71. (b)	72. (b)	73. (b)	74. (b)	75. (a)	76. (d)	77. (d)	78. (b)	79. (b)	80. (b)
81. (b)	82. (a)	83. (c)	84. (b)	85. (d)	86. (a)	87. (a)	88. (c)	89. (c)	90. (d)
91. (c)	92. (d)	93. (a)	94. (b)	95. (b)	96. (b)	97. (b)	98. (d)	99. (c)	100. (d)
101. (b)	102. (a)	103. (d)	104. (d)	105. (c)	106. (c)	107. (d)	108. (a)	109. (d)	110. (b)
111. (d)	112. (b)	113. (c)	114. (a)	115. (d)	116. (a)	117. (d)	118. (a)	119. (b)	120. (d)
121. (a)	122. (d)	123. (d)	124. (d)	125. (b)	126. (d)	127. (c)	128. (c)	129. (d)	130. (d)
131. (c)	132. (d)	133. (d)	134. (d)	135. (a)	136. (a)	137. (d)	138. (d)	139. (b)	140. (c)
141. (c)	142. (b)	143. (b)	144. (a)	145. (d)	146. (c)	147. (b)	148. (b)	149. (b)	150. (a)
151. (a)	152. (b)	153. (b)	154. (a)	155. (a)	156. (c)	157. (a)	158. (b)	159. (b)	160. (d)
161. (b)	162. (a)	163. (b)	164. (b)	165. (c)	166. (a)	167. (b)	168. (a)	169. (d)	170. (b)
171. (a)	172. (a)	173. (c)	174. (d)	175. (d)	176. (b)	177. (a)	178. (c)	179. (d)	180. (a)
181. (b)	182. (a)	183. (c)	184. (d)	185. (d)	186. (a)	187. (d)	188. (c)	189. (d)	190. (b)
191. (d)	192. (d)	193. (b)	194. (a)	195. (a)	196. (a)	197. (d)			

Numerical Type Questions

1. 5	2. 20 days	3. 3	4. T-80% B-20%	5. 5	6. 9,70,000
7. 20 days	8. 5	9. 6	10. 15,000	11. 110	

EXERCISE – II

MCQ Type questions

1. (b)	2. (b)	3. (c)	4. (c)	5. (d)	6. (d)	7. (a)	8. (d)	9. (d)	10. (b)
11. (b)	12. (a)	13. (b)	14. (b)	15. (d)	16. (a)	17. (b)	18. (d)	19. (a)	20. (b)
21. (a)	22. (d)	23. (c)	24. (d)	25. (b)	26. (d)	27. (b)	28. (c)	29. (c)	30. (d)
31. (b)	32. (c)	33. (c)	34. (c)	35. (a)	36. (a)	37. (c)	38. (b)	39. (d)	40. (d)
41. (a)	42. (c)	43. (a)	44. (d)						

Numerical Type Questions

1. (1.38 to 1.40)

EXPLANATIONS

EXERCISE - I

MCQ TYPE QUESTIONS

1. They are polypeptide growth factors secreted by leucocytes. More than 15 interleukins are isolated.

   IL-1 is secreted by Monocyte, leukocyte

2. Multiple myeloma (plasma cytoma)

   Ig secreting cells are transformed into malignant cells, one done is enormously proliferated. Thus Ig molecules of the very same type are produced in large quantities.

3. Ig's are secreted by plasma cells belonging to the B-cells of lymphocytes.

4. ELISA-Indirect ELISA method is used to detect Antibody (Ab).

   Method : Ag from HN is coated in the wells of a multiwell plate. Patient's serum is added & incubated. If it contains the Ab; it is fixed and the wells are washed. This is to remove excess Ab's in serum. Next a second Ab(Ab against human Ig) conjugated with HRP. Then colour reagent containing H_2O_2 & diaminobenzidine is poured over. If brown colour developed, means that the Ab was originally present in the parent's serum.

6. Ag – Ab reaction leads to activation of complement system, which destroys the foreign cells.

   The Ab's can destroy the target cells by the following mechanisms.

   (I) Classical complement pathway

   (II) Antigen dependent cell-mediated cytotoxicity

   (III) Agglutination

   (IV) Opsinization

7. IgE fixes to mast cells & basophils : They are cytophilic ab's and mediate allergy, hypersensitivity and anaphylaxis. They have property to fix on mast cells and basophils. When certain Ab's such as penicillin are injected few times, IgE class Ab's are produced which anchor on mast cells, when the same chemicals is injected next time, the Ag's fix on such Ab's, causing most cell degranulation and release of histamine and slow reacting substance.

8. The 'J' chain connects the dimens. The dimens are stabilized against proteolytic enzymes by the secretary piece. This secretary piece is produced in liver, nelches to the intestinal mucosal cells, where it combines with IgA dimmer to form the secretary IgA which is then released.

9. T-cells when stimulated by Ag's, liberate many soluble substances called aytokines/Lymphokines

10. The idiotypic determinants are located in the variable parts of Ab. (i.e., Ag reorganization site).

11. When tuberculin is injected intradermally to a patient with tuberculosis, an erythematous inducted lesion develops slowly reading it's maximum within 48-72 hours. This is called delayed type hyper sensitivity. Hypersensitivity is the over reaction of the immune system, often resulting in unwanted tissue destruction.

12. In this condition IgM level in blood is increased considerably with monoclonal peak. This is due to malignant proliferation of IgM clones.

13. MHC genes are present on chromosome 6 and they are A, C, B, D and DR loci. All these loci together contain more than 150 alleles.

14. IFN- α - secreted by leucocytes

    Function – Proliferation of Macrophages

    IFN - β - secreted by Fibroblast

    IFN-γ -T & NK (Natural killer)

    Function – Antiviral differentiates cells

15. IgG again subdivided into I_gG, $G_{2/}$ G_3 & G_4 depending on the amino acid sequence of the heavy chain constant region.

16. Indirect ELISA is useful to detect the small quantities of Ab's in the blood

    Ex :– Test for detection of HIV antibody

18. Here electrophoretic separation is followed by Ag-Ab reaction. The electrophoresis is carried out first by applying the patient's serum into the wells but out in the agar/agarose gel.

23. B cells and T cells recognize different substances as antigens and in a different form. The B cell uses cell surface-bound immunoglobulin as a receptor and the specificity of that receptor is the same as the immunoglobulin that it is able to secrete after activation.

24. Antigen processing and presentation are processes that occur within a cell.

25. Antigen processing and presentation are processes result in fragmentation (proteolysis) of proteins, association of the fragments with MHC molecules, and expression of the peptide-MHC molecules at the cell surface where they can be recognized by the T cell receptor on a T cell.

26. The three main types of antigen presenting cells are dendritic cells, macrophages and B cells, although other cells, that express class II MHC molecules, (e.g., thymic epithelial cells) can act as antigen presenting cells in some cases. Dendritic cells, which are found in skin and other tissues, ingest antigens by pinocytosis and transport antigens to the lymph nodes and spleen.

27. Super antigens are antigens that can polyclonally activate T cells to produce large quantities of cytokines that can have pathological effects. These antigens must be presented to T cells in association with class II MHC molecules but the antigen does not need to be processed.

28. Functional T cells in the periphery have to recognize foreign antigens associated with self MHC, because APC or target cells present foreign antigen associated with self MHC. However, an individual does not need functional T cells in the periphery that recognize antigen (self or foreign) associated with foreign MHC. An individual especially does not want functional T cells in the periphery that can recognize self antigens associated with self MHC because they could lead to damage of healthy, normal tissues.

29. Since B cells are not MHC-restricted there is no need for positive selection of B cells. However, negative selection (i.e., elimination of self-reactive clones) of B cells is required. This occurs during B cell development in the bone marrow. However, negative selection of B cells is not a critical as for T cells since, in most instances, B cells require T cell help in order to become activated. Thus, if a self reactive B cell does get to the periphery it will not be activated due to lack of T cell help.

30. A given T cell is restricted to recognize a peptide antigen only when it is bound to self-MHC molecules.

31. CD4 binding to Class II MCH, which ensures that Th cells only interact with APCs

32. The principal functions of B cells are to make antibodies against antigens, perform the role of antigen-presenting cells (APCs) and eventually develop into memory B cells after activation by antigen interaction. B cells are an essential component of the adaptive immune system.

33. The path leading to the association of protein fragments with MHC molecules differs for class I and class II MHC. MHC class I molecules present degradation products derived from intracellular (endogenous) proteins in the cytosol. MHC class II molecules present fragments derived from extracellular (exogenous) proteins that are located in an intracellular compartment.

34. I. Cell-cell interactions of the adaptive immune response are critically important in protection from pathogens, These interactions are orchestrated by the immunological synapse whose primary components are the T cell antigen receptor (TCR) and the Major histocompatibility complex (MHC) molecule. The major function of the TCR is to recognize antigen in the correct context of MHC and to transmit an excitatory signal to the interior of the cell. Since binding of peptide within the MHC is not covalent, there are many factors while help stabilize the immunological synapse.

    II. Class I MHC molecules are composed of two polypeptide chains, a long α chain and a short β chain called β2-microglobulin.

    III. The α chain has four regions.

35. The TCR is closely associated with a group of 5 proteins collectively called the CD3 complex The CD3 complex is composed of one 1γ, one 1δ, two 2ε and 2ε chains. All of the proteins of the CD3 complex are invariant and they do not contribute to the specificity in any way. The CD 3 complex is necessary for cell surface expression of the TCR during T cell development. In addition, the CD3 complex transduces activation signals to the cell following antigen interaction with the TCR.

36. Immature B cells are produced in the bone marrow of most mammals. Rabbits are an exception; their B cells develop in the appendix-sacculus rotundus. After reaching the IgM+ immature stage in the bone marrow, these immature B cells migrate to the spleen, where they are called transitional B cells, and some of these cells differentiate into mature B lymphocytes.

37. When the B cell fails in any step of the maturation process, it will die by a mechanism called apoptosis, here called clonal deletion. B cells are continuously produced in the bone marrow. When B cell receptors on the surface of the cell matches the detected antigens present in the body, the B cell proliferates and secretes a free form of those receptors (antibodies) with identical binding sites as the ones on the original cell surface. After activation, the cell proliferates and B memory cells would form to recognise the same antigen. This information would then be used as a part of the adaptive immune system for a more efficient and more powerful immune response for future encounters with that antigen.

38. B-1 cells express IgM in greater quantities than IgG receptors show polyspecificity, meaning that they have low affinities for many different antigens, but have a preference for other immunoglobulins, self antigens and common bacterial polysaccharides. B-1 cells are present in low numbers in the lymph nodes and spleen and are instead found predominantly in the peritoneal and pleural cavities.

39. I. B cells recognize their cognate antigen in its native form. They recognize free (soluble) antigen in the blood or lymph using their BCR or membrane bound-immunoglobulin.

II. In contrast, T cells recognize their cognate antigen in a processed form, as a peptide fragment presented by an antigen presenting cell's MHC molecule to the T cell receptor.

III. A critical difference between B cells and T cells is how each lymphocyte recognizes its antigen.

40. Antigens are usually proteins or polysaccharides. This includes parts (coats, capsules, cell walls, flagella, fimbrae, and toxins) of bacteria, viruses, and other microorganisms. Lipids and nucleic acids are antigenic only when combined with proteins and polysaccharides. Non-microbial exogenous (non-self) antigens can include pollen, egg white, and proteins from transplanted tissues and organs or on the surface of transfused blood cells. Vaccines are examples of immunogenic antigens intentionally administered to induce acquired immunity in the recipient.

41. Super antigen - A class of antigens which cause non-specific activation of T-cells resulting in polyclonal T cell activation and massive cytokine release.

42. The substance of a lymph node consists of lymphoid follicles in the outer portion called the "cortex", which contains the lymphoid follicles, and an inner portion called "medulla", which is surrounded by the cortex on all sides except for a portion known as the "hilum". The hilum presents as a depression on the surface of the lymph node, which makes the otherwise spherical or ovoid lymph node bean-Shaped. The efferent lymph vessel directly emerges from the lymph node here. The arteries and veins supplying the lymph node with blood enter and exit through the hilum.

43. The Humoral Immune Response (HIR) is the aspect of immunity that is mediated by secreted antibodies (as opposed to cell-mediated immunity, which involves T lymphocytes) produced in the cells of the B lymphocyte lineage (B cell).

44. Isograft

45. Avidity is influenced by both the valence of the antibody and the valence of the antigen.

46. Serum sickness is type-II hypersensitivity

Type of rejection	Type of HS	Target sites in transplantation
Hyper acute rejection	Type II	Small blood vessels
	Type III	in donor tissues
Acute rejection	Type II cytotoxic	Parenchymal cells
	Type III	small blood vessels
Chronic rejection	Type III Type IV	Small blood vessels

47. IgM.

The most widely complement fixing Ab

48. Schick test

Neutralization test

Virus : e.g. plague inhibition test

Bacteria : Invivo; e.g. Schick test; Invitro : ASO test.

50. RIA

51. Precipitation : When soluble Ag combines with its Ab in presence of electrolytes, it forms insoluble precipitates/floccules.

57. Lymphocytes

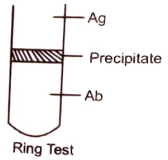
Type IV hypersensitivity is mediated by sensitized T-lymphocytes which can cause direct target cell-mediated lysis and the release of soluble lymphokines. Since these cells arrive at a skin site containing Ag 24 to 72 hrs after infection producing a typical hard swelling, this type of reaction is called a delayed reaction.

67. L-chain disease

When Ig secreting cells are transformed into malignant cells, one clone alone is enormolisly proliferated. This Ig molecules of the very same type are produced in large quantities. This is seen in electrophoresis as the myeloma band or monoclonal band or M-band with sharp narrow spike.

69. Aggiutination test used in laboratory diagnosis is the Weil-Felix reaction. This test is based on the fact that several of rickettias have common AG's with strains of proteins spp; therefore serum from patients with rickettisial infection agglutinates suspensions of the proteins organisms.

74. Precipitation : When soluble Ag combines with its Ab in presenc21hite disk at the interface between the clear solution for Ag and Ab.



Ring Test

75. Agglutination occurs when bacterial cell antigens are exposed to homologous antibodies.

76. A concentration and infectives filter of the virus

The amount/intensity of the agglutination depend on the concentration and infective titer of the virus. The procedure consists in making serial dilutions of the virus antigen.

77. The exact technique for conducting a hemagglutination test varies with the virus, i.e. antigen. This test is useful for serological diagnosis of many viruses including the influenza virus, mumps and the ECHO viruses.

78. In the case of the influenza / other viruses which cause hemagglutination, an increase in the Ab which inhibits this agglutination (neutralizes the viral agglutination) indicates the presence of the virus of the type of the antigen employed.

79, 80. Complement-fixation tests

Complement fixation tests are based on the presence of complement-fixing Ab's in serum. These Ab's are produced in an animal body stimulated by bacterial Ag's. In the presence of the Ab's complement causes lysis of the specific bacterial cells. The purpose of the complement fixation test is to determine whether specific bacterial Ab's are present in serum.

81. One is the bacteriolytic / complement – fixing system in which serum, bacterial suspension and complement are mixed. If the Ag and Ab in the serum are capable of union the complement is said to be fixed.

82, 83. Wassermann test

The complement fixation test is widely used in the laboratory diagnosis of many infectious diseases, best known applications of the complement-fixation test is the Wassermann test for syphills.

84. L-chain

Bence-Jones proteins

Monoclonal Ig consist of L-chain found typically in multiple myeioma

These proteins have special property of precipitation when heated between 45°C and 60°C, but redissolving at higher than 80°C and lower than 45°C.

85. Zone phenomenon (seen in agglutination and precipitation) consists of 3-parts

i. Prozone = Ab excess = weak / absent precipitation reaction = False – ve

ii. Zone of equivalence = peak amount of precipitation

iii. Post zone = Ag excess = weak / absent per reaction

86. 1. Measures enzyme labeled ag, hapten / ab

2. It may be homogenous / heterogenous

3. Major type of ELISA which involves the use of immunosorbent – an absorbing material specific for one of components of reaction, the Ag/Ab

e.g. : ELISA for detection of Retrovirus in fetus / HIV ab in serum.

87. C_2, C_4, C_3 decreased complements are activated by 3 pathways.

In classic pathway level of C_1 to C_9 decreases within alternate pathway all complement level decreases except C_1, C_2, C_4.

88. Ag, Ab reaction cannot occur in the absence of electrolytes.

89. The combining site of an antibody is located in the Fab portion of the molecule and is constructed from the hypervariable regions of the heavy and light chains.

91. IgA of 2 types

Serum IgA – monomeric, synthesized by plasma cells

Secretory IgA (SIgA) – 2 monomers joined by J-chain

J-chain is synthesized by plasma cells near mucosa / glandular epithelium while secretory component / secretory piece of SIgA is synthesized by mucosal / glandular epithelial cells not by lymphoid cells.

92. All Type-I : Immediate hypersensitivity reactions

Large amount of histamine released

Cytokines like IL-4, IL-5 and IL-6 and GMCSF are released

94. Agglutination Tests – The Widal test was devised specifically to aid in diagnosis of typhoid fever by agglutinating typh oid bacilli with patient's serum, but the term is sometimes loosely applied to other agglutination tests using heat – killed cultures of organisms other than Salmonella typhi.

98. Combined Immuno deficiencies is (B and T cell defect)

Syndrome

Celular Immunodeficiency with abnormal immunoglobulin synthesis

(Nezelof syndrome)

ataxia telangiectasia

Wiskott Aldrich syndrome

Severe combine immuno deficiency

Defect

Abnormal T-cell maturation in thymus with normal

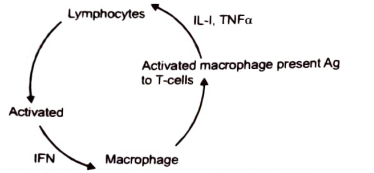
Increase or decrease immunoglobulin DNA repair defect WASP gene mutation

Adenosine deaminase deficiency

102. By Leukotriene is the adhesion factor for the neutrophill on the cell surface to attack to endothelium.

103. Graft rejection is not an immune complex disease.

108. T-lymphocyte activation



Main source	Major Function
IL-1 Macrophage	Proliferation and differentiation of T, B cells; pyrogenic; acute phase reactions; BM cell proliferation
IL-2 TH1 cells	Growth and differentiation T and B cells; cytotoxicity of T and NK cells
IL-3 T cells	Stimulation of hematopoletic grogenitors
IL-4 TH2 cells	Proliferation of B and cytotoxic T cells; IgG and IgE production; MHC class II and IgE receptor.
IL-5 TH2 cells	Proliferation of Eosinophils; IgA and IgE production
IL-6 TH1, macrophage	Promote B-Cell differentiation, IgA production Acute phase proteins.
IL-7 Spleen, BM Stromal cells	B and T cell growth factor

109. There is ab induced proliferation of NK cells, NK cell activity is non-immune; MHC unrestricted; non-ab mediated killer of target cells which are usually malignant cell types, transplanted foreign cells/virus infected cells.

110. Albumin.

Glycoprotein are proteins to which usually 2-10 oligosaccharides are covalently attached. Albumin is not a glycoprotein.

111. ABO group specific substances are found in high concentration in saliva, semen, vaginal secretion and gastric juice and while in low concentration in sweat, tears and wine, so it is possible to determine blood group from from an examination of these secretions.

Thus, they are present in all tissues except CSF.

112. Light chain

Bence-Jones proteins are identified in wine by its characteristic property of coagulation when heated at 50°C and dissolved at 70°C.

Bence-Jones protein (abnormal Ig) are light chain of Ig found typically in multiple myeloma.

Light chain consist of either K or λ, never both.

113. Kappa and Lambda chains.

Already explained, please see the answer No. 5

114. ↑ ed expression MHC class I molecules

Target cell killing by NK cells.

1. Inversely related to target cell expression of MHC class I molecules. So, if kills the cell that express little/no HLA class I molecule (Provide immune surveillance) such as virus infected cells, certain tumor cells and allogenic cells.
2. Non-Immune that is without previous sensitization, MHC unrestricted and non-ab mediated.
3. It kill host cell infected with intracellular bacteria Eg: Mycobacteria, TB.
4. Not kill cells which express class I MHC

115. Tumour necrosis factor-α, and Tumour necrosis factor - β. MHC class genes is classified as class I, class II, class-III

Class III includes
- soluble protein of complement system.
- Properdin factor β of alternative pathway.
- TNF alpha and β
- Heat shock protein.

116. Cytotoxic T-cells.

1. CD-8 cytotoxic killer cells secrete soluble mediators like granzymes and perforins (role in Type IV HSN).
2. Perforins cause drilling holes in the target cells through which water enter cells, cause osmotic lysis.
3. Pores also allow granzymes to enter target cells which activate apoptosis of target cells.

117. Half life of IgG is 23 days; IgD is heat stable; and IgE has highest carbohydrate content.

IgD has highest % of carbohydrate but according to Harrison – it is IgE.

Half life of Ig : IgG 23 days > IgA 6 days > IgM 5 days > IgD > IgE.

119. Inhibits viral replication in cells,.
- IFN has no direct action viruses but inhibit viral replication by selectively inhibiting translation of viral m-RNA without affecting cellular m-RNA.
- IFN are not virus specific but species specific.
- 3 types

IFN α - Antiviral activity

IFN β - Direct antitumor effects.

IFN γ - Regulate macrophage and NK cells activation

120. Antibodies appear in short time; and persist for long Secondary immune response is characterized by
1. Prompt, powerful, prolonged response.
2. Much higher level of ab's that last for long periods.
3. Short, negligible lag phase.
4. Ab is predominantly IgG.

121. 1. Most naturally Ag are proteins and polysaccharides.
2. Protein are better Antigen than polysaccharides.
3. All proteins except gelatin is antigenic.

122. Immunoglobulin species specific.

Reacts with cells and the affected cell then become persistent to a number of different viruses.

123. All of the above
1. Production of interferon
2. Lymphokine production
3. Rosette formation

124. IgM Binary (Secondary) exposure to an antigen results in sudden increase in IgM. It produced primary response to antigen.

125. When soluble antigen combines with its antibody in presence of electrotypes, it forms insoluble precipitates/floccules.

→ ppt reaction show "zone phenomenon" (also in agglutination) either in the form of pro zone (ab excess) or post zone (ag excess).

→ ppt occurs in zone of equivalent due to lattice form (also in hemagglutination).

126. Derived from plasma cells.

127. Thymus

T-Lymphocytes, - Thymus designed lymphocytes consist of 60-70% of peripheral lymphocyte.

128. T4/T8 ratio reversal seen in AIDS.

129. Platelets

Leucotrions are secreted by macrophages, T4, T8 cells.

130. Cases of paternal, dispute

HLA Antigens – The T-cell recognition is also closely related to with the HLA (human leucocytes antigen) system. When organs are transplanted, the donor and receiver are matched for HLA system.

131. Migration inhibition test.

132. Presence in spleen.

133. Mucosal epithelium.

IgA - 2 types.
1. Serum IgA
2. Secretory IgA; SIgA (Dimer joined by J chain present on respiratory/Intestinal mucosal and in secretions).

J chain is also produced by plasma cells but secretory pieces/secretory component of SIgA is not produced by lymphoid cells but mucosal/ glandular epithelial cells.

134. Plague.

135. Virus

NK cell / Target cell killing by NK cell is :

– Inversely related to target cell expression of MHC class I molecule

– so, it kills the cell that express little/no HLA class I molecule (provide immunosurveillance) such as virus infected cells, certain tumour cells and allogenic cells.

136. IgG1

IgG General purpose ab, enhances phagocytosis by Opsonization.

– It has 4 subclasses G1 > G2 > G3 > G4.

– IgG decreased from birth to reach immunity levels by 3rd month.

– Types of antibody in various conditions.

IgG-Ab against Rh factor (Anto Rh-D).

137. IgM

Types of ab in various conditions

IgM – Biological false positive Ab in syphilis.

– Rheumatold factor

– Ab against ABO

– Ab to typhoid O Ag.

138. All T-lymphocyte contain CD-3 molecule which are involved in signal transduction L. Other surface molecules/co-receptors includes CD2/CD4/ CD8, CD110, CD-40.

→ CD4 expressed on 60% T-cells, while 30% expressed CD8

139. I. Antibodies to the Rh factor generally do not agglutinate red blood cells.

II. The agglutination test can be modified to be used for the measurement of soluble antigens

III. In direct immunofluorescence, the antibody specific to the antigen is directly tagged with the fluorochrome

140. Ig transfer.

141. Opsonization is by IgG.

IgG can bind with microbes and the Fc portion can fix with macrophages, so that phagocytes is made easier this is called Opsonization effect of ab.

142. CSF.

143. Opsonization

IgG can bind with microbes and the Fc portion can fix with macrophages, so that phagocytes is made easier this is called Opsonization effect of ab.

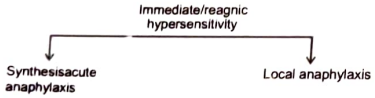
145. Secrets specific antibodies, release MIF

T – lymphocytes constitute 70-80% or peripheral lymphocyte.

T – Lymphocytes are involved Cell-mediated immunity.

MIF is secreted by T-Cells

146. IgE Type-I HS



147. IgG2 sub

IgG molecule is 4 types

IgG1, IgG2, IgG3 and IgG4

148. Cytotoxic type

Type-II (Cytotoxic/cytolytic) hypersensitivity

Mediated by IgM and IgG. It is complement dependent mediates cytolysis/ADCC.

Unfortunately type-II hypersensitivity can also damage the host because of the circulating ab's. Thus it also responsible for clinical transfusion reactions (mismatching), erythroblastosis fetalls (Rh isoimmune reactions) and some auto immune reactions.

149. C_3b.

C_3b-opsonins

C_3a and C_5a are anaphylatoxic (histamine releasing) and chemotactic. C567 is chemotactic and also brings about reactive lysis.

An important function of 'C' is to facilitate the uptake and destruction of pathogens by phagocytic cells. This opsonic effect is based on the presence on the surface of phagocytic cells.

150. IL1

- IL1 is secreted by monocyte/leukocyte.
- mwt 17000
- Target cell/function – Induces OL-2 receptor, induces acute phase proteins

151. Tears and saliva

IgA: IgA is 2nd most abundant class. It is the major immunoglobulin in the colostrums, saliva and tears.

152. IgM type

1. IgM: It is composed of 5 H_2L_2 and 1 J chain
2. It is presence in the serum indicates recent infection (Primary responses)
3. It has highes avidity among al Ig.
4. By 20th week fetus produce IgM, IgD and receives maternal IgG so IgA and IgE are not present.

153. Paratope

- It is the combining are of the ab molecule corresponding to the epitopic.
- Epitope and paratope determine specificity which is hallmark of immunological reaction.
- Bacteria/virus/ray contain ag mosaic (different epitopes) while same epitope on different ag may present causing antigenic cross reaction.
- Specific ag deteminants on paratope are called IDIOTOPES.

154. TNF

TNF-Tumar necrosis factor. It is 2 types

(1) INF-α and (2) INF-β

INF-α is secreted by macrophages. It's MW of is 17000

Function - Inflammation, fibrosis, pyrexia, acute phase proteins, necrosis.

155. IgM

IgM : IgM, a millionaire molecule composed of 5 H_2L_2 and 1 J-chain and is heaviest Ig.

IgM. Pentamer

156. Stromal cells

IL-11 is secreted by stromal cells.

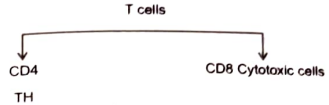
Target cell/Function - stem cell stimulator of B cells and megakaryocytes

157. CD-19

Leukocyte differentiation antigen

CD No	Cell type association
CD1	Thymocytes, Langerhans cells
CD2	T cells SRBC receptor
CD4	Helper T cell
CD8	Suppressor/Cytotoxic cells
CD19	B-Cells

158. NK has CD4



IFNγ favors differentiation of TH L cell so NK cell can influence CD4 and B-Cell

159. Epitope

Paratope

- It is the combining are of the ab molecule corresponding to the epitopic.
- Epitope and paratope determine specificity which is hallmark of immunological reaction
- Bacteria/virus/ray contain antigen mosalc (different epitopes) while same Epitope of different antigen may present causing antigenic cross reaction.
- Specific antigen determinants on paratope are called IDIOTOPES.

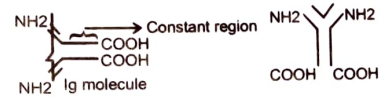
160. IgD is found on the surface of immune B lymphocytes and may be involved in their maturation and regulation

161. H-Chain

162. Mucosa barrier for infection

Secretory IgA confers immunity to infection by enteric bacterial and viral organisms. It presents attachments of bacterial and virus to mucous membrane.

163. N-terminal



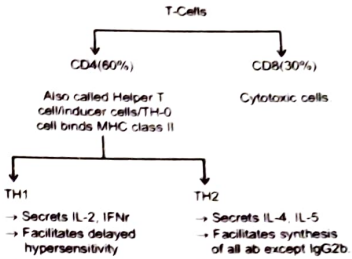
164. Complement

HLA II genes : Soluble protein of complement system (C_2 and C_4 of classical pathway : Properdin factor B of alternative pathway); heat shock proteins, TNF-α and β.

165. 2 Long and 2 short peptide chains

Ig molecule (glycoprotein) consist of 2 pairs of polypeptide chains (2H and 2L)

166. Helper T-Cells.



167. Found in some but not all members of species.

168. Dendritic cells (DCs), which have the broadest range of antigen presentation, and are probably the most important APC. Activated DCs are especially potent T_H cell activators because, as part of their composition, they express co-stimulatory molecules such as B7.

169. IgA

Heavy chain	IgG	IgA	IgM	IgD	IgE
Domains	4	4	5	4	5

170. IgM

IgG - IgG molecule is not having additional unit in their structure and also IgG and IgE. But IgA is having S and J additional units, IgM is having "J" piece

171. IgG constitutes about 75-80% of total immunoglobulin's in circulation. One to sedimentation co-efficient, it is sometimes referred to as Sig and also IgD molecule.

172.

Average electrophoretic Mobility	IgG	IgA	IgM	IgD	IgE
	γ	γ-β	γ-β	β-γ	γ

173. The cytolysis does not require complement activity

ADCC

ADCC requires ab, but the cytolysis does not require complement activity. The effector cells are neither T/B cells but are NK cells. The specificity of this reaction resides in the ab molecule.

174. Macrophages

The myeloperoxidase present inside the phagocytes which destroys the bacteria.

175. Disulfide bridge.
Immunoglobulin molecule is made up of a 2 heavy (H) chains and 2 light chains, combined through disulfide bridges.

176. IgG crosses placenta. It is the major antibody in newborns. These maternal ab's are seeing neonatal circulating up to 2-4 months.

177. IgM. Natural ab's are also in IgM nature.

178. $F(ab)^2$
Pepsin proteolyte enzyme clears the Ig molecule in ti tragments (2 Fab together and Fc)

179. Immunization with toxoid.
Killed/attenuated organisms.

180. MHC
The genes of major histocompatibility complexes (MHC) are involved in the recognition between self and non-self ag's. In humans the MHC genes are present on chromosome no. 6 there are A, B, C, D and DR loci.

181. HLA antigens when organs are transplanted, the donor and recelyr are matched for HLA system. These are main transplantation ag's responsible for rejection of allograft.

182. IgG.
IgG crosses placenta. It is the major proective antibody in new borns. These maternal antibodies are seen in neonatal circulation up to 2-4 months. Placental crossing of IgG also explains the Rh is immunisation. This occurs when mother is Rh negative and fetus is Rh positive and when ABD system ag's are similar to both mother and fetus. During maturation, fetal RBCs may enter into maternal arculation. Leading to formation of anti Rh ab's. Dering next pregnancy these ab's, being IgG in class can enter into fetal circulation causing fetal hemolysis, neonatal jaundice and in severe cases. Neonatal death/miscarriage. Passing injected anti Rh ab when injected within 24 hrs of delivery of 1st child, will revert the iso immunization and can protect future pregnancy

183. Activation of mobility of macrophage
Inhibition of mobility of macrophage
MIF macrophage migration inhibition factor.
It has molecular weight 50,000, secreted by T-cells. Target cell/function of MIF is activation and inhibition of mobility of macrophage.

184. Leucocytes
IL-6 molecular weight 20,000 and secreted by leucocytes
Function : B cell and myeloid differentiation, includes actute phase proteins.

185. IgD
→ IgD is found on the surface of immature B-lymphocytes and may be involved in their maturation and regulation.
→ It is a marker for mature B-Cells.

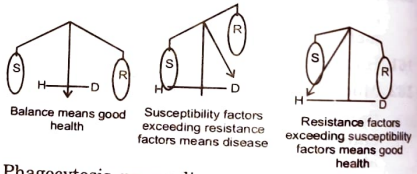
186. Monocytes
Monocytes are not very phagocytic untill they are stimulated by lymphokines, another type of protein that influences how the immune system responds. Monocytes develop into actively phagocytic macrophages.

187. None
Blood macrophages (Monocytes) are largest lymphoid cells.
Tissue macrophages (histocytes) are
– microglia in CNS, – Alveolar macrophages in legs,
– Kuffer cells in liner, – Osteoclast in bone,
– Sims histocytes spleen, lymph node.

188. Kuffer cells macrophages.
Two types of mature macrophages are recognized.
1. Those wondering in tissues and body spaces (ex; alveolar and peritoneal macrophages)
2. Those fixed to vascular endothelium (ex; Kuffer cells and fixed macrophages of the spleen and lymph nodes).

189. Null cells. Recently, a subpopulation of lymphocytes has been discovered which are referred to as null cells because they do not possess certain surface Ag's.

190. Disease.



Balance means good health
Susceptibility factors exceeding resistance factors means disease
Resistance factors exceeding susceptibility factors means good health

191. Phagocytosis, propredin.
In phagocytosis, the phagocyte extends small pseudopods around the microbe. After adherence, these pseudopods fuse and form a vacuole by means of imagination of the phagocyte plasma membrane engulfing / surrounding the bacterium. The vacuole is now called a phagosome. Subsequent events depend on the activity of the lysosomal granules. These more toward the phagosome, fuse with its membrane to form a phagolysome and discharge their contents (hydrolytic enzymes) into the vacuole, thus irritating and intracellular killing and digestion of the microbe. Within the phagocytosome, most microbe are killed in minutes, although the coplete degranulation/degradation may take a few hours.

192. All. Specificity, Discrimination.
Immune responses are processes in which animal form specific creative protein (Ab's) and cells in response to great variety of foreign organic macromolecules molecules. The generalized immune response has 4 primary characteristic.

Q. 193-194
Hapten.

Hapten
They are non-immunologic, but has immunological reactivity (Combine with Ab).
They are two types :
• Complex hapten (precipitate with specific ab)
• Simple hapten (non precipitating)

Hapten requires carrier for specific ab production

195. High molecular weight antigens.
Antigens: An antigen is any substance which, when introduced into the vertebrate host, stimulates the prod n of ab's and reacts with performed ab's.
High mwt Ag's which posses both immunogenicity and specificity are called complete Ag's.

196. Less than 1000 Deltons.
Antigens are generally proteins/polysaccharides with molecular weights greater than 10000 Daltons.
Antigenic determinants/happens generally have a mwt of less than 1000 Daltons.

197. Alloantigen/Isoantigen.
A type of tissue-specific ag which is present in one individual of a species but not in another is called as alloantigen/is antigen ex: Human blood group system.

NUMERICAL TYPE QUESTIONS

3. Component C_1 has 3 subunits; they are C_{1q}, $C1_r$ & C_{1s} which are stabilized by Ca^{+2}. C_{1q} binds with Ab.

4. T – 80% B – 20%

6. 9, 70, 000

mwt of IgG	IgA	IgM	IgD	IgE
1,46,000	3,85,000	9,70,000	1,85,000	1,90,000

7. 20 days

	IgG	IgA	IgM	IgD	IgE
Half life in days	20	6	10	3	2

8. 5%

Distribution (% intravascular)	IgG	IgA	IgM	IgD	IgE
	45	5	95	75	50

9. MHC
The genes of major histocompatibility complexes (MHC) are involved in the recognition between self and non-self ag's. In humans the MHC genes are present on chromosome no. 6 there are A, B, C, D and DR loci.

11. 110 amino acids.
The 1st 108 amino acids in k eight chain,
1st 109 amino acids in λ eight chain.
Heavy chains constitute the variable region 1st 110 amino acids.

EXERCISE – II

MCQ TYPE QUESTIONS

1. Generation of junctional diversity starts as the proteirs, recombination activating gene-1 and -2 (RAG1 and RAG2), along with DNA repair proteins, such as Artemis, are responsible for single-stranded cleavage of the hairpin loops and addition of a series of palindromic, 'P' nucleotides. Subsequent to this, the enzyme, terminal deoxynucleotidyl transferase (TdT), adds further random 'N' nucleotides. The newly synthesised strands anneal to one another, but mismatches are common.

2. Affinity maturation is the process by which THC cell-activated B cells produce antibodies with increased affinity for antigen during the course of an immune response. With repeated exposures to the same antigen, a host will produce antibodies of successively greater affinities once the memory has been generated. Whereas Isotope switching refers to the generation of IgA, IgD, IgM, IgE, IgG from IgM in response to a particular Ag. For e.g.: IgM will undergo isotope switching to IgE in response to Allergen reaction.

3. Activated effector T cells or CD4+ cells initially contact with the macrophages which possess its receptor and stimulate microbial activity. It also make contact with B cells and stimulate differentiation into plasma cells. On contrary CD8+ cells make contact with B cells and stimulate differentiation into plasma cells. CD4 is a co-receptor that assists the T cell receptor (TCR) in communicating with an antigen-presenting cell. Using its intracellular domain, CD4 amplifies the signal generated by the TCR by recruiting an enzyme, the tyrosine kinase Lck, which is essential for activating many molecular components of the signaling cascade of an activated T cell.

4. Inability of T cells to produce or responds to antigens.
5. Only IgG and IgA has subtypes. IgG has 4 subtypes: Gamma1, Gamma2, Gamma3, Gamma4
6. Main 4 complement proteins are the initiator of the membrane attack complex are (C5b, C6,C7 and C8).
7. Splenocytes obtained from Spleen.
8. Generaly oligosaccharide is present.
9. Y- specific IgG will be high on day 84
10.

Group I	Group II
P. Mast cells	4. Defense against helminthic infection
Q. Natural killer cells	2. Expression of CD 56
R. Neutrophils	3. Contains azurophilic granules
S. Dendritic cells	6. Contains long membranous projections

11.

Group I	Group II
P. Chloramphenicol	3. Inhibits protein synthesis by acting on 50S ribosomal subunit
Q. Rifampicin	4. Interferes with RNA polymerase activity
R. Tetracycline	1. Inhibits protein synthesis by acting on 30S ribosomal subunit
S. Quinolone	2. Interferes with DNA replication by inhibiting DNA gyrase

12. Systemic acquired resistance (SAR) is a mechanism of induced defense that confers long-lasting protection against a broad spectrum of microorganisms. SAR requires the signal molecule salicylic acid (SA) and is associated with accumulation of pathogenesis-related proteins, which are thought to contribute to resistance. Using the model plant Arabidopsis, it was discovered that the isochorismate pathway is the major source of SA during SAR. In response to SA, the positive regulator protein NPR1 moves to the nucleus where it interacts with TGA transcription factors to induce defense gene expression, thus activating SAR.
13. Immunodiffusion in gels encompasses a variety of techniques, which are useful for the analysis of antigens and antibodies. An antigen reacts with a specific antibody to form an antigen-antibody complex, the composition of which depends on the nature, concentration and proportion of the initial reactants. Immunodiffusion in gels are classified as single diffusion and double diffusion. In ouchterlony double diffusion, both antigen and antibody are allowed to diffuse into the gel. This assay is frequently used for comparing different antigen preparation.
14. Cholera toxin is produced by the CTXf bacteriophage residing within the bacteria. Cholera toxin consists of an A subunit coupled to a B subunit: the A subunit consists of an A1 domain containing the enzymatic active site, and an A2 domain that has an a-helical tail, while the B subunit contains five identical peptides that assemble into a pentameric ring surrounding a central pore. The A and B subunits of cholera toxin are produced in the cytosol by two genes that overlap by one base, and are then assembled into a toxin in the periplasm located between the inner membrane and the outer cell wall of the bacteria. When cholera toxin is released from the bacteria in the infected intestine, it binds to the intestinal cells known as enterocytes through the interaction of the pentameric B subunit of the toxin with the GM1 ganglioside receptor on the intestinal cell, triggering endocytosis of the toxin. Next, the A/B cholera toxin must undergo cleavage of the A1 domain from the A2 domain in order for A1 to become an active enzyme. Once inside the enterocyte, the enzymatic A1 fragment of the toxin A subunit enters the cytosol, where it activates the G protein GS? through an ADP-ribosylation reaction that acts to lock the G protein in its GTP-bound form, thereby continually stimulating adenylate cyclase to produce cAMP. The high cAMP levels activate the cystic fibrosis transmembrane conductance regulator (CFTR), causing a dramatic efflux of ions and water from infected enterocytes, leading to watery diarrhea.
15. Tyrosine kinase is used to receive insulin hormone signals and regulates the level of insulin in the body. Cycli GMP is a part of 7 helical transmembrane domain protein comprising of G-coupled receptors whose receptor protein is guanylyl cyclase. GTPase activating protein or GAP protein is used for the inactivation of G-proteins and stop the signaling. Nuclear receptor receives thyroid hormone as a signal for signal transduction.
16. The bulk of the immunoglobulins is found in the IgG fraction, which also contains most of the antibodies. The IgM molecules are apparently pentamers-aggregates of five of the IgG molecules. Electron microscopy shows their five subunits to be linked to each other by disulfide bonds in the form of a pentagon. The IgA molecules are found principally in milk and in secretions of the intestinal mucosa. Some of them contain, in addition to a dimer of IgG, a "secretory piece" that enables the passage of IgA molecules between tissue and fluid; the structure of the secretory piece is not yet known. The IgM and IgA immunoglobulins and antibodies contain 10 to 15 percent carbohydrate; the carbohydrate content of the IgG molecules is 2 to 3 percent.
17. Protein A is a cell wall component produced by several strains of *Staphylococcus aureus* that consists of a single polypeptide chain and contains little or no carbohydrate. The Protein A molecule contains four high-affinity binding sites capable of interacting with the Fc region from IgG of several species including human and rabbit. Optimal binding occurs at pH 8.2, although binding is also effective at neutral or physiological conditions (pH 7.0 to 7.6). The interaction between Protein A and IgG is not equivalent for all species. Even within a species, Protein A interacts with some subclasses of IgG and not others. For instance, human IgG1, IgG2 and IgG4 bind strongly, while IgG3 does not bind. There are also many instances in which monoclonal antibodies do not bind to Protein A, especially the majority of rat immunoglobulins and mouse IgG1.
18. Herceptin is approved for the treatment of early-stage breast cancer that is Human Epidermal growth factor Receptor 2-positive (HER2+) and has spread into the lymph nodes, or is HER2+ and has not spread into the lymph nodes. If it has not spread into the lymph nodes, the cancer needs to be estrogen receptor/progesterone receptor (ER/PR)-negative or have one high risk feature. Herceptin has two approved uses in metastatic breast cancer: i) Herceptin in combination with the chemotherapy drug Taxol (paclitaxel) is approved for the first line treatment of Human Epidermal growth factor Receptor 2-positive (HER2+) metastatic breast cancer and ii) Herceptin alone is approved for the treatment of HER2+ breast cancer in patients who have received one or more chemotherapy courses for metastatic disease.
19. Adjuvants are compounds that enhance the specific immune response against co-inoculated antigens. Adjuvants can be used for various purposes: (i) to enhance the immunogenicity of highly purified or recombinant antigens; (ii) to reduce the amount of antigen or the number of immunizations needed for protective immunity; (iii) to improve the efficacy of vaccines in newborns, the elderly or immunocompromised persons; or (iv) as antigen delivery systems for the uptake of antigens by the mucosa. Some of the features involved in adjuvant selection are: the antigen, the species to be vaccinated, the route of administration and the likelihood of side-effects.
20. T cell recognition of antigen requires that a complex form between peptides derived from the protein antigen and cell surface glycoproteins encoded by genes within the major histocompatibilty complex (MHC). MHC class II molecules present both extracellular (exogenous) and internally synthesized (endogenous) antigens to the CD4 T cells subset of lymphocytes. Endogenous antigens are internally generated molecules that become presented on the cell surface in the complex with class I histocompatibilty molecules (MHC I). Endogenous antigens may result from exogenous viral or bacterial infections that have altered the host cell. In autoimmune disorders, endogenous, self-molecules induce autoimmune attack by CD8+ Tc/CTLs that have escaped negative selection in the thymus.
21. The nine multidrug resistance proteins (MRPs) represent the major part of the 12 members of the MRP/CFTR subfamily belonging to the 48 human ATP-binding cassette (ABC) transporters. Cloning, functional characterization, and cellular localization of most MRP subfamily members have identified them as ATP-dependent efflux pumps with broad substrate specificity for the transport of endogenous and xenobiotic anionic substances localized in cellular plasma membranes. Multidrug resistance protein (MRP) is a broad specificity, primary active transporter of organic anion conjugates that confers a multidrug resistance phenotype when transfected into drug-sensitive cells. The protein was the first example of a subgroup of the ATP-binding cassette superfamily whose members have three membrane-spanning domains (MSDs) and two nucleotide binding domains.
22. Sulfonamides were the first drugs acting selectively on bacteria which could be used systemically. Today they are infrequently used, in part due to widespread resistance. The target of sulfonamides, and the basis for their selectivity, is the enzyme dihydropteroate synthase (DHPS) in the folic acid pathway. Quinolones are molecules structurally derived from the

heterocyclic aromatic compound quinoline, the name of which originated from the oily substance obtained after the alkaline distillation of quinine. They have antimicrobial activity because of their action against the topoisomerase enzyme. Erythromycin is a macrolide antibiotic used to treat bacterial infections. By binding to the 50s subunit of the bacterial 70s rRNA complex, protein synthesis and subsequent structure and function processes critical for life or replication are inhibited. They prevent peptidoglycan synthesis. Cephalosporins are medicines that kill bacteria or prevent their growth. Cephalosporins disrupt the synthesis of the peptidoglycan layer of bacterial cell walls. The peptidoglycan layer is important for cell wall structural integrity. The final transpeptidation step in the synthesis of the peptidoglycan is facilitated by transpeptidases known as penicillin-binding proteins (PBPs). PBPs bind to the D-Ala-D-Ala at the end of muropeptides (peptidoglycan precursors) to crosslink the peptidoglycan. Beta-lactam antibiotics mimic the D-Ala-D-Ala site, thereby competitively inhibiting PBP crosslinking of peptidoglycan.

23. Immunoglobulin idiotypes are serologically defined determinants associated with the variable (V) region of antibody molecules

24. All the blood grouping reactions are agglutination reactions. In these the antibodies against A and B antigens when added, they bind to the cell surface and result in dumping of cells. These clumps or settling down of cells is referred to as agglutination.

25. All T cells originate from haematopoietic stem cells in the bone marrow. B cells undergo their complete maturation in the thymus while, T cells during their early stages of maturation migrate to the thymus and then undergo a process of maturation. They are released as competent and mature T cells from Thymus.

26. A humanized chain is a chain in which the complementarity determining regions (CDR) of the variable domains are foreign (originating from one species other than human, or synthetic) whereas the remaining chain is of human origin.

27. **A nude mouse** is a laboratory mouse from a strain with a genetic mutation that causes a deteriorated or absent thymus, resulting in an inhibited immune system due to a greatly reduced number of T cells.

28. Heat-inactivation (heating to 56°C for 30 minutes) of serum is done to inactivate complement, a group of proteins present in sera that are part of the immune response.

29. Digoxin- cardiovascular disorder
Stevioside- sweetener
Atropine- Muscle relaxant
Vinblastine- anticancer agent

30. Immunoglobulins (Ig) are glycoproteins molecules that are produced by plasma cells in response to an immunogen and which functions as antibodies. The immunoglobulins derived their name from the findings that they migrate with globular proteins when antibody containing serum is placed in an electric field. Immunoglobulins generally assume as one of the two roles-It acts as plasma membrane bound antigen receptor on the surface of B cells.

31. The sequence of oligosaccharides determines whether the antigen is A, B, or A1. The ABO blood group antigens are attached to oligosaccharide chains that project above the RBC surface. These chains are attached to proteins and lipids that lie in the RBC membrane. The ABO locus has three main allelic forms: A, B, and O. The A allele encodes a glycosyltransferase that produces the A antigen (N-acetylgalactosamine is its immunodominant sugar), and the B allele encodes a glycosyltransferase that creates the B antigen (D-galactose is its immunodominant sugar). The O allele encodes an enzyme with no function, and therefore neither A or B antigen is produced, leaving the underlying precursor (the H antigen) unchanged. These antigens are incorporated into one of four types of oligosaccharide chain, type 2 being the most common in the antigen-carrying molecules in RBC membranes. Thus, in ABO system of blood grouping, the antigenic determinants are of carbohydrate origin.

32. Although similar to Class I, the MHC Class II molecule is composed of two membrane spanning proteins. Each chain is approximately 30 kDa in size, and made of two globular domains. The domains are named α-1, α-2, β-1 and β-2. The two regions farthest from the membrane are α-1 and β-1. The two chains associate without covalent bonds. The MHC molecules ability to present a wide range of antigenic peptides for T cell recognition requires a compromise between broad specificity and high affinity. Class II molecules are dimers consisting of an α and β polypeptide chain. Each chain contains an immunoglobulin like region, next to the cell membrane. The antigen binding cleft, composed of two α-helices above a β-pleated sheet, specifically binds short peptides, about 15 to 24 residues long. The amino acid sequence around the binding site, which specifies the antigen binding properties, is the most variable site in the MHC molecule.

33. Hepatitis A virus (HAV), an atypical Picornaviridae that causes acute hepatitis in humans, usurps the HAV cellular receptor 1 (HAVCR1) to infect cells. HAVCR1 is a class 1 integral membrane glycoprotein that contains two extracellular domains: a virus-binding immunoglobulin-like (IgV) domain and a mucin-like domain that extends the IgV from the cell membrane. One of the early hallmarks of HIV infection is the impairment of a variety of CD4+ T-cell functions including T-cell colony formation, autologous mixed lymphocyte reactions, expression of interleukin-2 (IL-2) receptors, and IL-2 production. The nicotinic acetylcholine receptor (nAChR) is the first identified receptor for rabies virus. Rabies virus antigen was detected at sites coincident with nAChR. The nicotinic acetylcholine receptor (nAchR) is located at the postsynaptic muscle membrane. Heparan sulfate is a ubiquitously distributed polysulfated polysaccharide that is involved in the initial step of herpes simplex virus type 1 (HSV-1) infection. The virus interacts with cell-surface heparan sulfate to facilitate host-cell attachment and entry. 3-O-Sulfonated heparan sulfate has been found to function as an HSV-1 entry receptor.

34. Immunoglobulin M, or IgM for short, is a basic antibody that is produced by B cells. IgM is by far the physically largest antibody in the human circulatory system. It is the first antibody to appear in response to initial exposure to antigen. The spleen is the major site of specific IgM production.the same time because the large size of most antigens hinders binding to nearby sites.

35. Alzheimer's is a form of dementia which happens due to the deposition of amyloid beta (Aß). This happen in neural juncture of brain. Mad cow is a fatal neurodegenative disease caused in cattles due to a misfolded protein called Prions. Sickle-cell disease is a recessive genetic blood disorder. It causes abnormal, rigid, sickle shape of red blood cells due to mutation of the haemoglobin gene. Swine flu is a infection caused by various types of swine influenza virus. The known SIV strains include influenza C and the sub-types of influenza A known as H1N1, H1N2, H3N1, H3N2 and H2N3.

36. Hybridoma technology is used to combine the properties of site specific antigen activity of B cells with the immortality property of Myeloma M (cancerous) cells. Thus, the cells from a successful hybridoma experiment have single specificity of B cells with indefinite life span property. Anitbodies therefore produced are all of single specificity and hence are monoclonal in nature. Using monoclonal Ab's, cells are categorised on the basis of expression of certain defined markers. Prostate specific Ag and other organ-associated Ag are used with the produced monoclonal Ab for determining nature of a primary tumour.

37. Interferon beta is produced mainly by fibroblasts and some epithelial cell types. The synthesis of IFN beta can be induced by common inducers of interferons, including viruses, double- stranded RNA, and bacteria. It is also induced by some cytokines such as TNF (Tumor Necrosis Factor) and IL1 (Inter Leukin1).

38. A neonatally thymectomized mouse, appeared healthy until about 2 to 4 months of age. Thymectomy after 3 weeks of age was not associated with any significant impairment of homograft immunity.

39. Lymphocytes constitute 20- 40% of the body's white blood cells and 99% of the cells in the lymph. Lymphocytes continually circulate in the blood and lymph and are capable of migrating into the tissue spaces and lymphoid organs, thereby integrating the immune system to a high degree. Lymphocytes interact with foreign antigens in lymph nodes. Lymph nodes are the sites where immune responses are mounted to antigens in lymph. They are encapsulated bean shaped structures containing a reticular network packed with lymphocytes, macrophages, dendritic cells. Lymph nodes are the first organized lymphoid structure to encounter antigens that enter the tissue spaces.

40. A twin is one of two offspring produced in the same pregnancy and developed from one oocyte (monozygotic) or from two oocytes (dizygotic) fertilized at the same time. Dizygotic twins or fraternal twins are heterologous twins developed from two separate oocytes fertilized at the same time. They may be of the same or opposite sex, differ both physically and genetically, and have two separate and distinct placentas and membranes, both amnion and chorion. These twins can accept grafts from each other as they have identical MHC haplotypes and identical T_H cells.

41. HGH is used in dwarfism, Erythropoetin used in Annemia treatment, Factor VIII is been used in Haemophilia treatment and Plalet derived growth factor used in chemotherapeutic drug induced thrombocytopenia treatment.

42. A zinc finger protein is a DNA-binding protein domain consisting of zinc fingers ranging from two in the *Drosophila* regulator ADR1, the more common three in mammalian Sp1 up to nine in $TF_{111}A$. They occur in nature as the part of transcription factors conferring DNA sequence specificity as the DNA-binding domain.

43. PUFAs are long-chain fatty acids containing two or more double bonds.

They provide structural & functional characteristics, and are involved in a wide range of biological components including membranes (in phospholipids). Equally important PUFAs serve as precursors for conversion into metabolites that regulate critical biological functions.Biodiesel is a non-petroleum and environmentally-friendly alternative to petroleum-based diesel fuel and not fatty acid.PUFA is fatty acid it is not used as it forms epoxide in presence of oxygen.

44. A nude mouse is a laboratory mouse from a strain with a genetic mutation that causes a deteriorated or absent thymus, resulting in an inhibited immune system due to a greatly reduced number of T cells. The phenotype, or main outward appearance of the mouse is a lack of body hair, which gives it the "nude" nickname. The nude mouse is valuable to research because it can receive many different types of tissue and tumor grafts, as it mounts no rejection response.

NUMERICAL TYPE QUESTIONS

1. Given, Number of functional gene segments, $V_\alpha = 79$

Number of functional gene segments, $J_\alpha = 38$

Number of functional gene segments, $V_\beta = 21$

Number of functional gene segments, $D_\beta = 2$

Number of functional gene segments, $J_\beta = 11$

So, total number of possible combinations for α and β chains are:

$({}^{79}C \times {}^{38}C) + ({}^{21}C \times {}^{2}C \times {}^{11}C) = 1.375 \times 10^6$

■■